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**Predictive Modeling for Insurance Claim Probability -
DS81N25A**

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STATEMENT OF ORIGINALITY

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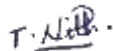
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ABSTRACT

The rise in motor vehicle usage has significantly increased the demand for car insurance services, compelling insurers to adopt robust data-driven mechanisms for risk assessment and management. This project aims to develop a predictive model capable of estimating the probability of a policyholder filing a car insurance claim within the next six months. Utilizing a publicly available dataset from Kaggle comprising customer demographics, vehicle specifications, and policy-related features this study incorporates data preprocessing, exploratory data analysis, feature engineering, and multiple machine learning algorithms were applied to generate robust, interpretable predictions.

Several classification models, including Logistic Regression, Random Forest, Gradient Boosting, and Neural Networks, were evaluated using key metrics such as precision, recall, F1-score, and ROC-AUC to address the class imbalance typically present in insurance claim datasets. The best-performing model underwent additional robustness checks via cross-validation and interpretability analysis using SHAP values.

The outcomes of this research offer valuable insights for insurance providers to enhance risk stratification, detect potential fraudulent claims, and support personalized pricing strategies. Ultimately, the developed predictive framework can aid in informed decision-making, reduce financial losses, and improve customer satisfaction across the insurance domain.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	Artificial Intelligence
API	Application Programming Interface
AUROC	Area Under the Receiver Operating Characteristic Curve
BERT	Bidirectional Encoder Representations from Transformers
CSV	Comma-Separated Values
CV	Cross-Validation
CRISP-DM	Cross-Industry Standard Process for Data Mining
DL	Deep Learning
EDA	Exploratory Data Analysis
ESC	Electronic Stability Control
ECW	Emergency Collision Warning
FP	False Positive
FN	False Negative
GLM	Generalized Linear Model
GNN	Graph Neural Network
IoT	Internet of Things
ID	Identifier
IQR	Interquartile Range
KNN	K-Nearest Neighbors
LSTM	Long Short-Term Memory (a type of recurrent neural network)
ML	Machine Learning
MI	Mutual Information
MLP	Multi-Layer Perceptron
NLP	Natural Language Processing
PCA	Principal Component Analysis
PC1	Principal Component one
PR-AUC	Precision-Recall Area Under Curve

ROC-AUC	Receiver Operating Characteristic – Area Under Curve
RF	Random Forest
SMOTE	Synthetic Minority Over-sampling Technique
SHAP	SHapley Additive exPlanations
SGD	Stochastic Gradient Descent
SVM	Support Vector Machine
TP	True Positive
TN	True Negative
TPMS	Tire Pressure Monitoring System
LIME	Local Interpretable Model-agnostic Explanations
XAI	Explainable Artificial Intelligence
XGBoost	eXtreme Gradient Boosting
XGB	XGBoost

GLOSSARY

Actuarial Science - A discipline using statistical and mathematical models to assess risk in insurance and finance.

AUROC (Area Under ROC Curve) - A performance metric showing a classifier's ability to distinguish between classes.

BERT - A deep learning model by Google for natural language processing, based on transformer architecture.

Claim - A request by a policyholder to the insurance company for compensation under the policy terms.

Claim Rate - The proportion of instances predicted as claims in the dataset

Cross-Validation - A method to evaluate model performance by splitting the dataset into K parts.

Differential Privacy - A privacy-preserving technique that adds noise to data or results to prevent individual identification.

Deep Learning (DL) - A class of machine learning that uses neural networks with multiple layers to extract complex patterns.

Data Leakage - Unintended sharing of information between training and test datasets, leading to misleading model performance.

Explainable AI (XAI) - A set of methods and tools that make machine learning models transparent and understandable.

F1-Score - Harmonic mean of precision and recall; balances false positives and false negatives.

F1-score (macro) - Average F1-score across classes, treating all classes equally.

Federated Learning - A machine learning technique where models are trained across multiple decentralized devices or servers.

Feature Engineering - The process of transforming raw data into informative features that improve model performance.

GNN (Graph Neural Network) - A neural network that operates on graph structures to capture relational data and interactions.

GLM (Generalized Linear Model) - A traditional statistical model used in insurance for classification and regression.

Hyperparameter tuning - Optimization of model parameters to enhance performance.

IQR (Interquartile Range) - A measure of statistical dispersion used to detect outliers.

Label Encoding - Encoding categorical variables as integers for use in machine learning models.

Multicollinearity - A statistical phenomenon where two or more predictor variables are highly correlated, potentially skewing model interpretation.

Model Drift - A degradation in model performance due to changes in the data over time.

Nominal categorical columns - Contain categories with no inherent order or ranking (e.g., fuel type, vehicle model).

Overfitting - A model learns noise instead of the signal, performing well on training but poorly on unseen data.

Ordinal categorical columns - Represent categories with a meaningful order or hierarchy (e.g., safety ratings like low, medium, high).

One-Hot Encoding - Encoding method that represents each category as a binary vector.

Outlier - A data point significantly different from others, potentially indicating noise or rare events.

Power BI - A Microsoft business analytics tool used for interactive data visualization and business intelligence.

Principal Component Analysis - Principal Component Analysis (PCA) is a statistical technique used to reduce the dimensionality of a dataset by identifying the most important features (principal components) that capture the most variation in the data

PR AUC - Area under the Precision-Recall Curve.

Policyholder - An individual or entity that owns an insurance policy.

Random Forest (RF) - An ensemble learning method that constructs multiple decision trees for classification or regression.

ROC AUC - Area under the Receiver Operating Characteristic Curve.

SMOTE - Synthetic Minority Over-sampling Technique: A method to balance class distribution by creating synthetic samples of the minority class.

StandardScaler - A normalization technique that transforms features to have zero mean and unit variance.

Threshold – The cutoff probability above which the model predicts a positive class.

XGBoost - An optimized gradient boosting algorithm that improves speed and accuracy through regularization.

CHAPTER 1 - INTRODUCTION

1.1. Background

The insurance industry faces increasing challenges in accurately predicting policyholder behavior, particularly regarding risk assessment and claim probability estimation. Accurate claim prediction is fundamental for optimizing pricing strategies, mitigating financial losses, detecting fraudulent activity, and enhancing customer retention. Traditionally, insurers have relied on actuarial models and expert judgment. However, advancements in data science and machine learning now enable leveraging large volumes of structured and unstructured data, improving claim outcome predictions (Wuthrich, 2018[\[11\]](#)).

With the growing availability of data from policyholders, vehicles, and external sources such as weather conditions, market dynamics, and regional risk indicators data-driven predictive models have become indispensable tools for forecasting insurance claims. These models enable insurance companies to identify high-risk customers, detect anomalous and potentially fraudulent claims, and offer personalized services. Consequently, this not only fosters stronger customer relationships but also provides competitive differentiation. As Zhang et al. (2020) [\[19\]](#), demonstrate, integrating predictive models with visualization tools like Power BI, machine learning models offer actionable business intelligence, optimize operational processes, and enhance strategic decision-making across departments.

1.2. Problem Statement

Insurance providers frequently encounter difficulty in accurately predicting whether a policyholder will file a claim within a specific period. Traditional methods often fail to model the complex nature of claim data, leading to poor risk profiling and pricing. This is made worse by issues such as class imbalance and high dimensionality. There is thus a pressing need to develop a reliable and interpretable predictive model that can assess the probability of a claim being filed.

The binary classification objective (0 = no claim, 1 = claim) enables focused risk assessment and informs resource allocation, underwriting decisions, and early fraud detection mechanisms. Furthermore, integrating such a model with a Power BI dashboard facilitates interactive analysis and real-time monitoring for insurance stakeholders and decision-makers (Dal Pozzolo et al., 2015) [\[12\]](#).

1.3. Objectives

- **To develop a machine learning-based predictive model** that estimates the likelihood of a policyholder filing an insurance claim within a six-month period.
- **To explore and preprocess a comprehensive dataset** containing internal variables (e.g., customer demographics, claims history) and external variables (e.g., weather trends, market data).
- **To engineer relevant and informative features, address class imbalance, and optimize predictive performance** through appropriate machine learning techniques.
- **To evaluate the performance of the predictive model and enhance its interpretability** by applying model explanation techniques such as SHAP (Lundberg & Lee, 2017 [\[13\]](#)) and LIME (Ribeiro et al., 2016 [\[14\]](#)).
- **To construct an interactive Power BI dashboard** that integrates model predictions, visualizations, and summary insights to support stakeholder engagement and decision making.

1.3.1 Business Objective

The overarching business objective of this project is to support insurance companies in enhancing their risk assessment capabilities, optimizing pricing strategies, detecting fraudulent claims more efficiently, and enabling customer segmentation through advanced data-driven analytics. The solution aims to contribute to operational efficiency, revenue optimization, and improved service personalization.

1.4. Research Questions

This project aims to answer the following key research questions:

1. What are the most significant factors influencing the likelihood of a policyholder filing a claim?
2. Which machine learning models yield the best predictive performance for insurance claim forecasting?

3. How can class imbalance in the dataset be effectively addressed to improve model reliability?
4. Which features - such as customer demographics, vehicle specifications, or policy history - are most predictive of claim behavior?
5. What visualization strategies in Power BI best enhance the interpretability and presentation of model outputs?
6. How can the predictive system be deployed to allow for real-time use, periodic retraining, and seamless integration into business workflows?

1.5. Scope and Significance

This project encompasses the full lifecycle of predictive modeling for insurance claim probability from initial problem definition and environment setup to data acquisition, preprocessing, exploratory data analysis, feature engineering, model training and validation, performance evaluation, and deployment. Emphasis is also placed on developing a Power BI dashboard to visualize critical insights, including predicted risk scores and claim probabilities.

1.5.1. Significance

For Insurance Companies:

This work offers insurance providers a comprehensive framework for enhancing decision-making and operational efficiency. By leveraging machine learning to identify high-risk policyholders and potential fraud, insurers can significantly reduce claim-related losses, improve customer satisfaction, and offer personalized pricing models based on more accurate risk assessments.

For Research:

This study contributes to the evolving field of predictive analytics in insurance by demonstrating practical machine learning applications. It addresses key challenges such as class imbalance and model interpretability, while integrating analytics with business intelligence tools like Power BI. Additionally, the deployment of the model using a web-based API provides insights into operationalizing machine learning systems for industry use.

1.6. Report Structure Overview

The report is structured as follows:

In Chapter 2 – Literature Review: Reviews prior research on insurance claim prediction using machine learning, focusing on model selection, class imbalance challenges, feature engineering techniques, and evaluation metrics. It identifies current gaps and establishes the foundation for this study.

Chapter 3 – Data Collection & Preprocessing: Describes the data acquisition process and preprocessing steps, including missing value handling, duplicate removal, encoding of categorical variables, feature scaling, and outlier treatment.

Chapter 4 – Exploratory Data Analysis (EDA): explores the data using visual and statistical techniques.

Chapter 5 – Methodology: Details the end-to-end machine learning workflow, including model selection, hyperparameter tuning, feature importance analysis, interpretability techniques, and integration with Power BI.

Chapter 6 – Results and Discussion: Presents model performance metrics, insights from SHAP and LIME explainability tools, and findings visualized through the Power BI dashboard.

Chapter 7 – Conclusion and Future Work: Summarizes the key contributions of the project, outlines practical implications and limitations, and suggests potential future improvements for real-time deployment and retraining mechanisms.

Subsequent chapters elaborate on each stage of this research to develop a robust, interpretable, and deployable insurance claim prediction system.

CHAPTER 2 - LITERATURE REVIEW

2.1. Introduction

Predictive modeling has become essential in the insurance industry, enhancing risk assessment, fraud detection, and claim prediction. Traditional actuarial models, such as Generalized Linear Models (GLMs), remain widely used but struggle with high dimensional and complex datasets.

While ML models improve accuracy by capturing complex relationships, challenges remain in balancing performance with fairness, interpretability, and regulatory compliance. A critical research gap lies in integrating ML based predictive analytics into real time decision making while ensuring fairness, transparency, and adherence to regulations. This study evaluates the strengths and limitations of ML for insurance claims, compares traditional and modern approaches, and identifies key challenges and future directions.

2.2. Theoretical Background & Key Concepts

2.2.1. Actuarial Science and Risk Assessment in Insurance

Conventional actuarial methods struggle to model high-dimensional data, leading insurers to adopt machine learning approaches capable of handling such complexity.

2.2.2. Machine Learning Techniques for Insurance Claims

Recent studies have applied various machine learning algorithms in the insurance domain:

- **Logistic Regression:** A simple, interpretable approach but limited by linear assumptions.
- **Random Forest (RF):** Handles feature interactions and missing values, making it effective for fraud detection (Jain et al., 2023) [\[4\]](#).
- **XGBoost:** A gradient boosting technique that reduces bias and variance, outperforming RF in imbalanced datasets (Harjai et al., 2019) [\[3\]](#).
- **Deep Learning (DL):** Models like LSTMs and CNNs process structured and unstructured data (e.g., claim descriptions, telematics) (Xu et al., 2022) [\[6\]](#).
- **Graph Neural Networks (GNNs):** Emerging models that detect fraud by analyzing relationships between policyholders and claim networks (Williams et al., 2022) [\[5\]](#).

A visual comparison of these models is presented in **Figure 1 (Performance Comparison of ML Models in Insurance Claims)**, highlighting accuracy, interpretability, and computational efficiency across datasets.

2.2.3.Challenges in Insurance Claim Prediction and Fraud Detection

- **Imbalanced Claims Data:** Fraud cases are rare; techniques like SMOTE and cost-sensitive learning help mitigate this (Harjai et al., 2019) [3].
- **Interpretability:** Black-box models (e.g., deep learning) lack transparency, making regulatory compliance difficult. SHAP/LIME methods enhance explainability (Williams et al., 2022)[5].
- **Regulatory Compliance:** Laws such as GDPR mandate fairness in claims processing; biased training data may lead to discriminatory outcomes (Xu et al., 2022) [6].

Understanding why certain models outperform others - considering data distribution and feature relevance - enhances the robustness of model evaluation. One key factor is data distribution, as imbalanced claims data can lead to biased model predictions, requiring techniques like SMOTE or cost-sensitive learning (Harjai et al., 2019) [3] to improve performance. However, synthetic oversampling might introduce noise, affecting generalization in real-world applications.

Additionally, feature importance plays a crucial role in model effectiveness. Some models, such as tree-based algorithms, can naturally capture interactions between features, whereas linear models may struggle with complex relationships. Identifying the most influential features through methods like SHAP or permutation importance (Williams et al., 2022)[5] could provide valuable insights into claim patterns and fraud detection.

Moreover, model limitations must be considered. Black-box models like deep learning offer high accuracy but lack interpretability, making them difficult to use in regulatory environments. While techniques like SHAP and LIME enhance explainability, they may not fully address concerns regarding trust and transparency (Xu et al., 2022) [6]. Ensuring fair and unbiased decision-making requires fairness-aware learning methods and periodic audits to mitigate risks associated with biased training data.

By integrating these aspects into the analysis, the research would offer a more comprehensive evaluation of model performance, providing not just quantitative results but also qualitative insights into their real-world applicability.

2.3. Comparative Analysis: ML vs. Traditional Actuarial Techniques

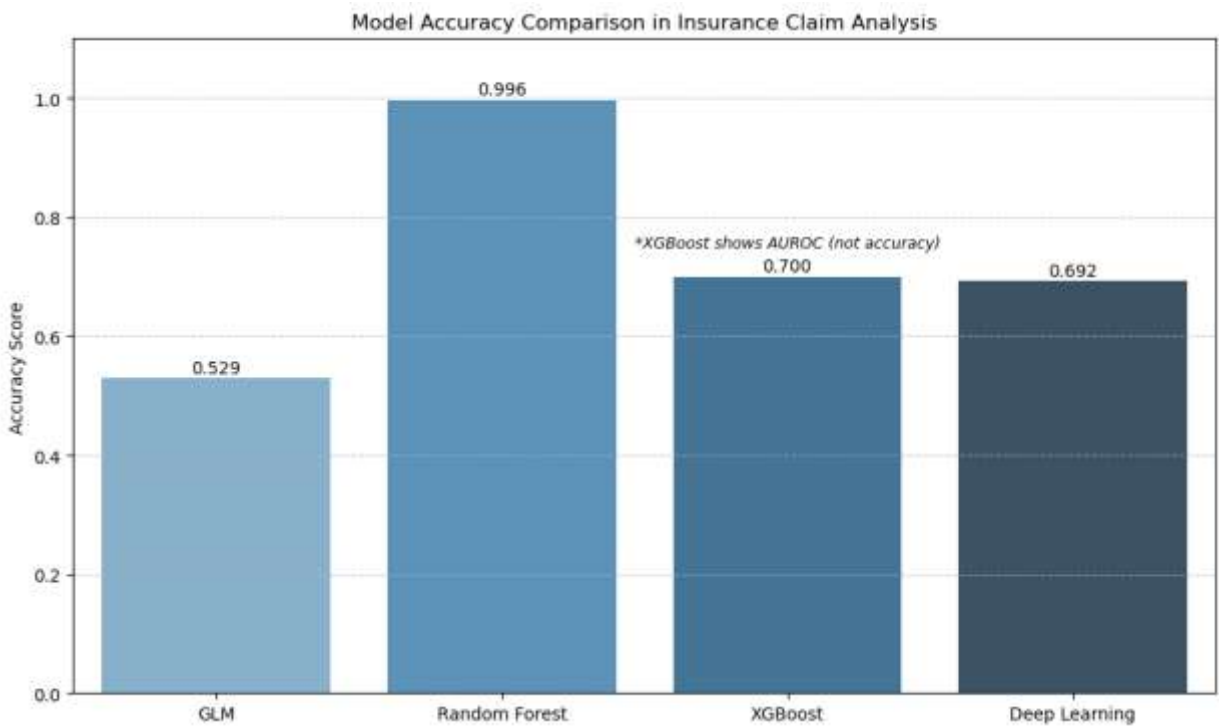


Figure 1 Performance Comparison of ML Models in Insurance Claims

Key Observations from Figure 1 Performance Comparison of ML Models in Insurance Claims

- Generalized Linear Models (GLMs), while interpretable, struggle with non-linearity and complex classifications (52.9% accuracy). They perform competitively in structured telematics-based prediction (AUROC = 0.69).
- Random Forest (RF) achieves high accuracy in binary fraud detection (99.6%) but faces performance drops in multiclass claim classification (93.72% accuracy, 0.619 F1-score). It is computationally moderate and interpretable.
- XGBoost balances accuracy and robustness in handling imbalanced datasets, showing strong discriminative power (AUROC = 0.70). However, it requires careful hyperparameter tuning to prevent overfitting.

- Deep Learning (BERT-based models) effectively process textual claim descriptions but exhibit moderate accuracy (69.2%) and high computational costs. Their "black-box" nature limits interpretability.

Model Comparison Summary

Table 1 presents a comparative analysis of various machine learning models.

Sources: (Jain et al., 2023) [\[4\]](#), (Williams et al., 2022)[\[5\]](#) , and (Xu et al., 2022) [\[6\]](#).

Model	Accuracy (%)	AUROC	Computational Complexity	Interpretability	Handling of Imbalance
GLMs	52.9	0.69	Low	High	Poor
Random Forest	99.6 (binary fraud detection)	0.619 (multiclass claims)	Moderate	Moderate	Good (handles missing data well)
XGBoost	93.72	0.70	High	Low	Excellent (handles imbalance effectively)
Deep Learning (BERT)	69.2	Not reported	Very High	Low ("black-box")	Moderate (depends on architecture)

Table 1 Comparative Analysis ML

2.4. Review of Existing Studies

2.4.1. Comparative Analysis of Studies

The **Table 2**, below summarizes key studies in insurance claim prediction and fraud detection.

Author(s)	Year	Methodologies	Datasets Used	Findings	Limitations
Car insurance claims time distribution function and bounds (Liu et al.,2012) [2]					
• Yan Liu • Qianying Gong • Guangjun Zai	Jul 2012	Time distribution modeling	Proprietary	Identified claim patterns over time	No ML integration
Fraud Detection and Analysis System for Car Insurance Claim Using Random Forest Classifier (Jain et al., 2023) [4]					
• Aryan Jain • Tanmay Dutta • T Senthil Kumar	Nov 2023	Random Forest	Proprietary	Improved fraud detection rates	Lacked interpretability tools
Detecting Fraudulent Insurance Claims Using Random Forests and Synthetic Minority Oversampling Technique (Harjai et al., 2019) [3]					
• Sonakshi Harjai • Sunil Kumar Khatri • Gurinder Singh	Nov 2019	Random Forest + SMOTE	Kaggle	Addressed class imbalance	Focused on synthetic data, not real-world deployment
BERT-based NLP techniques for classification and severity modeling in basic warranty data study (Xu et al., 2022) [6]					
• Shuzhe Xu • Chuanlong Zhang • Don Hong	Jul 2022	Deep Learning (LSTM)	Proprietary	Improved text-based fraud detection	High computational cost

Nightly Automobile Claims Prediction from Telematics-Derived Features: A Multilevel Approach (Williams et al., 2022) [5]					
• Allen R. Williams • Yoolim Jin • Anthony Duer • Tuka Alhani	2022	XGBoost + GNN	IoT sensor data	Improved driving behavior analysis	Privacy concerns with IoT data

Table 2 Comparative analysis of existing studies

The landscape of insurance claim prediction has seen substantial evolution, with studies applying both traditional actuarial methods and modern machine learning (ML) techniques. However, many existing studies exhibit methodological trade-offs that impact real-world applicability.

For example, Liu et al. (2012) [\[2\]](#) modeled the time distribution of car insurance claims, providing valuable insight into temporal patterns. However, their study did not integrate ML methods, limiting predictive accuracy in complex data scenarios. Jain et al. (2023) [\[4\]](#) implemented Random Forests (RF) for fraud detection, significantly improving classification performance. Yet, the lack of interpretability tools in their pipeline raises concerns for practical deployment. Similarly, Harjai et al. (2019) [\[3\]](#) addressed class imbalance using SMOTE alongside RF. While their results improved recall, the reliance on synthetic data may reduce generalization to real-world contexts.

Deep learning-based approaches, such as those by Xu et al. (2022) [\[6\]](#), employed LSTM and BERT architectures to analyze textual claims. Despite their accuracy, deep learning models incur high computational overhead and limited interpretability, restricting their applicability in regulated environments.

In contrast, Williams et al. (2022) [\[5\]](#) explored XGBoost and Graph Neural Networks (GNNs) on IoT sensor data, offering promising results in behavior-based risk modeling. However, their reliance on telematics data raises privacy concerns and limits scalability.

In light of these studies, this project aims to bridge performance and interpretability by employing an ensemble of RF and XGBoost, complemented by SHAP and LIME for transparency. This approach prioritizes both predictive strength and practical utility, addressing key limitations in previous research.

2.4.2. Computational Challenges & Optimizations

A major consideration in ML deployment is computational cost. *Figure 2 (Computational cost VS Accuracy)* illustrates how deep learning models, while accurate, impose significant computational overhead.

- **Latency vs. Accuracy Trade-Off:** Deep learning models (e.g., LSTMs, CNNs) require significant training time and GPU resources, making real-time deployment difficult.
- **Memory and Processing Overhead:** XGBoost can outperform RF in many cases, but it requires fine-tuning of hyperparameters to avoid excessive training times.
- **Optimization Strategies:** Approaches such as model pruning, quantization, and federated learning serve to minimize model complexity without significantly compromising predictive performance.

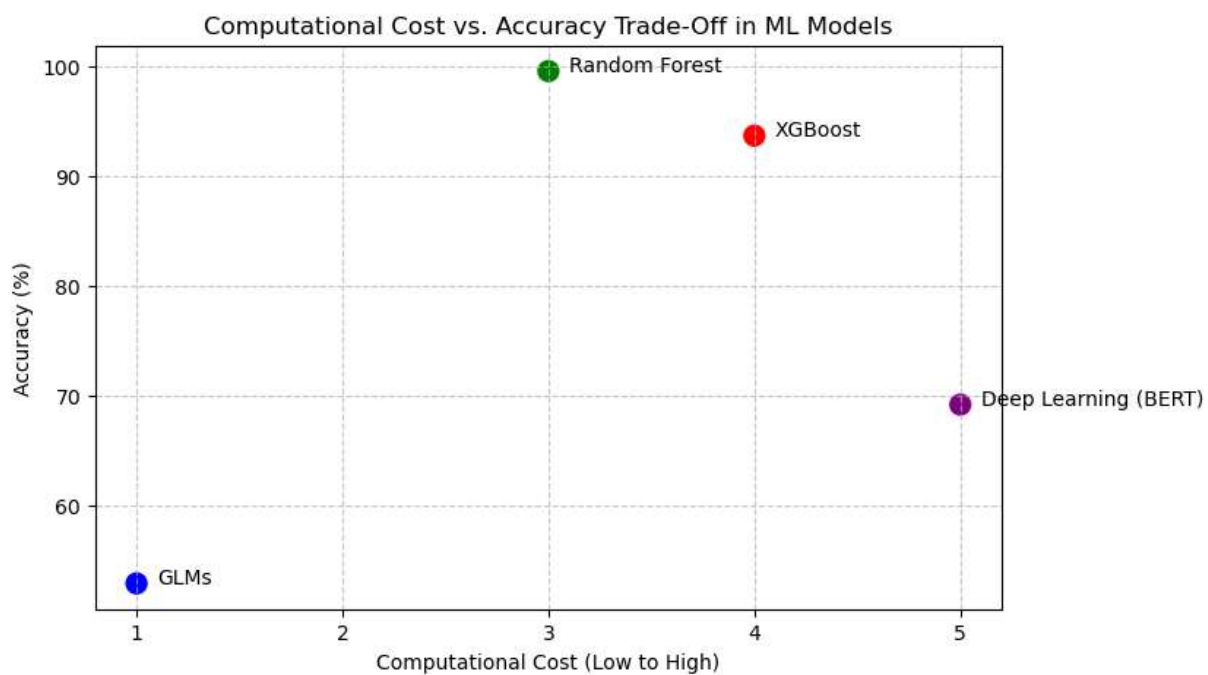


Figure 2 Computational cost VS Accuracy

2.4.3.Real-World Deployment Challenges

A critical aspect of implementing ML-based fraud detection in insurance is the challenge of **real-time deployment**.

Challenge	Impact	Potential Solution
Model Drift	Degradation in accuracy due to evolving fraud patterns	Periodic retraining and monitoring using online learning
Latency	High computation time can delay fraud detection	Deploying lightweight models or edge AI
Interpretability	Black-box models raise compliance issues	Using SHAP, LIME for explainability
Regulatory Constraints	GDPR and data privacy concerns	Federated learning and differential privacy

Table 3 Deployment Challenges

2.4.4.Ethical & Fairness Considerations: Case Studies/Industry Adoption

The integration of machine learning (ML) in insurance claim prediction has significantly enhanced fraud detection and risk assessment. However, these advancements also introduce ethical, fairness, and regulatory challenges that must be addressed for responsible deployment.

One notable case is **Allstate’s telematics-based pricing**, which improves risk assessment but raises concerns over personal data security. Similarly, **Lemonade’s AI-driven** claims processing enhances efficiency yet risks bias in decision-making, potentially leading to unfair claim denials. These concerns underscore the necessity for regulatory oversight to ensure fair and transparent AI-driven insurance applications.

Industry reports from **McKinsey and Deloitte** highlight the increasing adoption of ML in fraud detection and claims optimization while emphasizing the importance of ethical considerations. Compliance with data privacy laws, such as the **General Data Protection Regulation (GDPR) (European Commission, 2018)** [\[23\]](#), is crucial to maintaining transparency and fairness in automated decision-making.

To mitigate ethical risks, insurers are adopting federated learning and differential privacy:

- **Federated learning** enables predictive model training across decentralized datasets without exposing sensitive customer data, preserving both privacy and accuracy.
- **Differential privacy** introduces controlled noise into datasets, ensuring robust fraud detection while preventing personal data leakage.

By **integrating fairness-aware algorithms, explainable AI (XAI) techniques, and privacy-preserving methodologies**, insurers can leverage ML while maintaining **ethical responsibility, regulatory compliance, and stakeholder trust**.

2.5. Gaps in Existing Research

Despite the advancements in ML-driven insurance claim prediction, several key research gaps persist:

- **Limited Use of Unstructured Data:** Most studies focus on structured datasets, overlooking the potential of social media, telematics, and claim narratives for fraud detection.
- **Challenges in Real-time Deployment:** ML models often require high computational resources, making their deployment in low-latency, real-world insurance systems a challenge.
- **Lack of Standardized Evaluation Frameworks:** The absence of universally accepted benchmarking metrics complicates the comparison of ML models across different insurance datasets.
- **Bias in ML Models:** Algorithmic biases can lead to discriminatory claim approvals or denials, necessitating fairness-aware ML approaches.
- **Regulatory Constraints:** GDPR and similar data protection laws limit data accessibility, affecting model training and predictive performance.

This study addresses real-time deployment challenges by proposing an RF-XGBoost ensemble model for low-latency fraud detection, bridging the gap between accuracy and computational efficiency.

2.5.1.Future Directions

To advance ML applications in insurance, future research should focus on integrating structured and unstructured data (e.g., NLP from claim descriptions), enhancing interpretability, and ensuring ethical compliance.

Potential directions include:

- **Hybrid Modeling:** Combining GLMs with ML models to retain interpretability while improving accuracy.
- **Bias Mitigation:** Implementing adversarial debiasing or fairness constraints to prevent discriminatory predictions.
- **Ethical Deployment:** Introducing real-world case studies (e.g., Lemonade AI bias incident) can raise awareness of the implications of automated decisions.
- **Model Efficiency:** Applying quantization and pruning for real-time fraud detection on edge devices.

Such measures will support the development of ML models that are accurate, fair, transparent, and operationally feasible.

2.6. Conclusion

Machine learning has transformed claim prediction and fraud detection but faces challenges:

- **Class Imbalance:** Fraud cases require SMOTE and cost-sensitive learning.
- **Model Interpretability:** Black-box models hinder regulatory compliance.
- **Fairness Concerns:** Bias risks discriminatory claim denials, requiring fairness-aware solutions.
- **Real-time Deployment:** Optimized models improve efficiency and scalability.

Future work should focus on integrating unstructured data (NLP & computer vision), standardizing evaluation metrics (precision, recall, AUROC), and enhancing transparency via XAI. Bridging these gaps will drive the next wave of intelligent, transparent, and regulatory-compliant insurance solutions.

CHAPTER 3 - DATA COLLECTION & PREPROCESSING

3.1. Introduction

In machine learning workflows, data preprocessing is a vital step that converts raw, unrefined inputs into a structured and reliable format suitable for model development. Inadequate preprocessing can degrade model accuracy and result in misleading outcomes. This chapter details the data acquisition process and the systematic procedures undertaken to cleanse and prepare the dataset for predictive modeling within the scope of car insurance claim prediction.

3.2. Data Collection

The dataset was sourced from Kaggle's "Car Insurance Claim Prediction - Classification"[\[1\]](#) repository, which contains detailed records on policies, customers, and vehicles. The primary goal is to predict whether a policyholder will file an insurance claim, represented as a binary target variable `is_claim`.

The dataset is split into two parts:

- Training set: 58,592 records with 44 features including the target variable.

```
Index(['policy_id', 'policy_tenure', 'age_of_car', 'age_of_policyholder',  
      'area_cluster', 'population_density', 'make', 'segment', 'model',  
      'fuel_type', 'max_torque', 'max_power', 'engine_type', 'airbags',  
      'is_esc', 'is_adjustable_steering', 'is_tpms', 'is_parking_sensors',  
      'is_parking_camera', 'rear_brakes_type', 'displacement', 'cylinder',  
      'transmission_type', 'gear_box', 'steering_type', 'turning_radius',  
      'length', 'width', 'height', 'gross_weight', 'is_front_fog_lights',  
      'is_rear_window_wiper', 'is_rear_window_washer',  
      'is_rear_window_defogger', 'is_brake_assist', 'is_power_door_locks',  
      'is_central_locking', 'is_power_steering',  
      'is_driver_seat_height_adjustable', 'is_day_night_rear_view_mirror',  
      'is_ecw', 'is_speed_alert', 'ncap_rating', 'is_claim'],  
      dtype='object')
```

Figure 3 lists all column names - Train Dataset

- Test set: 39,063 records with 43 features (excluding the target variable).

```
Index(['policy_id', 'policy_tenure', 'age_of_car', 'age_of_policyholder',  
      'area_cluster', 'population_density', 'make', 'segment', 'model',  
      'fuel_type', 'max_torque', 'max_power', 'engine_type', 'airbags',  
      'is_esc', 'is_adjustable_steering', 'is_tpms', 'is_parking_sensors',  
      'is_parking_camera', 'rear_brakes_type', 'displacement', 'cylinder',  
      'transmission_type', 'gear_box', 'steering_type', 'turning_radius',  
      'length', 'width', 'height', 'gross_weight', 'is_front_fog_lights',  
      'is_rear_window_wiper', 'is_rear_window_washer',  
      'is_rear_window_defogger', 'is_brake_assist', 'is_power_door_locks',  
      'is_central_locking', 'is_power_steering',  
      'is_driver_seat_height_adjustable', 'is_day_night_rear_view_mirror',  
      'is_ecw', 'is_speed_alert', 'ncap_rating'],  
      dtype='object')
```

Figure 4 lists all column names - Test Dataset

Both datasets include a mix of

- Numerical features (e.g., age of car, policy tenure, engine displacement),
- Categorical features (e.g., fuel type, model, transmission type),
- Binary features (e.g., presence of parking sensors, ESC, power steering),
- And a unique identifier *policy_id*.

3.3. Data Understanding

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 58592 entries, 0 to 58591
Data columns (total 44 columns):
```

#	Column	Non-Null Count	Dtype
0	policy_id	58592 non-null	object
1	policy_tenure	58592 non-null	float64
2	age_of_car	58592 non-null	float64
3	age_of_policyholder	58592 non-null	float64
4	area_cluster	58592 non-null	object
5	population_density	58592 non-null	int64
6	make	58592 non-null	int64
7	segment	58592 non-null	object
8	model	58592 non-null	object
9	fuel_type	58592 non-null	object
10	max_torque	58592 non-null	object
11	max_power	58592 non-null	object
12	engine_type	58592 non-null	object
13	airbags	58592 non-null	int64
14	is_esc	58592 non-null	object
15	is_adjustable_steering	58592 non-null	object
16	is_tpms	58592 non-null	object
17	is_parking_sensors	58592 non-null	object
18	is_parking_camera	58592 non-null	object
19	rear_brakes_type	58592 non-null	object
20	displacement	58592 non-null	int64
21	cylinder	58592 non-null	int64
22	transmission_type	58592 non-null	object
23	gear_box	58592 non-null	int64
24	steering_type	58592 non-null	object
25	turning_radius	58592 non-null	float64
26	length	58592 non-null	int64
27	width	58592 non-null	int64
28	height	58592 non-null	int64
29	gross_weight	58592 non-null	int64
30	is_front_fog_lights	58592 non-null	object
31	is_rear_window_wiper	58592 non-null	object
32	is_rear_window_washer	58592 non-null	object
33	is_rear_window_defogger	58592 non-null	object
34	is_brake_assist	58592 non-null	object
35	is_power_door_locks	58592 non-null	object
36	is_central_locking	58592 non-null	object
37	is_power_steering	58592 non-null	object
38	is_driver_seat_height_adjustable	58592 non-null	object
39	is_day_night_rear_view_mirror	58592 non-null	object
40	is_ecw	58592 non-null	object
41	is_speed_alert	58592 non-null	object
42	ncap_rating	58592 non-null	int64
43	is_claim	58592 non-null	int64

```
dtypes: float64(4), int64(12), object(28)
memory usage: 19.7+ MB
None
```

Figure 5 Train Data information

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 39063 entries, 0 to 39062
Data columns (total 43 columns):
```

#	Column	Non-Null Count	Dtype
0	policy_id	39063 non-null	object
1	policy_tenure	39063 non-null	float64
2	age_of_car	39063 non-null	float64
3	age_of_policyholder	39063 non-null	float64
4	area_cluster	39063 non-null	object
5	population_density	39063 non-null	int64
6	make	39063 non-null	int64
7	segment	39063 non-null	object
8	model	39063 non-null	object
9	fuel_type	39063 non-null	object
10	max_torque	39063 non-null	object
11	max_power	39063 non-null	object
12	engine_type	39063 non-null	object
13	airbags	39063 non-null	int64
14	is_esc	39063 non-null	object
15	is_adjustable_steering	39063 non-null	object
16	is_tpms	39063 non-null	object
17	is_parking_sensors	39063 non-null	object
18	is_parking_camera	39063 non-null	object
19	rear_brakes_type	39063 non-null	object
20	displacement	39063 non-null	int64
21	cylinder	39063 non-null	int64
22	transmission_type	39063 non-null	object
23	gear_box	39063 non-null	int64
24	steering_type	39063 non-null	object
25	turning_radius	39063 non-null	float64
26	length	39063 non-null	int64
27	width	39063 non-null	int64
28	height	39063 non-null	int64
29	gross_weight	39063 non-null	int64
30	is_front_fog_lights	39063 non-null	object
31	is_rear_window_wiper	39063 non-null	object
32	is_rear_window_washer	39063 non-null	object
33	is_rear_window_defogger	39063 non-null	object
34	is_brake_assist	39063 non-null	object
35	is_power_door_locks	39063 non-null	object
36	is_central_locking	39063 non-null	object
37	is_power_steering	39063 non-null	object
38	is_driver_seat_height_adjustable	39063 non-null	object
39	is_day_night_rear_view_mirror	39063 non-null	object
40	is_ecw	39063 non-null	object
41	is_speed_alert	39063 non-null	object
42	ncap_rating	39063 non-null	int64

```
dtypes: float64(4), int64(11), object(28)
memory usage: 12.8+ MB
None
```

Figure 6 Test Data information

As shown in *Figure 5 & Figure 6*, the initial inspection revealed 28 categorical/object-type features and 16 numerical features across both datasets. Additionally, the training dataset includes a binary target variable, `is_claim`.

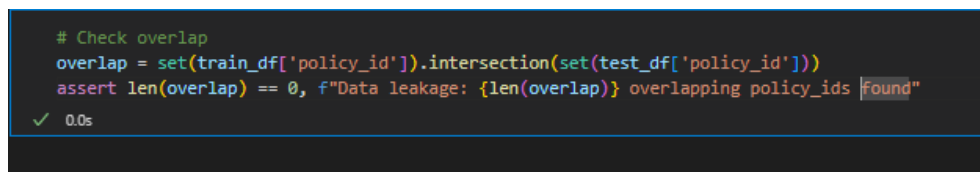
(Refer to [Appendix A.1, A.2](#) for a detailed description of each feature.)

3.3.1. Data Leakage Prevention

To ensure the integrity of model evaluation, a data leakage check was performed. First, it was confirmed that both training and test datasets had the same feature columns and matching data types to ensure consistency during preprocessing and model training.

A crucial step involved verifying that there were no overlapping `policy_id` values between the datasets. As `policy_id` uniquely identifies each record, any duplication could cause the model to learn from the test set indirectly. This confirms the absence of data leakage between the training and test sets. The `policy_id` column was removed later during feature engineering to prevent indirect leakage or unintended learning.

The screenshot (*Figure 7*) shows that the assertion passed without raising an error, indicating **zero overlapping** `policy_id` values. This confirms there is **no data leakage** between the training and test sets based on the unique identifier.



```
# Check overlap
overlap = set(train_df['policy_id']).intersection(set(test_df['policy_id']))
assert len(overlap) == 0, f"Data leakage: {len(overlap)} overlapping policy_ids found"
✓ 0.0s
```

Figure 7 Output of Data Leakage Check for overlapping policy_id

3.3.2. Categorical Feature Overview

The dataset includes a mix of nominal, binary, and textual categorical features capturing key vehicle and policy characteristics.

Nominal Categorical Features: `area_cluster`, `segment`, `model`, `fuel_type`, `engine_type`, `transmission_type`, `steering_type`, and `rear_brakes_type` are nominal features that lack intrinsic ordering. These were label encoded to ensure compatibility with tree-based models.

Binary Features: Features such as `is_power_steering`, `is_parking_camera`, `is_speed_alert`, `is_esc`, `is_tpms`, and similar binary variables indicate the presence or absence of specific functionalities. These were converted to numerical format by mapping ‘Yes’ to 1 and ‘No’ to 0.

Textual Performance Features: The max_power and max_torque features were originally recorded as text values with units (e.g., "88.77bhp@4000rpm"). These were parsed into two separate numerical components: the value and the corresponding RPM, for use in modeling.

```
Unique values in area_cluster: ['C1' 'C2' 'C3' 'C4' 'C5' 'C6' 'C7' 'C8' 'C9' 'C10' 'C11' 'C12' 'C13'
'C14' 'C15' 'C16' 'C17' 'C18' 'C19' 'C20' 'C21' 'C22']
Unique values in segment: ['A' 'C1' 'C2' 'B2' 'B1' 'Utility']
Unique values in model: ['M1' 'M2' 'M3' 'M4' 'M5' 'M6' 'M7' 'M8' 'M9' 'M10' 'M11']
Unique values in fuel_type: ['CNG' 'Petrol' 'Diesel']
Unique values in max_torque: ['60Nm@3500rpm' '113Nm@4400rpm' '91Nm@4250rpm' '250Nm@2750rpm'
'200Nm@3000rpm' '82.1Nm@3400rpm' '200Nm@1750rpm' '85Nm@3000rpm'
'170Nm@4000rpm']
Unique values in max_power: ['40.36bhp@6000rpm' '88.50bhp@6000rpm' '67.06bhp@5500rpm'
'113.45bhp@4000rpm' '88.77bhp@4000rpm' '55.92bhp@5300rpm'
'97.89bhp@3600rpm' '61.68bhp@6000rpm' '118.36bhp@5500rpm']
Unique values in engine_type: ['F8D Petrol Engine' '1.2 L K12N Dualjet' '1.0 SCe' '1.5 L U2 CRDi'
'1.5 Turbocharged Revotorq' 'K Series Dual jet' '1.2 L K Series Engine'
'K10C' 'i-DTEC' 'G12B' '1.5 Turbocharged Revotron']
Unique values in is_esc: ['No' 'Yes']
Unique values in is_adjustable_steering: ['No' 'Yes']
Unique values in is_tpms: ['No' 'Yes']
Unique values in is_parking_sensors: ['Yes' 'No']
Unique values in is_parking_camera: ['No' 'Yes']
Unique values in rear_brakes_type: ['Drum' 'Disc']
Unique values in transmission_type: ['Manual' 'Automatic']
Unique values in steering_type: ['Power' 'Electric' 'Manual']
Unique values in is_front_fog_lights: ['No' 'Yes']
Unique values in is_rear_window_wiper: ['No' 'Yes']
Unique values in is_rear_window_washer: ['No' 'Yes']
Unique values in is_rear_window_defogger: ['No' 'Yes']
Unique values in is_brake_assist: ['No' 'Yes']
Unique values in is_power_door_locks: ['No' 'Yes']
Unique values in is_central_locking: ['No' 'Yes']
Unique values in is_power_steering: ['Yes' 'No']
Unique values in is_driver_seat_height_adjustable: ['No' 'Yes']
Unique values in is_day_night_rear_view_mirror: ['No' 'Yes']
Unique values in is_ecw: ['No' 'Yes']
Unique values in is_speed_alert: ['Yes' 'No']
```

Figure 8 Unique Values of Categorical data

As shown in *Figure 8*, understanding the distribution and unique values of categorical features is crucial for accurate encoding and effective analysis.

3.4. Data Preprocessing

3.4.1. Handling Missing Values

Although the datasets initially appeared to have no missing values based on info() output, a detailed analysis was conducted to confirm this. Functions like isnull().sum() and visualization using the missingno library were employed.

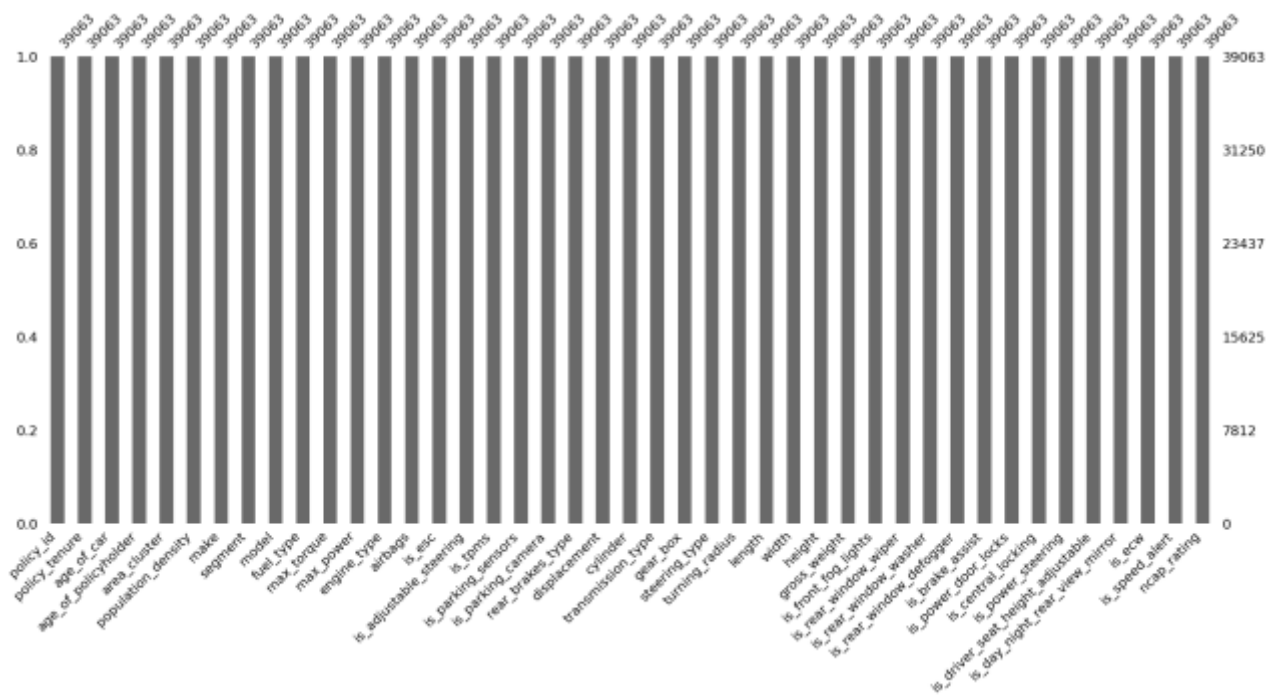


Figure 9 Bar Chart of missing values - Train dataset

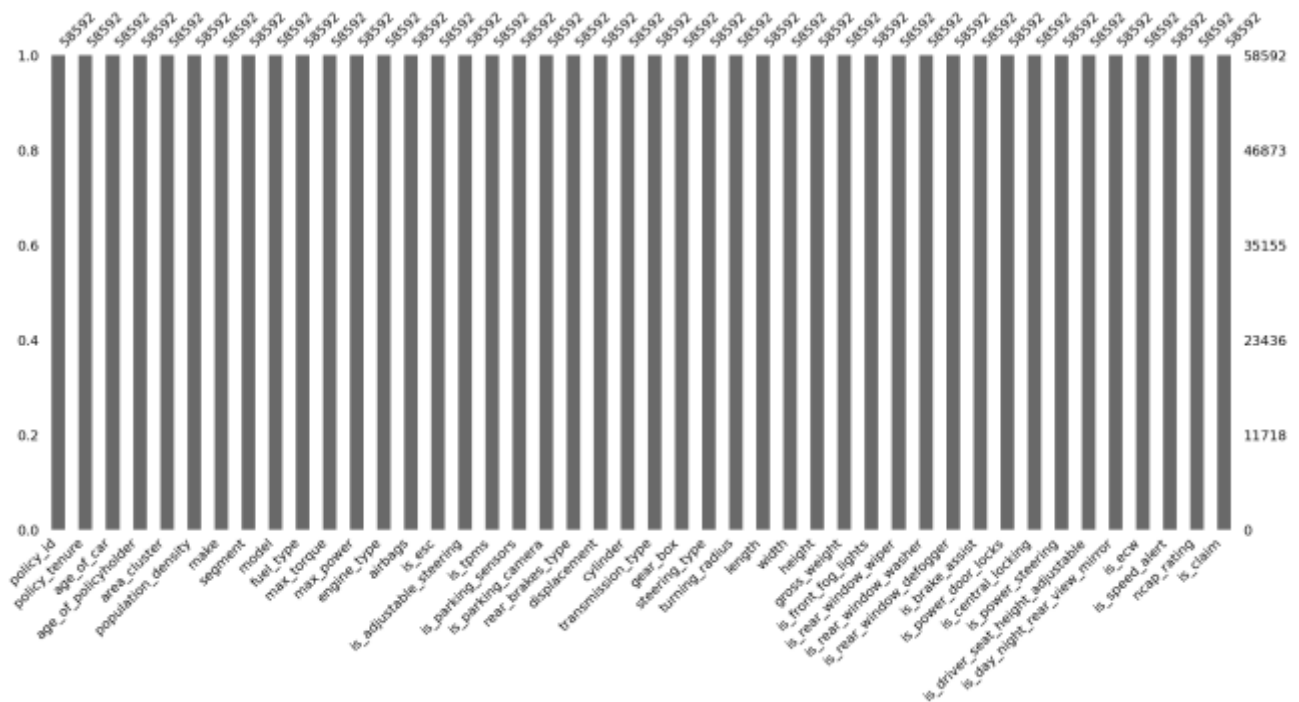


Figure 10 Bar Chart of missing values - Test dataset

The number of values in each column is illustrated by the bar charts in *Figures 9 and 10*. It was observed that all columns contain the same number of values, indicating no missing data in either the training or test datasets.

```

policy_id 0
policy_tenure 0
age_of_car 0
age_of_policyholder 0
area_cluster 0
population_density 0
make 0
segment 0
model 0
fuel_type 0
max_torque 0
max_power 0
engine_type 0
airbags 0
is_esc 0
is_adjustable_steering 0
is_tpms 0
is_parking_sensors 0
is_parking_camera 0
rear_brakes_type 0
displacement 0
cylinder 0
transmission_type 0
gear_box 0
steering_type 0
turning_radius 0
length 0
width 0
height 0
gross_weight 0
is_front_fog_lights 0
is_rear_window_wiper 0
is_rear_window_washer 0
is_rear_window_defogger 0
is_brake_assist 0
is_power_door_locks 0
is_central_locking 0
is_power_steering 0
is_driver_seat_height_adjustable 0
is_day_night_rear_view_mirror 0
is_ecw 0
is_speed_alert 0
ncap_rating 0
is_claim 0
dtype: int64

```

Figure 11 Number of missing values of train dataset

```

policy_id 0
policy_tenure 0
age_of_car 0
age_of_policyholder 0
area_cluster 0
population_density 0
make 0
segment 0
model 0
fuel_type 0
max_torque 0
max_power 0
engine_type 0
airbags 0
is_esc 0
is_adjustable_steering 0
is_tpms 0
is_parking_sensors 0
is_parking_camera 0
rear_brakes_type 0
displacement 0
cylinder 0
transmission_type 0
gear_box 0
steering_type 0
turning_radius 0
length 0
width 0
height 0
gross_weight 0
is_front_fog_lights 0
is_rear_window_wiper 0
is_rear_window_washer 0
is_rear_window_defogger 0
is_brake_assist 0
is_power_door_locks 0
is_central_locking 0
is_power_steering 0
is_driver_seat_height_adjustable 0
is_day_night_rear_view_mirror 0
is_ecw 0
is_speed_alert 0
ncap_rating 0
dtype: int64

```

Figure 12 Number of missing Values of test dataset

Figures 11 and 12, although no missing values were found, an imputation function was developed for robustness, using **mean** for numerical features and **mode** for categorical features.

3.4.2. Removing Duplicates

Both datasets were checked for duplicate entries using the. duplicated().sum() method. While no duplicates were detected, the removal function was retained to ensure robustness.

```

Number of duplicate rows: 0
Dataset shape after removing duplicates: (58592, 44)
Number of duplicate rows after removing rows: 0

```

Figure 13 Removing Duplicates of train dataset

```

Number of duplicate rows: 0
Dataset shape after removing duplicates: (39063, 43)
Number of duplicate rows after removing rows: 0

```

Figure 14 Removing Duplicates of test dataset

3.4.3. Outlier Detection and Treatment



Figure 15 Boxplots of numerical features highlighting outliers in policyholder age and vehicle weight, with potential implications for model robustness.

Outliers in numerical variables were visualized using boxplots in *Figure 15*. Although a few outliers were identified (particularly in age of car and age of policy holder), they were retained to preserve rare but valid cases which are important in fraud and claim prediction tasks.

The function `handle_outliers()` was implemented to detect. However, since no excessively anomalous values were found, no rows were removed.

3.4.4. Feature Engineering

To improve the quality of the data, several transformations were applied:

- 1) **Removal of policy_id:** The `policy_id` column was excluded as it functioned solely as a unique identifier, offering no predictive value and potentially introducing noise in models sensitive to high-cardinality features.
- 2) **Parsing of max_power and max_torque:** Textual entries embedding both values and RPMs were parsed into separate numeric features (`power_value`, `power_rpm`, `torque_value`, `torque_rpm`), enabling accurate modeling of engine performance metrics.
- 3) **Engineering of car_volume:** A new feature was derived to create a meaningful representation of vehicle size while reducing feature redundancy and combines three correlated dimensions into a single composite feature. `car_volume` is intuitive and aligns with common understanding, making it more interpretable for business users.
`car_volume` was engineered as a derived feature:

$$Car_volume = length \times width \times height$$

Equation 1 Car Volume

- 4) **Dropping Redundant Columns (Length, Width, and Height):** These original dimensions were dropped post-engineering to minimize multicollinearity and simplify the model.
- 5) **Several binary "Yes/No" features were converted to 1/0 integers:** Encoding binary features as 0 and 1 enables compatibility and computational efficiency with machine learning models, particularly tree-based classifiers.

Figures 16 and 17 illustrate the dataset following the completion of feature engineering steps.

```
'Categorical features:'
['area_cluster',
 'segment',
 'model',
 'fuel_type',
 'engine_type',
 'rear_brakes_type',
 'transmission_type',
 'steering_type']
'Boolean features:'
['is_esc',
 'is_adjustable_steering',
 'is_tpms',
 'is_parking_sensors',
 'is_parking_camera',
 'is_front_fog_lights',
 'is_rear_window_wiper',
 'is_rear_window_washer',
 'is_rear_window_defogger',
 'is_brake_assist',
 'is_power_door_locks',
 'is_central_locking',
 'is_power_steering',
 'is_driver_seat_height_adjustable',
 'is_day_night_rear_view_mirror',
 'is_ecw',
 'is_speed_alert']
'Numeric features:'
['policy_tenure',
 'age_of_car',
 'age_of_policyholder',
 'population_density',
 'make',
 'airbags',
 'displacement',
 'cylinder',
 'gear_box',
 'turning_radius',
 'gross_weight',
 'ncap_rating',
 'is_claim',
 'torque_value',
 'torque_rpm',
 'power_value',
 'power_rpm',
 'car_volume']
```

Figure 16 Final list of features after feature engineering, including newly derived and transformed variables.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 58592 entries, 0 to 58591
Data columns (total 43 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   policy_tenure                        58592 non-null  float64
1   age_of_car                          58592 non-null  float64
2   age_of_policyholder                 58592 non-null  float64
3   area_cluster                        58592 non-null  object
4   population_density                  58592 non-null  int64
5   make                                58592 non-null  int64
6   segment                             58592 non-null  object
7   model                               58592 non-null  object
8   fuel_type                           58592 non-null  object
9   engine_type                         58592 non-null  object
10  airbags                             58592 non-null  int64
11  is_esc                              58592 non-null  int8
12  is_adjustable_steering              58592 non-null  int8
13  is_tpms                             58592 non-null  int8
14  is_parking_sensors                  58592 non-null  int8
15  is_parking_camera                   58592 non-null  int8
16  rear_brakes_type                    58592 non-null  object
17  displacement                        58592 non-null  int64
18  cylinder                             58592 non-null  int64
19  transmission_type                   58592 non-null  object
20  gear_box                            58592 non-null  int64
21  steering_type                       58592 non-null  object
22  turning_radius                      58592 non-null  float64
23  gross_weight                        58592 non-null  int64
24  is_front_fog_lights                 58592 non-null  int8
25  is_rear_window_wiper                58592 non-null  int8
26  is_rear_window_washer               58592 non-null  int8
27  is_rear_window_defogger             58592 non-null  int8
28  is_brake_assist                     58592 non-null  int8
29  is_power_door_locks                 58592 non-null  int8
30  is_central_locking                  58592 non-null  int8
31  is_power_steering                   58592 non-null  int8
32  is_driver_seat_height_adjustable     58592 non-null  int8
33  is_day_night_rear_view_mirror        58592 non-null  int8
34  is_ecw                              58592 non-null  int8
35  is_speed_alert                      58592 non-null  int8
36  ncap_rating                         58592 non-null  int64
37  is_claim                            58592 non-null  int64
38  torque_value                        58592 non-null  float64
39  torque_rpm                          58592 non-null  int64
40  power_value                         58592 non-null  float64
41  power_rpm                           58592 non-null  int64
42  car_volume                          58592 non-null  int64
dtypes: float64(6), int64(12), int8(17), object(8)
memory usage: 12.6+ MB
```

Figure 17 Data types of variables post feature engineering

3.4.4.1. Summary of Feature Engineering

Table 4 summarizes the key feature engineering steps applied during preprocessing, outlining the specific actions taken, their rationale, and the expected contribution of each transformation to the overall model performance and interpretability.

Feature	Action Taken	Rationale	Contribution to Model
<u>policy_id</u>	Removed	Unique identifier with no predictive value	Prevents <u>overfitting</u> and noise
<u>max_power</u> , <u>max_torque</u>	Parsed into value and RPM components	Converts text to usable numeric form	Improves interpretability and captures engine behavior
<u>length</u> , <u>width</u> , <u>height</u>	Combined into <u>car_volume</u>	Reduces dimensionality and <u>multicollinearity</u>	Simplifies model input and captures vehicle size
Binary features	Encoded to 1/0	Ensures model compatibility	Maintains semantic meaning and enables modeling
Redundant dimensions	Dropped	Avoid redundancy with <u>car_volume</u>	Increases model efficiency and avoids <u>collinearity</u>

Table 4 Summary of Feature Engineering

3.4.5. Categorical Encoding

All categorical features were transformed into numerical formats to be compatible with machine learning models.

- Nominal categorical columns (e.g., fuel_type, model, segment) were label-encoded as they were intended for use in tree-based models, which are not sensitive to numerical ordering.
- Ordinal column ncap_rating was preserved as a numeric ordinal feature due to its inherent safety rating scale.
- Although features such as make, gear_box, cylinder, and airbags are represented as numerical values, they were treated as categorical variables. This is because these values represent distinct categories rather than quantities with meaningful arithmetic relationships. For example, the number of cylinders or airbags indicates type, not magnitude, making them more appropriate for categorical encoding in the modeling process.

A multicollinearity analysis was conducted to identify highly correlated features from **Figure 18**, which could introduce redundancy and affect model interpretability or performance. Pearson correlation coefficients were used to assess pairwise relationships across different feature groups.



- 1. Engine & Performance Features:** Strong multicollinearity was identified among engine and performance features, indicating redundancy. Features like displacement, torque_value, cylinder, gear_box, and gross_weight were removed. power_value and car_volume were retained as the most predictive and representative features of engine capacity.

Feature 1	Feature 2	Correlation	Action/Note
displacement	power_value	0.96	Drop displacement (keep power_value)
displacement	car_volume	0.94	Drop displacement
power_value	torque_value	0.95	Drop torque_value (retain power_value)
power_value	car_volume	0.95	Both retained if highly predictive
cylinder	displacement	0.85	Drop cylinder
gross_weight	car_volume	0.91	Drop gross_weight

Table 5 Multicollinearity Check - Engine & Performance Features

- 2. Make/Model/Segment Clustering:** Moderate correlation was observed among make, model, and segment, suggesting that model may serve as a proxy for both. This insight may inform encoding or clustering strategies.

Feature 1	Feature 2	Correlation	Action/Note
make	model	~0.75–0.79	model may implicitly represent make and segment
make	segment	~0.68	Consider combining or dimensionality reduction
model	segment	~0.73	

Table 6 Multicollinearity Check - Make/Model/Segment Clustering

- 3. Gear & Turning:** Among gear and turning features, gear_box was dropped due to its high correlation with turning_radius.

Feature 1	Feature 2	Correlation	Action/Note
gear_box	turning_radius	~0.95	Drop gear_box if turning_radius is more predictive

Table 7 Multicollinearity Check - Gear & Turning

- 4. Safety Features:** Several binary safety indicators showed soft multicollinearity. While not immediately dropped, they are candidates for future transformation into composite safety scores.

Feature 1	Feature 2	Correlation	Action/Note
is_parking_camera	is_parking_sensors	~0.70	Consider combining into a composite "parking safety" score
is_brake_assist	is_ecw	~0.73	Soft multicollinearity; may combine or retain based on importance
is_power_door_locks	is_central_locking	~0.77	Strongly related; consider dropping one or combining
is_power_steering	is_driver_seat_height_adjustable	~0.72	Consider composite "comfort" or "drive-assist" score
is_speed_alert	is_ecw	~0.69	

Table 8 Multicollinearity Check - Safety Features

3.4.7. Feature Scaling

To ensure consistency across the modeling pipeline, feature scaling was implemented to standardize numerical variables. While tree-based algorithms such as Random Forest and XGBoost are generally unaffected by the scale of input features - since they rely on threshold-based splitting - scaling remains essential for algorithms like Logistic Regression and Neural Networks, which are influenced by differences in feature variance.

Standardization was performed using StandardScaler, which transforms features to have a mean of 0 and standard deviation of 1, according to the formula:

$$z = \frac{x - \mu}{\sigma}$$

Where x is the original value, μ is the feature mean, and σ is the standard deviation

Equation 2 Standard Scaler

This standardization offers several benefits:

- **Accelerates convergence** in gradient-based models such as logistic regression and neural networks by normalizing feature scales (Geron, A. (2019) [\[20\]](#)).
- **Prevents dominance** of features with larger numeric ranges in models that compute distances or gradients (Kuhn, M., & Johnson, K. (2013) [\[21\]](#)).
- **Improves model interpretability** by making the effect of each feature more directly comparable in linear models (Geron, A. (2019) [\[20\]](#)).

Alternative methods considered included:

- **MinMaxScaler**: Effective for bounded or distance-based models (e.g., KNN, SVM) but sensitive to outliers.
- **RobustScaler**: Resistant to outliers but less effective for normally distributed features.
- **MaxAbsScaler**: Useful for sparse data; less suitable here.

StandardScaler was chosen due to its broad compatibility and consistent performance across models sensitive to feature scale.

3.4.8. Handling Class Imbalance

The target variable in the training data set demonstrated a significant class imbalance, with only approximately 6% of records labeled as positive (claims made). This imbalance can lead to models that are biased toward the majority class (non-claims), resulting in poor recall for the minority class the actual claims which are often the most critical to detect in insurance applications.

Among the resampling techniques considered, ADASYN raised recall to 0.91 but reduced precision to 0.05, while class weighting in XGBoost without SMOTE gave F1 = 0.14. SMOTE offered the best precision-recall trade-off and was therefore selected. Extended descriptions of alternative techniques are detailed in [Appendix A.3](#).

In this project, SMOTE was used during cross-validation within the training folds, and class weights were tuned as part of hyperparameter optimization for classifiers that support it. This hybrid approach helps improve recall on the minority class while minimizing overfitting or information loss.

```
is_claim
0.0    0.936032
1.0    0.063968
Name: proportion, dtype: float64
```

Figure 19 Claim Rate

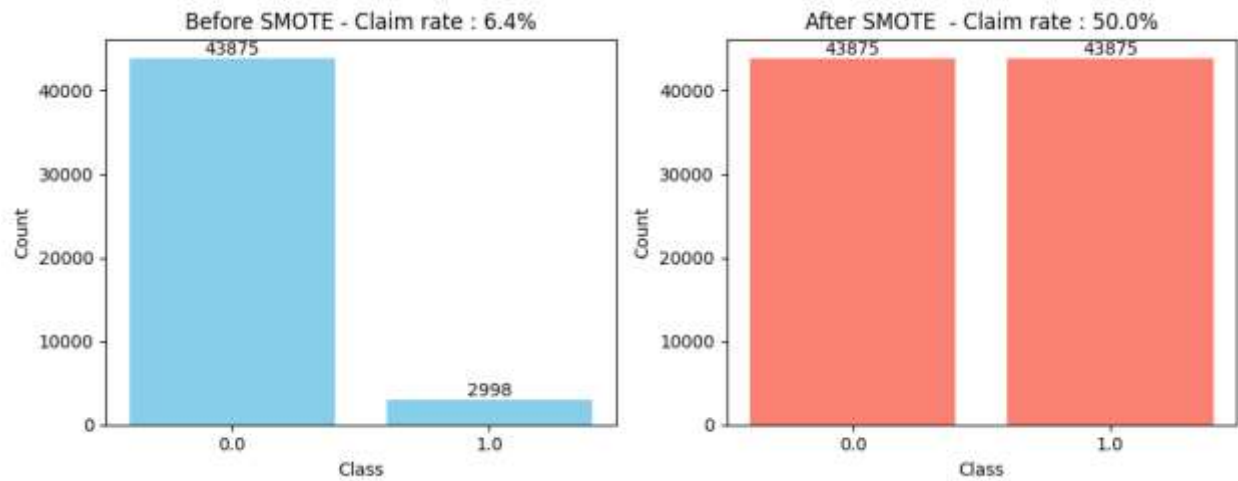


Figure 20 Claim rate before & after SMOTE

Final model evaluations compared the effectiveness of each strategy using performance metrics like F1-score and ROC-AUC.

3.5. Train-Test Split Strategy

In this study, the dataset was pre-split into training and test sets by the data provider. Therefore, no additional splitting (e.g., `train_test_split`) was performed. The training set was used for preprocessing, feature engineering, model training, and cross-validation, while the test set lacking the target variable was reserved strictly for final evaluation. This approach aligns with the provider's design and prevents data leakage

3.6. Final Prepared Dataset

After all preprocessing steps, the cleaned datasets were saved.

- `cleaned_train_data.csv` – Used for model training and evaluation.
- `cleaned_test_data.csv` – Used for final prediction/testing.

These datasets contain consistent, well-structured, and encoded features suitable for machine learning modeling, including binary classification, feature selection, and interpretability.

3.7. Summary of Data Preprocessing steps

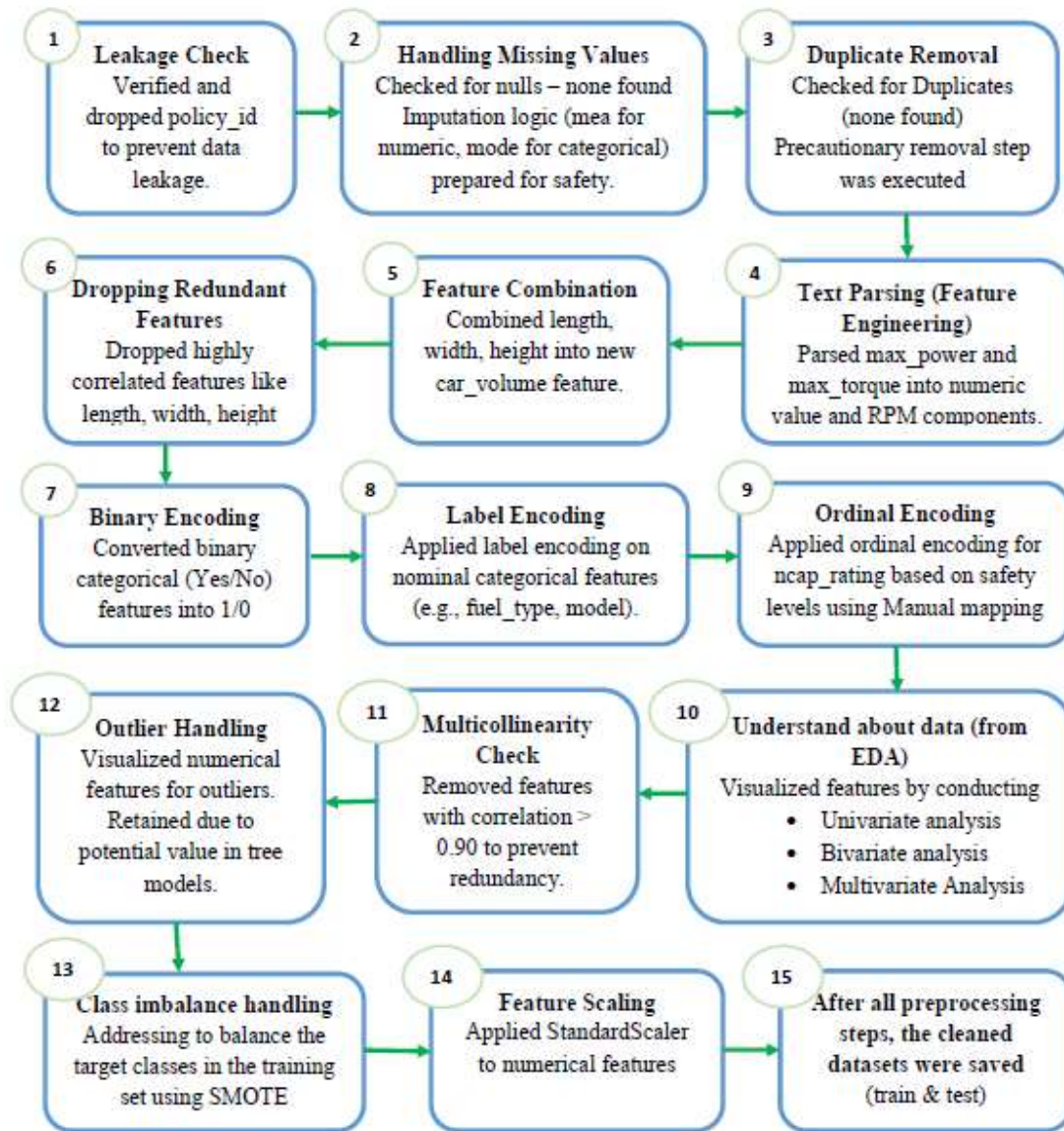


Figure 21 Summary of Data Preprocessing steps

The *Figure 21* presents a concise summary of the preprocessing steps applied to the dataset, including the sequence of operations, the specific techniques used, and the technologies implemented at each stage.

This comprehensive preprocessing ensures that the dataset is optimized for reliable, interpretable, and effective machine learning modeling in car insurance claim prediction.

For a comprehensive list and explanation of the Python libraries used during preprocessing, refer to [Appendix B](#).

CHAPTER 4 - EXPLORATORY DATA ANALYSIS (EDA)

4.1. Introduction

Exploratory Data Analysis (EDA) is a critical phase in the data science process, aimed at understanding the underlying structure, trends, and anomalies in the data before applying predictive modeling. In this project, EDA helped uncover feature distributions, correlations, and potential relationships with the target variable (`is_claim`), guiding subsequent steps in preprocessing, feature selection, and modeling.

4.2. Univariate Analysis

Univariate analysis was performed to study the individual distributions of numerical, categorical, and binary features.

4.2.1. Numerical Features

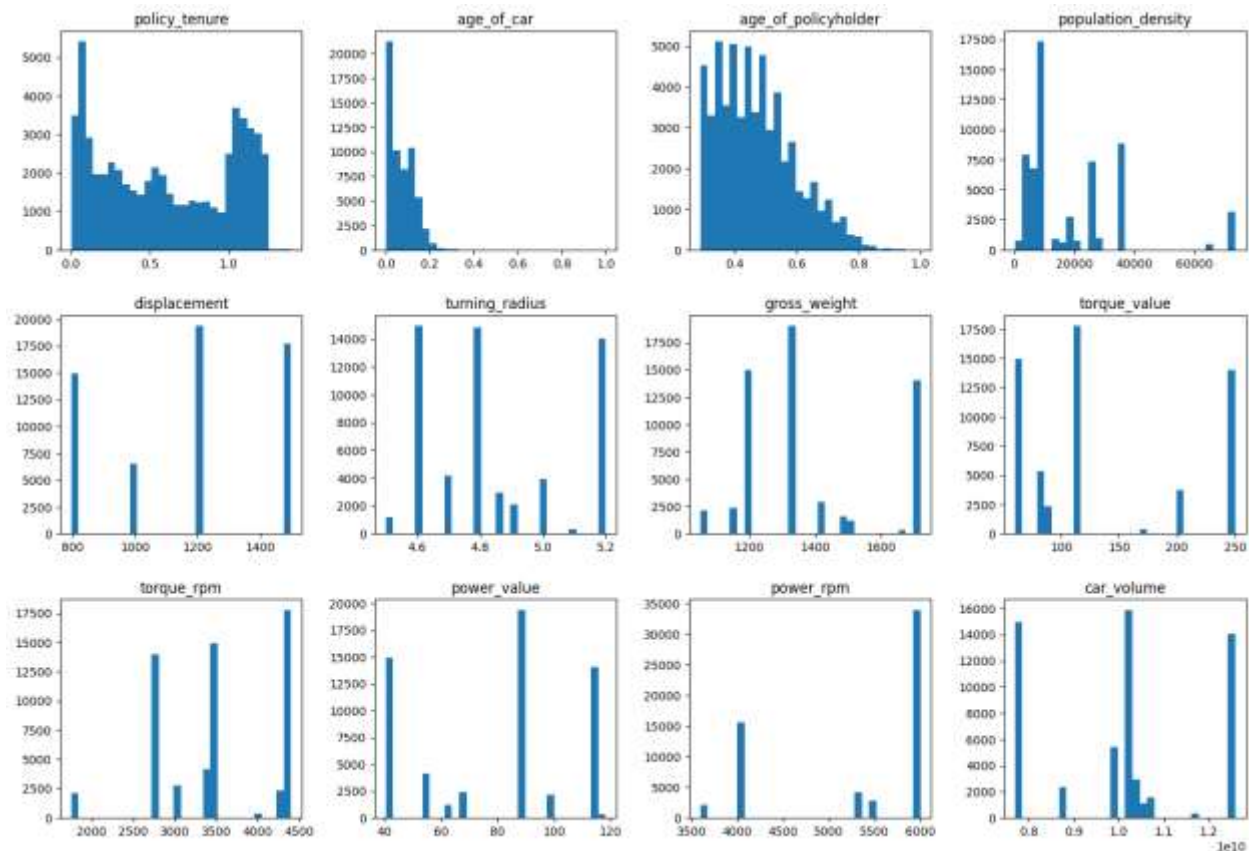


Figure 22 Histograms of Numerical Features

Key numerical features such as `age_of_car`, `age_of_policyholder`, `policy_tenure`, `displacement`, `gross_weight`, `power_value`, and `car_volume` were examined through histograms in *Figure 22*.

Most policyholders were between 20 and 70 years old, with a concentration around 35 to 50. Vehicle age peaked at 2 years, and other variables such as power, volume, and weight showed right-skewed distributions, suggesting the presence of high-end vehicles.

Most policyholders had tenure between 1 to 5 years, which is typical in the insurance domain.

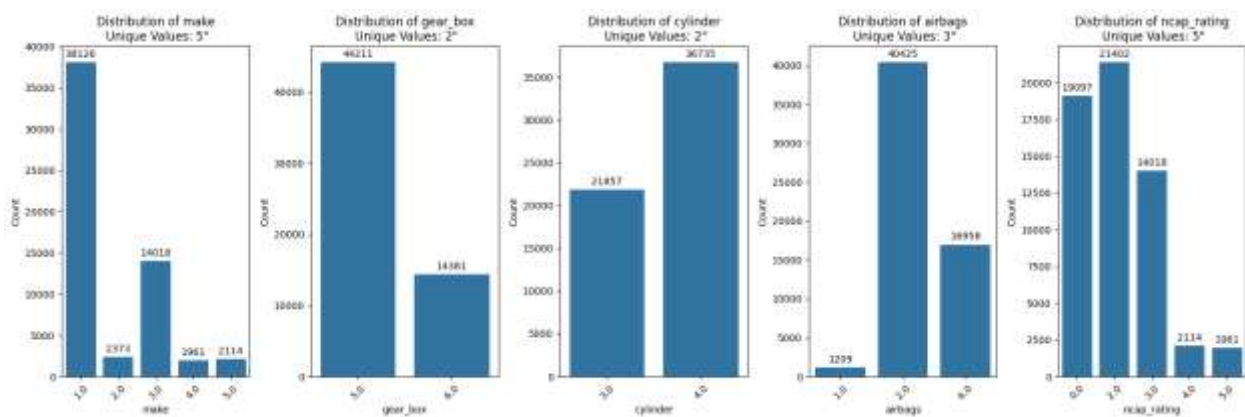


Figure 23 Histograms of make, gear_box, cylinder, airbags, and ncapi_rating

From *Figure 23*, this dataset contains several features that appear numerical but require special consideration due to their categorical or ordinal nature.

‘Make’ represents categorical vehicle manufacturers encoded with arbitrary numbers, carrying no mathematical significance. In contrast, features like ‘gear_box’ and ‘cylinder’ are ordinal, reflecting a natural order (e.g., 6 cylinders > 4), though the intervals between values are not uniform.

Similarly, ‘airbags’ and ‘ncapi_rating’ are ordinal safety indicators, where higher values imply greater protection but not necessarily in equal steps. These features often cluster around common values, reinforcing their discrete, ordered nature.

Proper handling such as ordinal encoding or treating them as categorical is essential to preserve their semantics and prevent distortion in model training.

4.2.2. Categorical Features

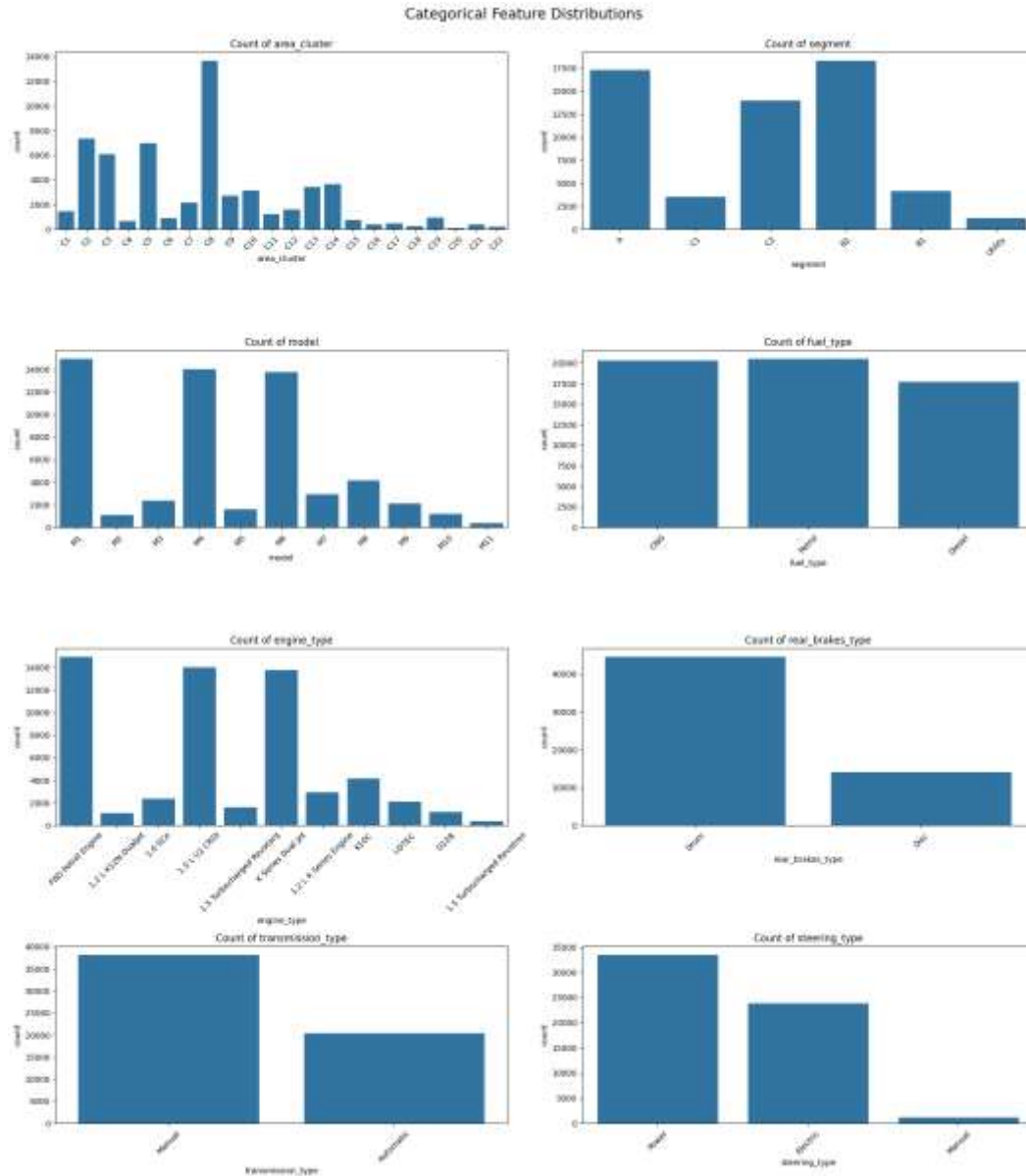


Figure 24 Distribution of categorical features including fuel_type, segment, and area_cluster, revealing dominance of petrol vehicles and geographic skew.

Bar c (Figure 24) were employed to visualize the frequency distributions of key categorical features in the dataset. For the **fuel_type** column, Petrol emerged as the most common fuel type, followed by Diesel and CNG. The **segment** feature showed that segments C1 and B2 were the most prevalent, indicating a concentration of vehicles in these popular market categories. In the **model** column, certain car models appeared with much higher frequency, revealing a skewed distribution and potential overrepresentation of specific models.

Lastly, the **area_cluster** feature, which includes 22 distinct customer regions, showed noticeable imbalance, with clusters such as C1 and C5 having significantly higher customer counts than others. These visualizations highlight categorical imbalances that could influence model behavior and require consideration during preprocessing.

4.3. Bivariate analysis

Bivariate analysis explored relationships between features and the target variable (*is_claim*), helping identify potentially predictive variables.

4.3.1. Numerical vs Target

Boxplots revealed how numerical variables influenced claim probability:

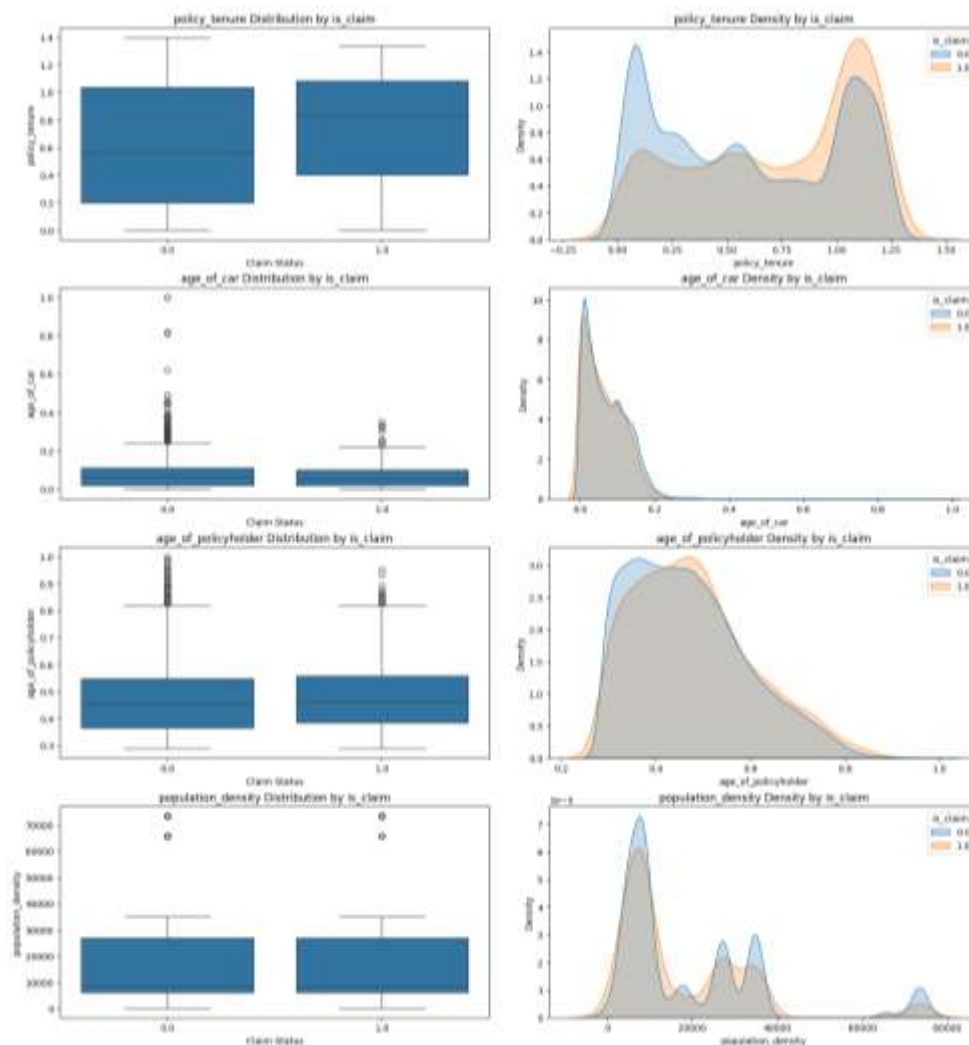


Figure 25 Claim rate across numerical feature -Part 01

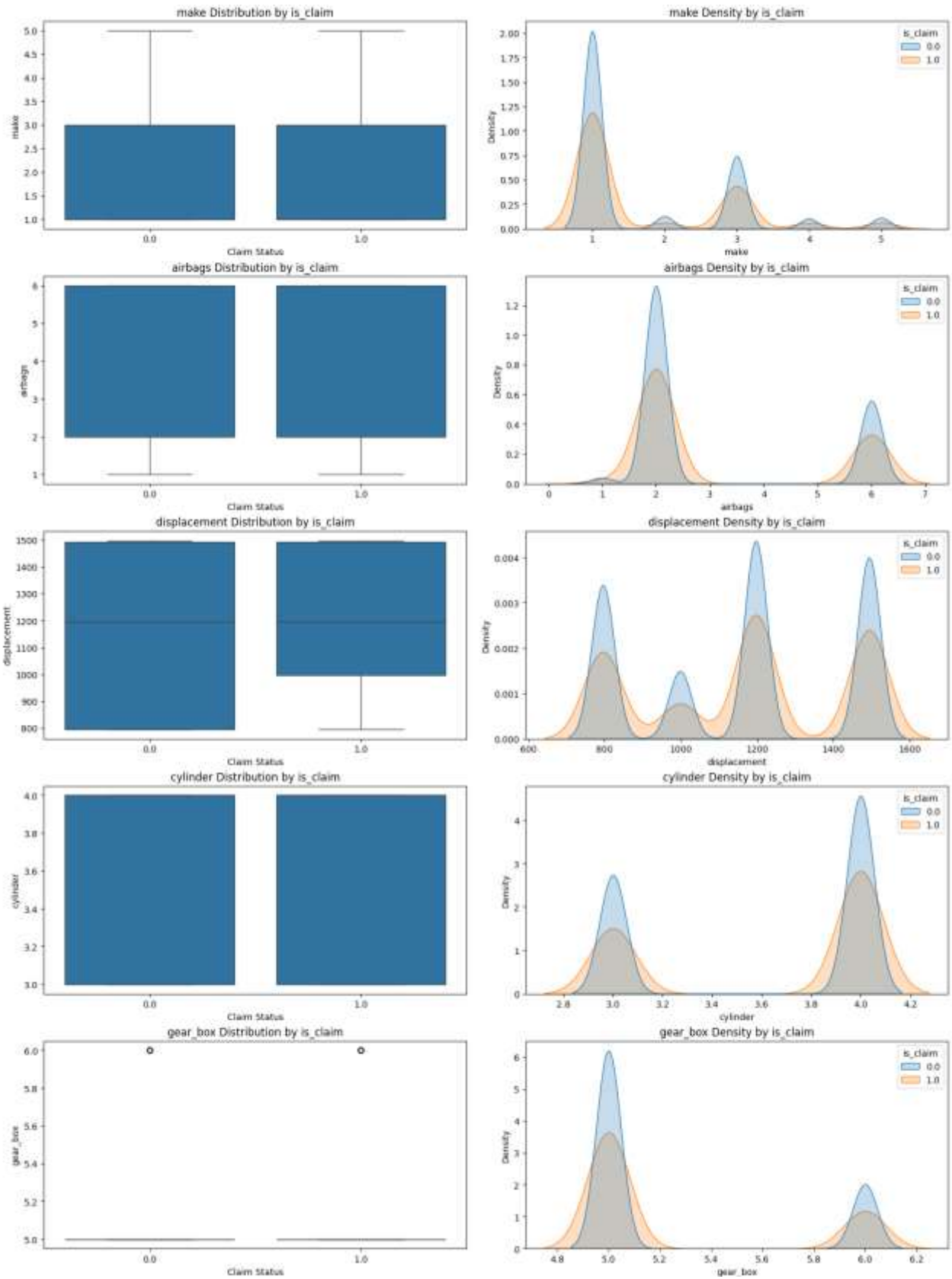


Figure 26 Claim rate across numerical feature -Part 02

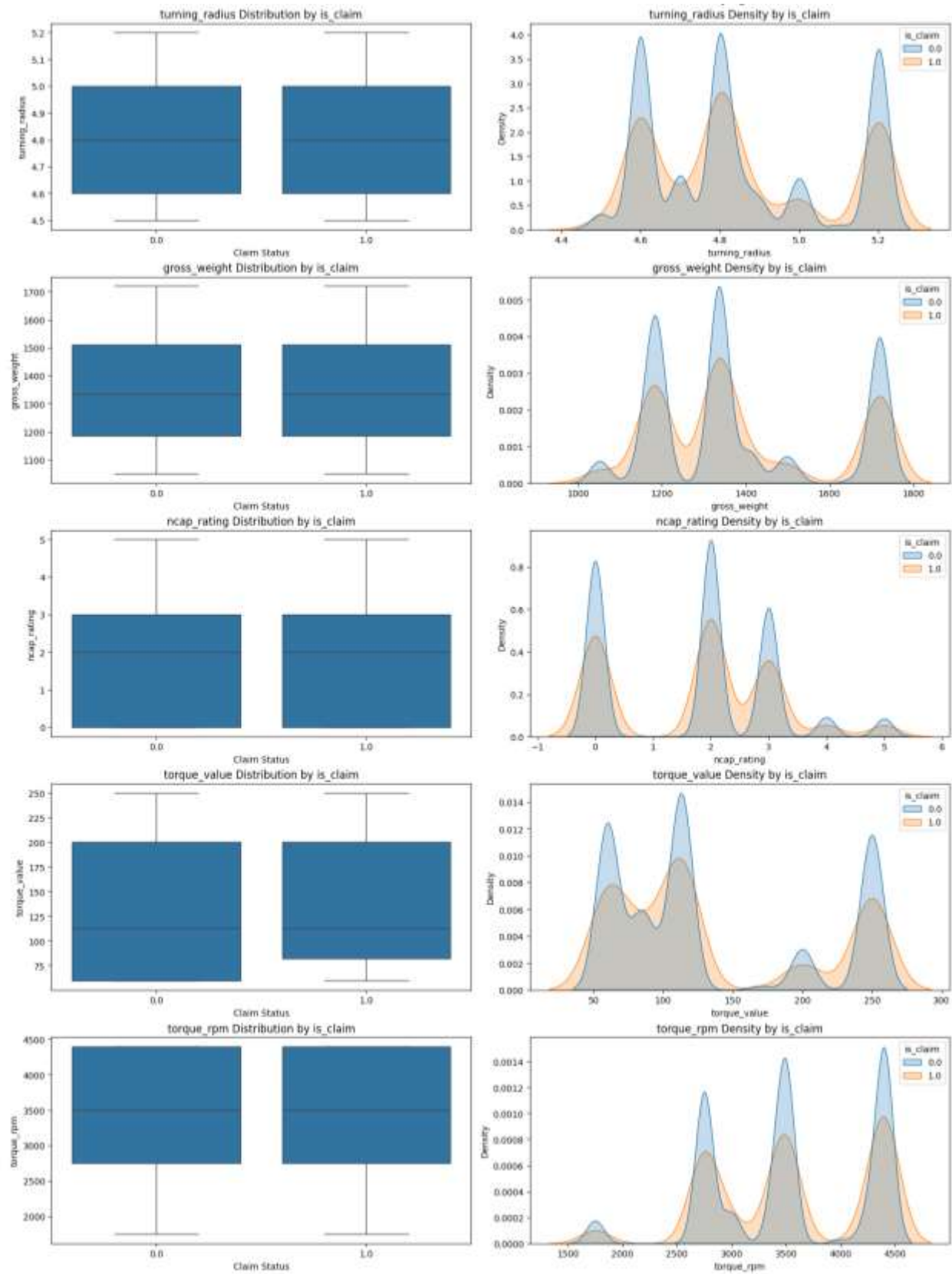


Figure 27 Claim rate across numerical feature -Part 03

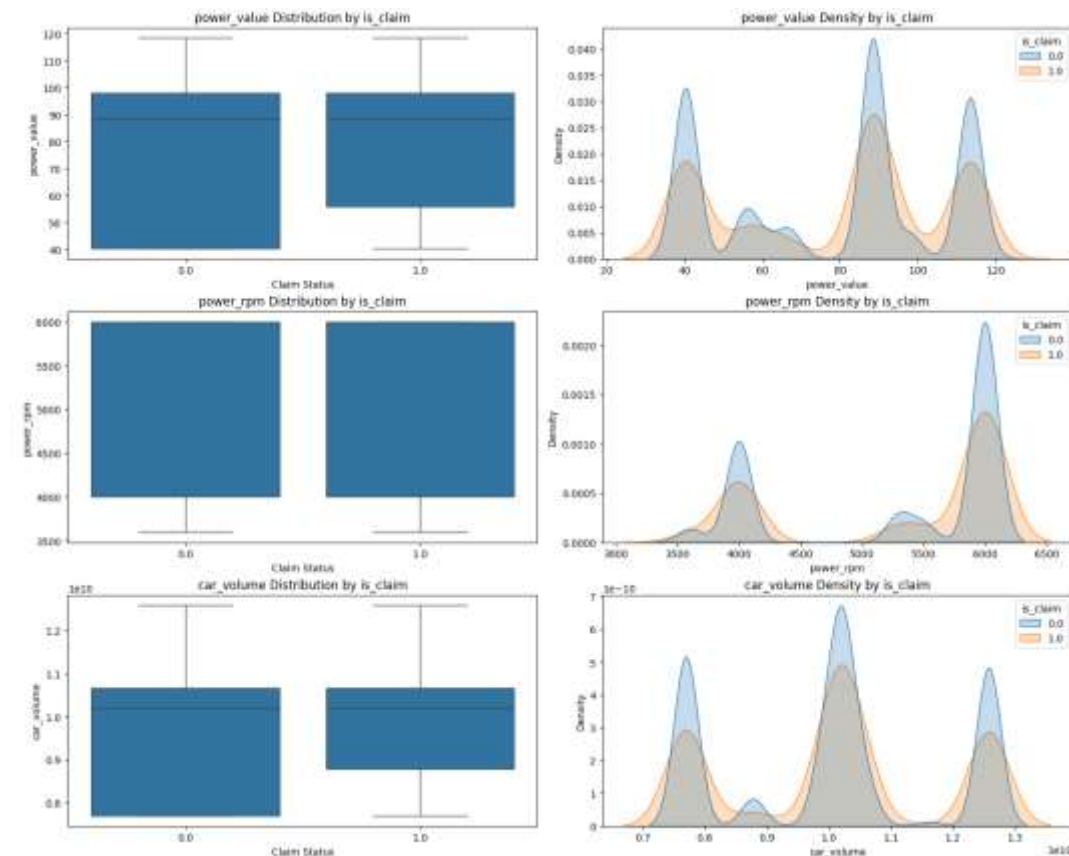


Figure 28 Claim rate across numerical feature -Part 04

Figures 25 to 28 illustrate the bivariate analysis between various **numerical features** and the **target variable (claim status)**. `policy_tenure` had an inverse relation: shorter tenure slightly correlated with more claims, potentially indicating higher risk among new or less loyal customers. This aligns with insurance domain insights where new policyholders may be less familiar with policy terms or represent a transient, higher-risk group.

Further exploration of numerical features reveals additional relationships between vehicle characteristics and claim behavior. Policyholders with **higher annual premiums** reflected newer or more valuable cars exhibited a slightly higher claim rate. This may reflect the increased road presence or more assertive driving behavior associated with higher-value vehicles.

In contrast, features like **make, number of gears, and cylinders** showed no meaningful difference between claim statuses, suggesting limited predictive utility in this context.

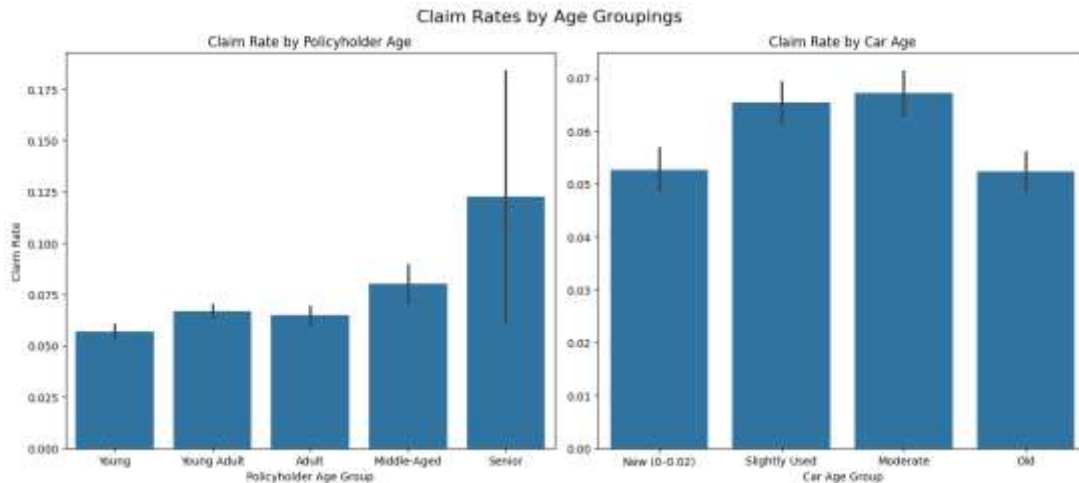


Figure 29 Age analysis

From **Figure 29**, the relationship between policyholder age and claim rate reveals a non-linear trend. While younger policyholders (categorized as "Young" and "Young Adult") show slightly elevated claim rates, the highest claim frequency is observed among **senior** policyholders. This contrasts with conventional expectations that older, more experienced drivers pose a lower risk. The elevated claim rate among seniors may be attributed to factors such as **age-related declines in vision, reflexes, or physical health**, which can impair driving performance. Additionally, seniors who continue to drive might do so out of necessity rather than comfort, potentially increasing their exposure to challenging driving conditions.

Similarly, the analysis of vehicle age shows that **claim rates peak among “Slightly Used” and “Moderate” cars**, while being lower for both the newest and oldest vehicles. This may be due to the increased usage intensity of cars that are no longer brand new but still in excellent working condition. Drivers may be more confident and drive more frequently or assertively with such vehicles, perceiving them as reliable yet not fragile. Conversely, very new vehicles might be driven more cautiously to avoid early damage, while older cars may be used less frequently or more conservatively. These patterns highlight that **both policyholder and vehicle age have non-linear relationships with claim likelihood**, influenced by behavioral, experiential, and mechanical factors.

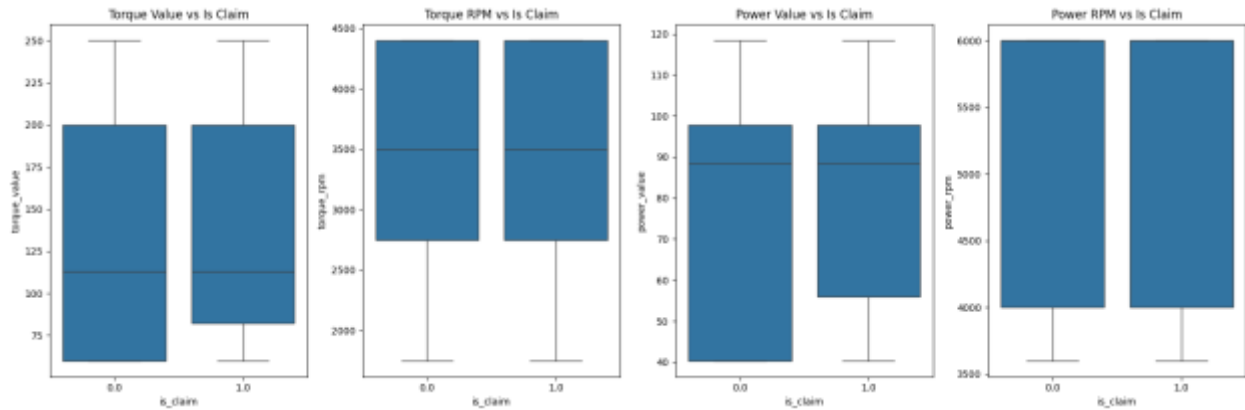


Figure 30 Box plot of Power & Torque

Engine-related attributes such as **power, displacement, and torque** also showed minor shifts towards higher values among claimants, suggesting that **more powerful vehicles might be correlated with increased accident likelihood**, due to performance capabilities encouraging faster or riskier driving. These patterns align with known underwriting insights, where vehicle performance is factored into risk scoring.

Based on the analysis, the most predictive numerical features for claim likelihood include **policy_tenure, age_of_policyholder, population_density**, and vehicle-related attributes such as **displacement, gross_weight, power, turning_radius, airbags, torque_value, and power_value**. These features show noticeable differences between claimants and non-claimants. In contrast, variables like **cylinder count, gear_box, make, car_volume, torque_rpm, and power_rpm** show minimal variation and are **less predictive** for modeling purposes.

4.3.2. Categorical vs Target

Bar plots (*Figure 31, 32*) showed how claim rates varied across categories:

- **fuel_type:** Diesel vehicles had slightly higher claim rates than Petrol and CNG.
- **segment and model:** Certain segments and models had noticeably higher claim frequencies, indicating potential risk differences by vehicle class. Larger or premium vehicles may be costlier to repair and more prone to theft or high-speed driving, which could increase claim incidence.

- `area_cluster`: Some clusters had higher claim proportions, suggesting regional risk variation. This may be attributed to differences in traffic density, accident-prone zones, or urban vs rural driving patterns, which are often cited in spatial risk assessments.

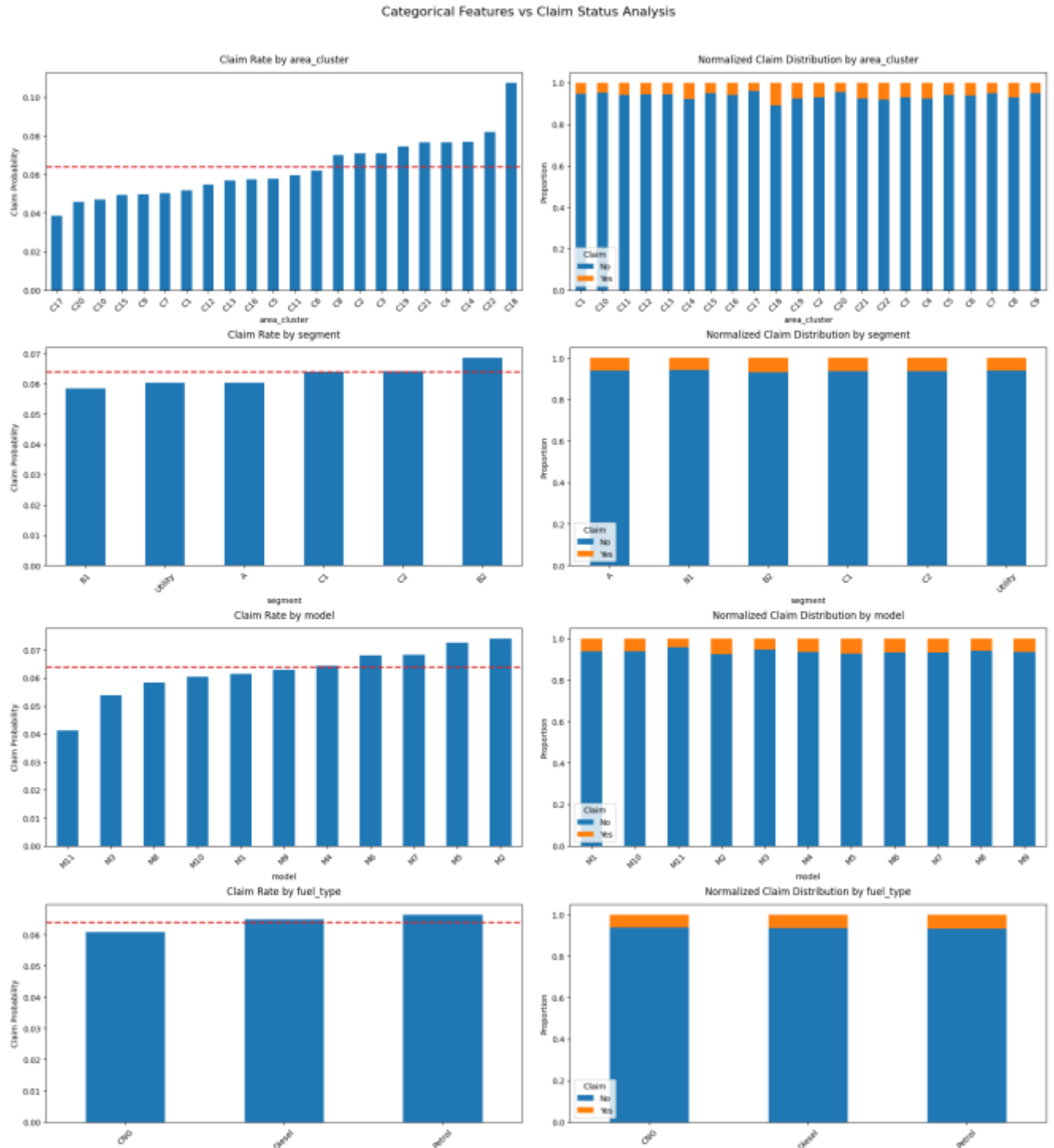


Figure 31 Claim rate across categorical feature -Part 01

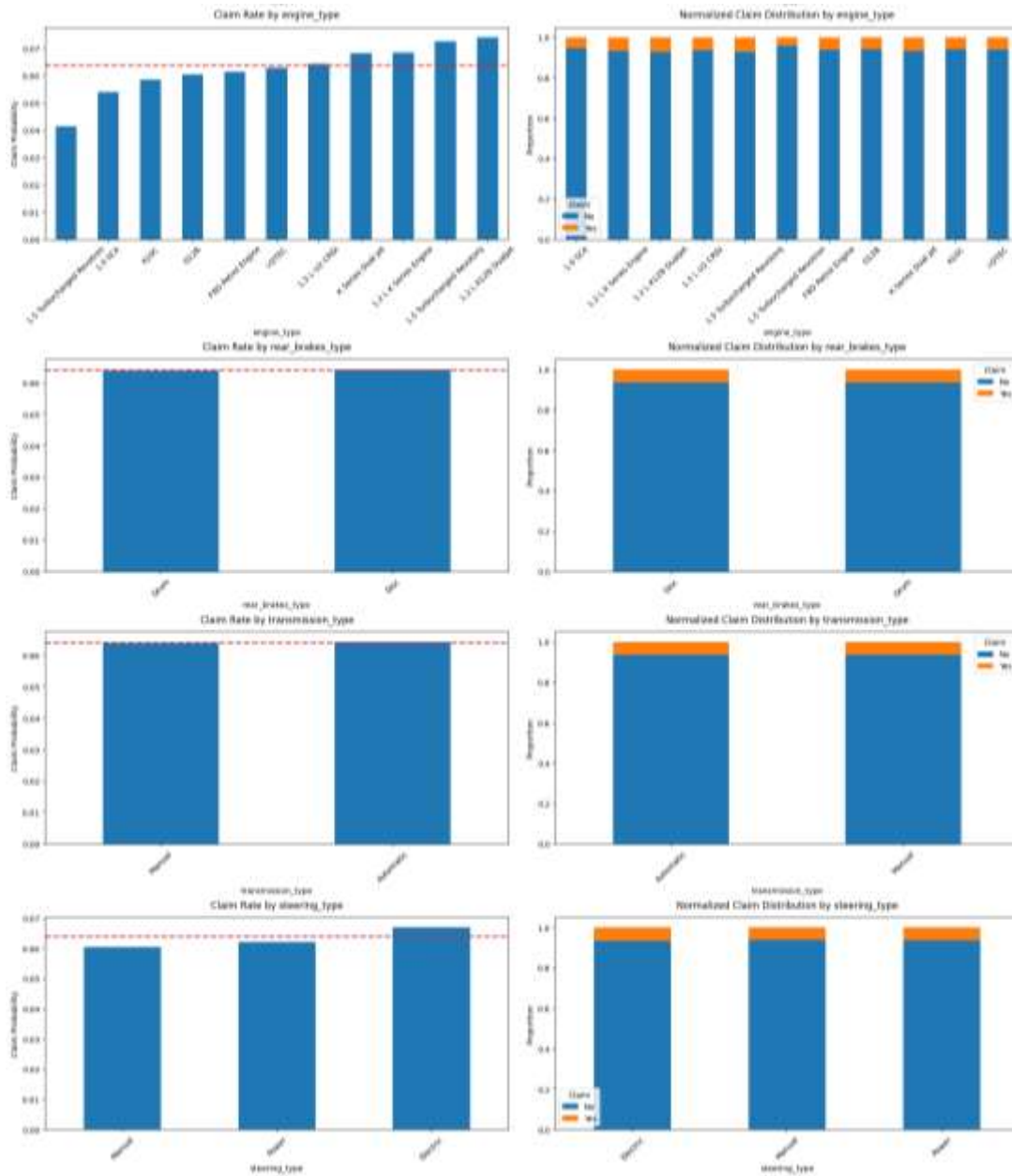


Figure 32 Claim rate across categorical feature- Part 02

Among the categorical features, **area_cluster**, **model**, **segment**, and **engine_type** appear to be the most predictive, showing clearer differences in claim rates across categories. These variables capture regional risk patterns, vehicle class, and engine characteristics that influence driving behavior or exposure. On the other hand, features like **steering_type**, **transmission_type**, **fuel_type**, and **rear_brakes_type** show little variation in claim distribution and are **less predictive** for claim modeling.

4.3.3. Binary vs Target

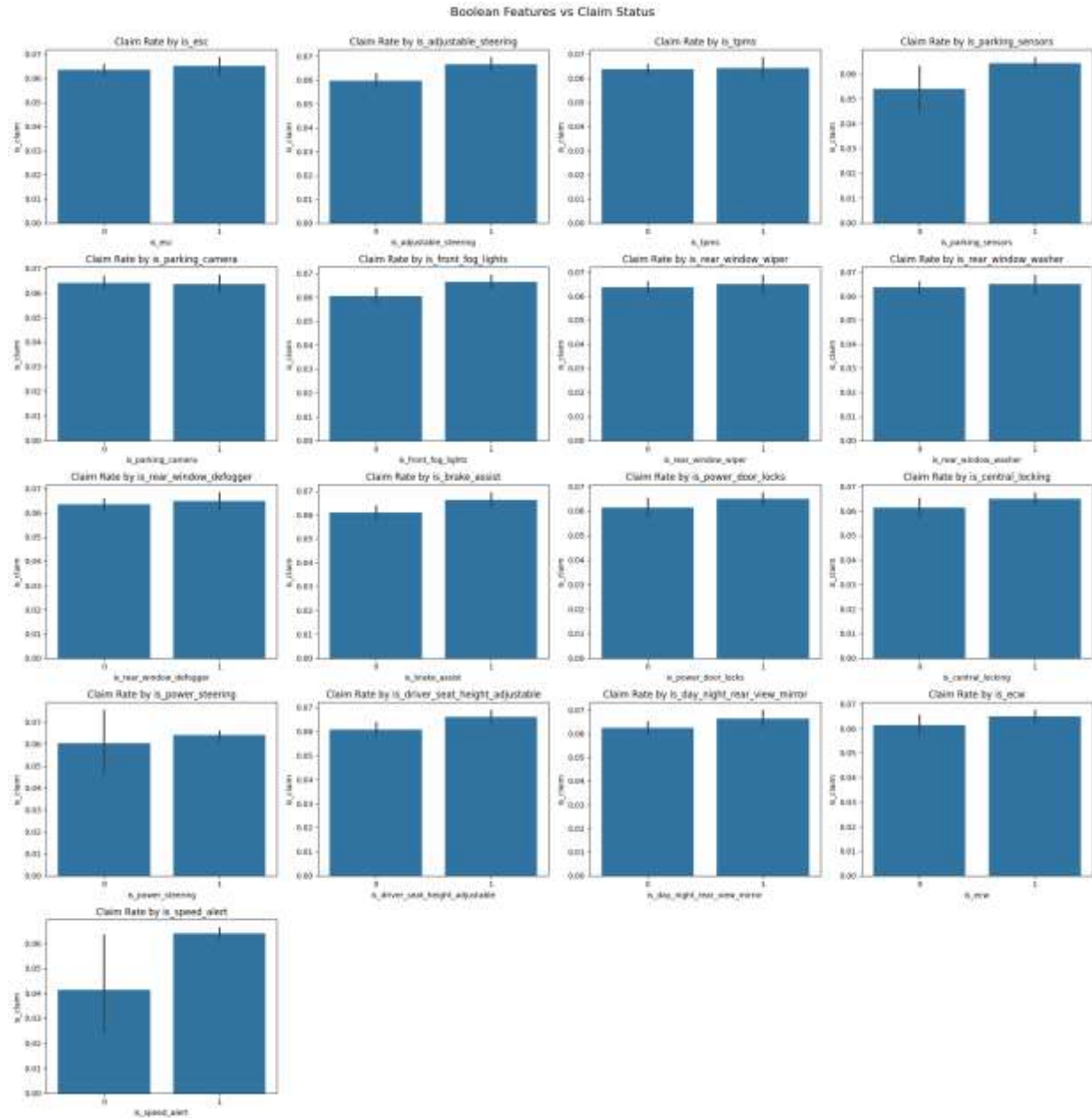


Figure 33 Claim rate across binary features, illustrating higher risk associations with absence of safety features.

Claim rates were compared across binary features using grouped bar charts:

- Vehicles equipped with features like is_brake_assist, is_driver_seat_height_adjustable, is_speed_alert, and were associated with lower claim rates, indicating their potential importance in risk mitigation.

- Conversely, the absence of certain safety features corresponded with slightly elevated claim risk. This highlights the potential role of passive and active safety systems in reducing accident severity and claim frequency, consistent with industry findings.

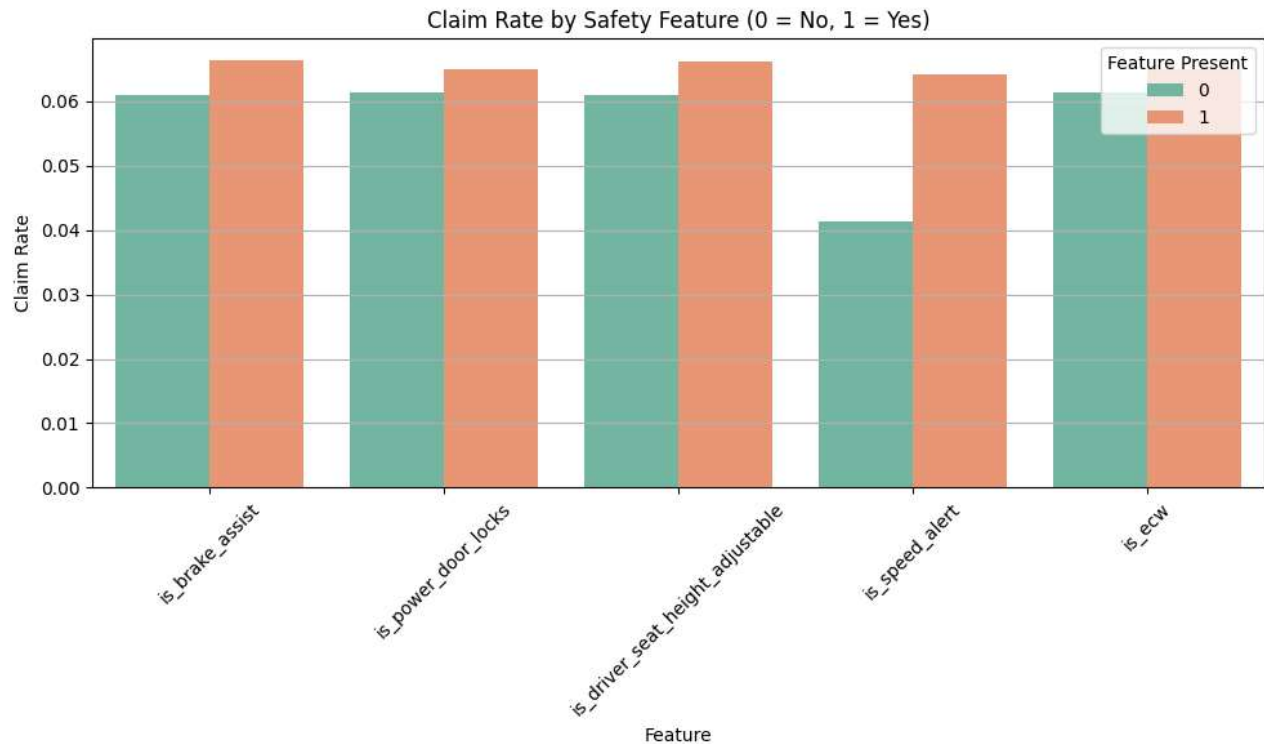


Figure 34 Bar Chart of Important Binary Feature

Among the Binary safety and feature-related variables, **is_brake_assist**, **is_power_door_locks**, **is_driver_seat_height_adjustable**, **is_speed_alert**, and **is_ecw** appear to be the most predictive, as they show clearer associations with claim occurrence possibly indicating driver behavior, vehicle safety emphasis, or model-specific characteristics. In contrast, features like **is_esc**, **is_adjustable_steering**, **is_tpms**, **is_parking_sensors**, **is_parking_camera**, and others related to minor comfort or convenience, such as **rear window functions** and **interior mirrors**, show little variation across claim statuses and are **less predictive** in assessing claim risk.

4.4. Multivariate Analysis

4.4.1. Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was applied to the standardized numeric features of the dataset to explore its underlying structure and to assess the dimensional complexity. The first two principal components (PC1 and PC2) together explain approximately **62.4%** of the total variance in the data, with PC1 alone accounting for **47.6%**. This indicates that a substantial amount of information is preserved in just two dimensions, making PCA a useful tool for visualizing the feature space.

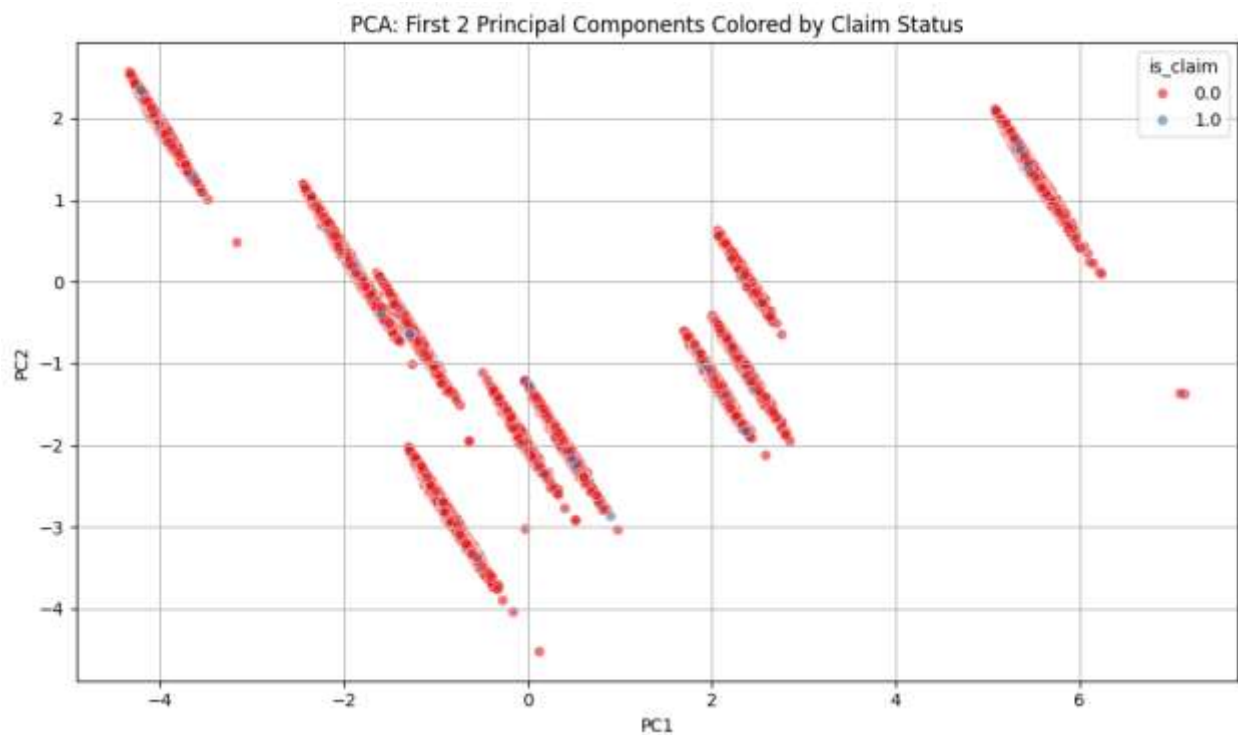


Figure 35 Scatter plot of PC1 vs. PC2

A scatter plot (see [Figure 35](#)) of PC1 vs. PC2, colored by the target variable (`is_claim`), revealed moderate class separation, suggesting potential multivariate structure related to claim status. While not linearly separable, the presence of subtle clustering patterns may indicate interaction effects between features that are not evident in univariate plots.

```

Cumulative variance by PC1 to PC1: 0.4761
Cumulative variance by PC1 to PC2: 0.6239
Cumulative variance by PC1 to PC3: 0.6998
Cumulative variance by PC1 to PC4: 0.7630
Cumulative variance by PC1 to PC5: 0.8129
Cumulative variance by PC1 to PC6: 0.8490
Cumulative variance by PC1 to PC7: 0.8834
Cumulative variance by PC1 to PC8: 0.9129
Cumulative variance by PC1 to PC9: 0.9389
Cumulative variance by PC1 to PC10: 0.9581
Cumulative variance by PC1 to PC11: 0.9724
Cumulative variance by PC1 to PC12: 0.9847
Cumulative variance by PC1 to PC13: 0.9928
Cumulative variance by PC1 to PC14: 0.9975
Cumulative variance by PC1 to PC15: 0.9994
Cumulative variance by PC1 to PC16: 1.0000
Cumulative variance by PC1 to PC17: 1.0000
Cumulative variance by PC1 to PC18: 1.0000
Cumulative variance by PC1 to PC19: 1.0000
Cumulative variance by PC1 to PC20: 1.0000
Cumulative variance by PC1 to PC21: 1.0000
Cumulative variance by PC1 to PC22: 1.0000
Cumulative variance by PC1 to PC23: 1.0000
Cumulative variance by PC1 to PC24: 1.0000
Cumulative variance by PC1 to PC25: 1.0000
Cumulative variance by PC1 to PC26: 1.0000

```

Figure 36 Cumulative variance report

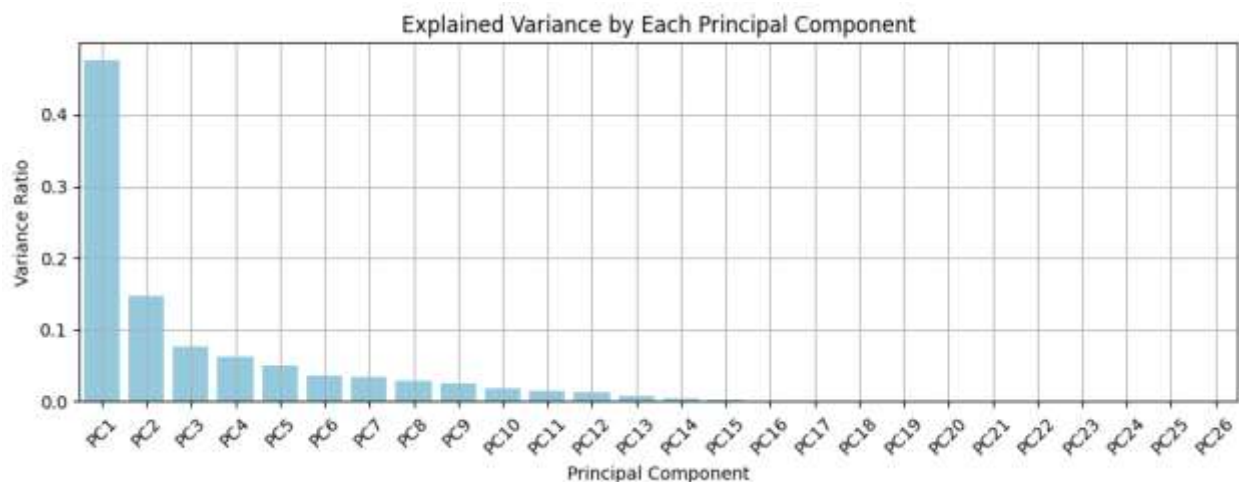


Figure 37 Variance plot

The explained variance plot (*Figure 37*) further shows that approximately **95.8%** of the total variance is captured by the first **10 principal components**, while over **99%** is captured within **14 components**. This indicates that the dataset has some degree of redundancy or correlation among features, and dimensionality could be reduced if necessary (e.g., for modeling with linear methods or for interpretability).

2. DBScan Clustering

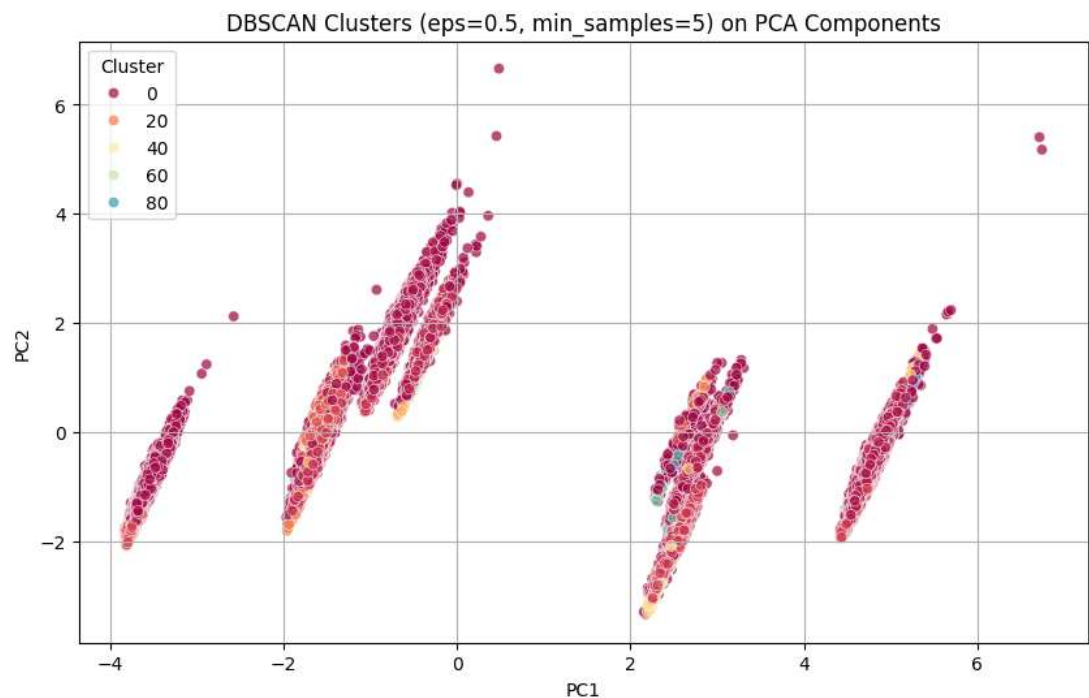


Figure 39 Scatterplot DBSCAN clusters on PCA components

```
Cluster vs Target Crosstab (noise = -1):
is_claim  0.0  1.0
Cluster
-1         2295  128
0         12522  823
1         12956  987
2         12312  868
3          1102   95
...         ...   ...
88           3    0
89           5    0
90           4    0
91           5    0
92           5    0

[94 rows x 2 columns]

(DBSCAN(),
 PCA(n_components=2),
 array([0, 0, 0, ..., 0, 1, 2]),
      PC1      PC2  Cluster  is_claim
0   -3.531371 -0.727487      0      0.0
1   -3.593446 -1.140202      0      0.0
2   -3.530941 -0.855389      0      0.0
3   -0.580851  2.061299      1      0.0
4   -1.533074  0.076951     10      0.0
...         ...      ...      ...
58587 -1.482932  0.244479     10      0.0
58588 -3.492795 -0.569938      0      0.0
58589 -3.476702 -0.647752      0      0.0
58590 -0.505793  2.610127      1      0.0
58591  4.588598 -1.260753      2      0.0

[58592 rows x 4 columns])
```

Figure 40 DBSCAN clusters on PCA components

The DBSCAN clustering algorithm identified 94 clusters, including noise points labeled as Cluster -1. Most data points were assigned to a few large clusters (e.g., Clusters 0, 1, and 2), while the remaining clusters contain very few points. For instance, Cluster 0 includes 12,522 non-claim and 823 claim instances, while Cluster 1 and Cluster 2 also have substantial counts with similar class distributions. The noise cluster (-1) contains 2,295 non-claim and 128 claim cases, representing outliers or points that do not belong to any dense region.

PCA was used to reduce the dimensionality to 2 principal components (PC1 and PC2) for visualization and clustering. While DBSCAN can find arbitrarily shaped clusters and outliers, the high number of small clusters (including many with fewer than 5 samples) suggests that the chosen DBSCAN parameters may have resulted in over-segmentation. Despite this, larger clusters (0–3) capture a significant portion of the data and show varying proportions of claims, indicating some level of separation between groups.

4.4.3. Interaction between Variables

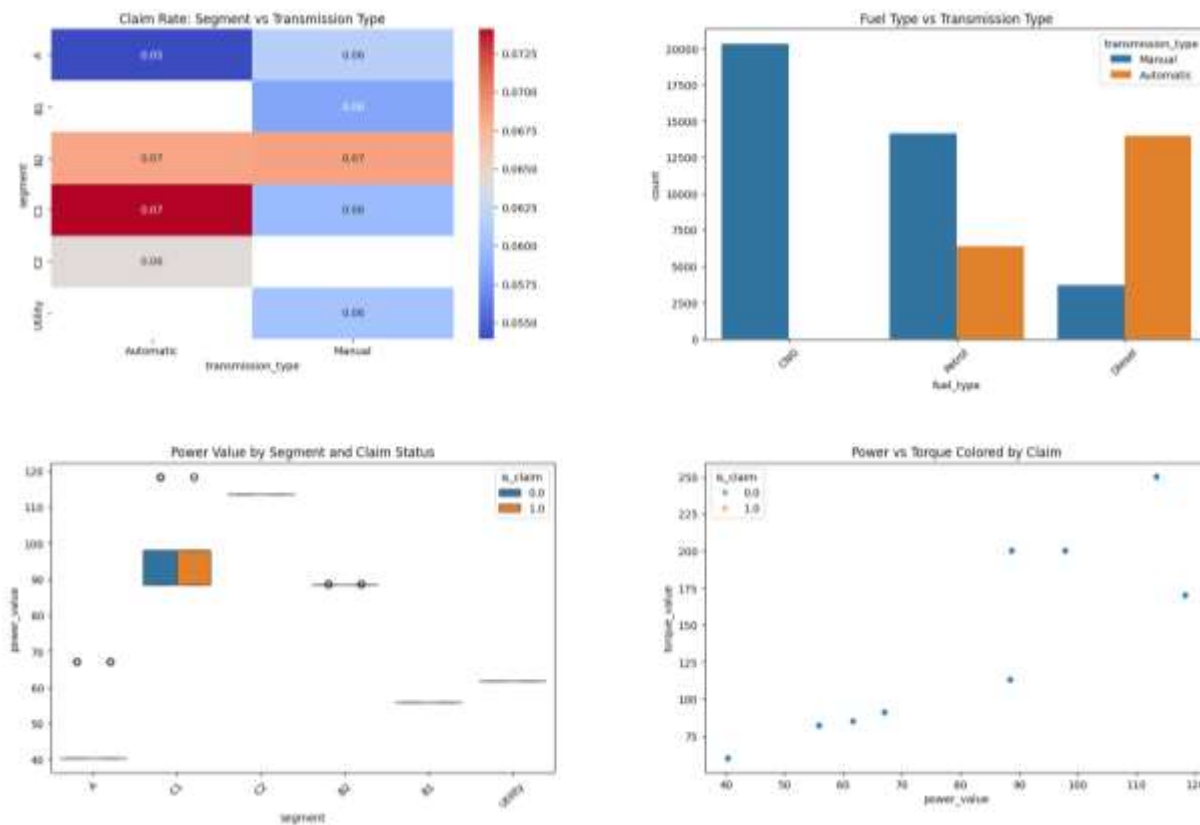


Figure 41 Variable Interactions -Part 01



Figure 42 Variable Interactions -Part 02

The visualizations (**Figure 41, 42**) provide a comprehensive overview of how different vehicle attributes interact with the insurance claim status. The claim rate heatmaps (top-left, bottom-right, and middle-right) reveal that certain segments particularly Segment B2 and B1 consistently show slightly higher claim rates across different variables such as transmission type, NCAP rating, and engine type. This suggests segment-based risk stratification may be useful for modeling.

The bar chart in the top-right shows the relationship between fuel type and transmission, highlighting that diesel vehicles are more commonly associated with automatic transmissions, while petrol is more evenly split. This distribution may influence claim likelihood indirectly through driving patterns.

Box plots for power and torque (middle-left and bottom-left) show how these technical specifications vary across segments and fuel types, with little distinction in claim status indicating these features may not strongly discriminate between claims and non-claims in isolation.

The scatter plot of power vs. torque (middle-right) shows clustering of data points, but no strong visual separation between claim and non-claim groups. Similarly, mean power by model (bottom-left) suggests limited variation across models with respect to claim status, though a few models do show higher power levels.

In summary, the strongest patterns emerge in categorical variable interactions (segment vs. transmission, engine type, and NCAP), where claim rates show mild but consistent differences potentially useful for risk modeling or underwriting strategy.

4.5. Summary

Key Insights from EDA:

- **Feature Importance Guidance:** EDA highlighted several candidate features (e.g., `policy_tenure`, `segment`, `age_of_policyholder`, and `safety` features) as potentially predictive.
- **Multicollinearity Identified:** Informed feature reduction and engineering.
- **Class Imbalance Detected:** Supported use of resampling strategies.
- **Target Association:** Differences in claim likelihood were visible across categories like `fuel_type`, `segment`, and `safety equipment`.

EDA provided essential insights into the structure and behavior of the dataset. It revealed important feature-target relationships, potential multicollinearity issues, and a class imbalance problem, all of which influenced the feature selection and modeling strategies described in the subsequent chapters. (Refer [Appendix C.1](#) for statistical analysis)

CHAPTER 5 METHODOLOGY

5.1. Introduction

This chapter outlines the methodological framework adopted for developing a predictive model to forecast whether a car insurance policyholder will file a claim. The methodology integrates various stages of a typical data science workflow.

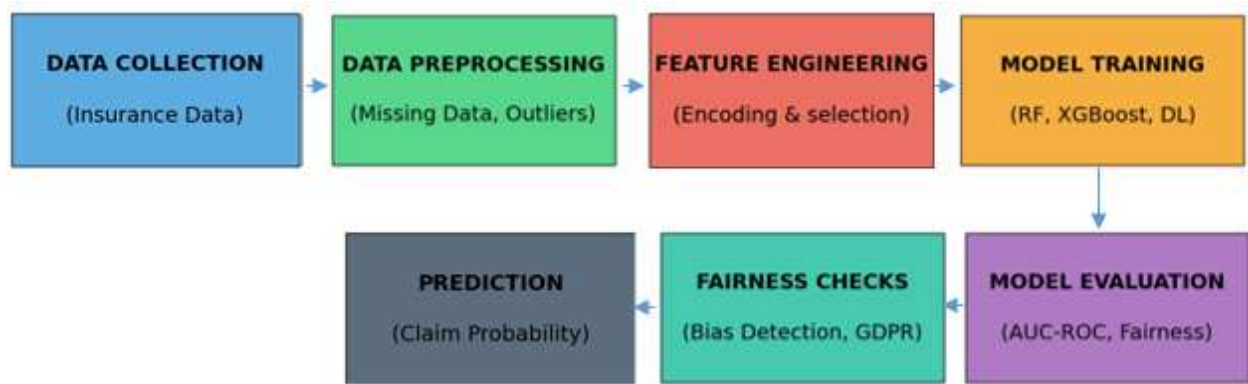


Figure 43 Basic Machine Learning Pipeline

This *Figure 43*, illustrates the transformation of raw data into predictive insights through structured steps.

5.2. CRISP-DM Framework and Project Mapping

The study follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework (Shearer, 2000) [7], widely adopted due to its structured approach to real-world data science projects. Each phase of the CRISP-DM methodology has been tailored to suit the context of insurance claim prediction, incorporating data preprocessing, feature engineering and selection, model development, evaluation, interpretation, and deployment. The ultimate objective is to develop a predictive pipeline aligned with real-world insurance operations, supporting tasks like fraud detection and risk assessment.

The entire process integrates both statistical and machine learning techniques, with special attention paid to the challenges posed by the dataset, such as class imbalance and mixed data types. The final model is operationalized through a Power BI dashboard, facilitating dynamic visualization and actionable insights for stakeholders.

Figure 44, breaks down each CRISP-DM phase and aligns it with specific tasks executed in this project. For example, the Business Understanding phase focused on defining project goals, while the Data Understanding phase involved exploratory data analysis.

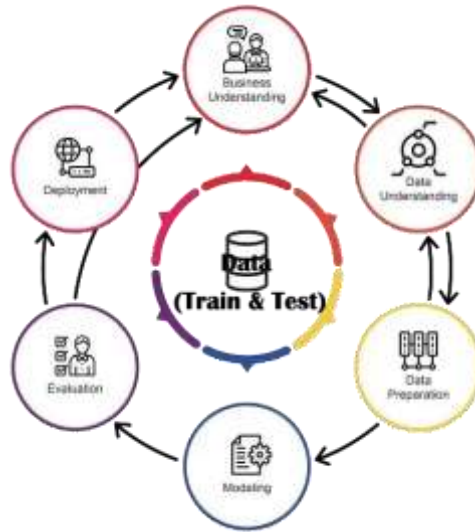


Figure 44 Phases of the CRISP DM Methodology

A summary of CRISP-DM phases mapped to this project is presented in **Table 9**.

CRISP-DM Phase	Mapped Project Activity
1. Business Understanding	Defined project goals: predict insurance claims to improve risk assessment and fraud detection.
2. Data Understanding	Explored the Kaggle dataset, assessed feature distributions, data types, and class imbalance.
3. Data Preparation	Handled missing values, outliers, categorical encoding, feature scaling, and class imbalance.
4. Modeling	Applied and tuned multiple classification algorithms (Logistic Regression, RF, XGBoost, NN).
5. Evaluation	Assessed model performance using F1-score, ROC-AUC, confusion matrix, and cross-validation.
6. Deployment	Deployed final model insights through a Power BI dashboard for decision support.

Table 9 CRISP-DM Framework and Project Mapping

5.3. Flowchart of methodology

The methodology employed in this study is illustrated in *Figure 45*, which outlines the step-by-step process followed to build the car insurance claim prediction model. The flowchart includes key stages such as data preprocessing, exploratory data analysis (EDA), feature engineering, feature selection, model training and evaluation, interpretability, and deployment.

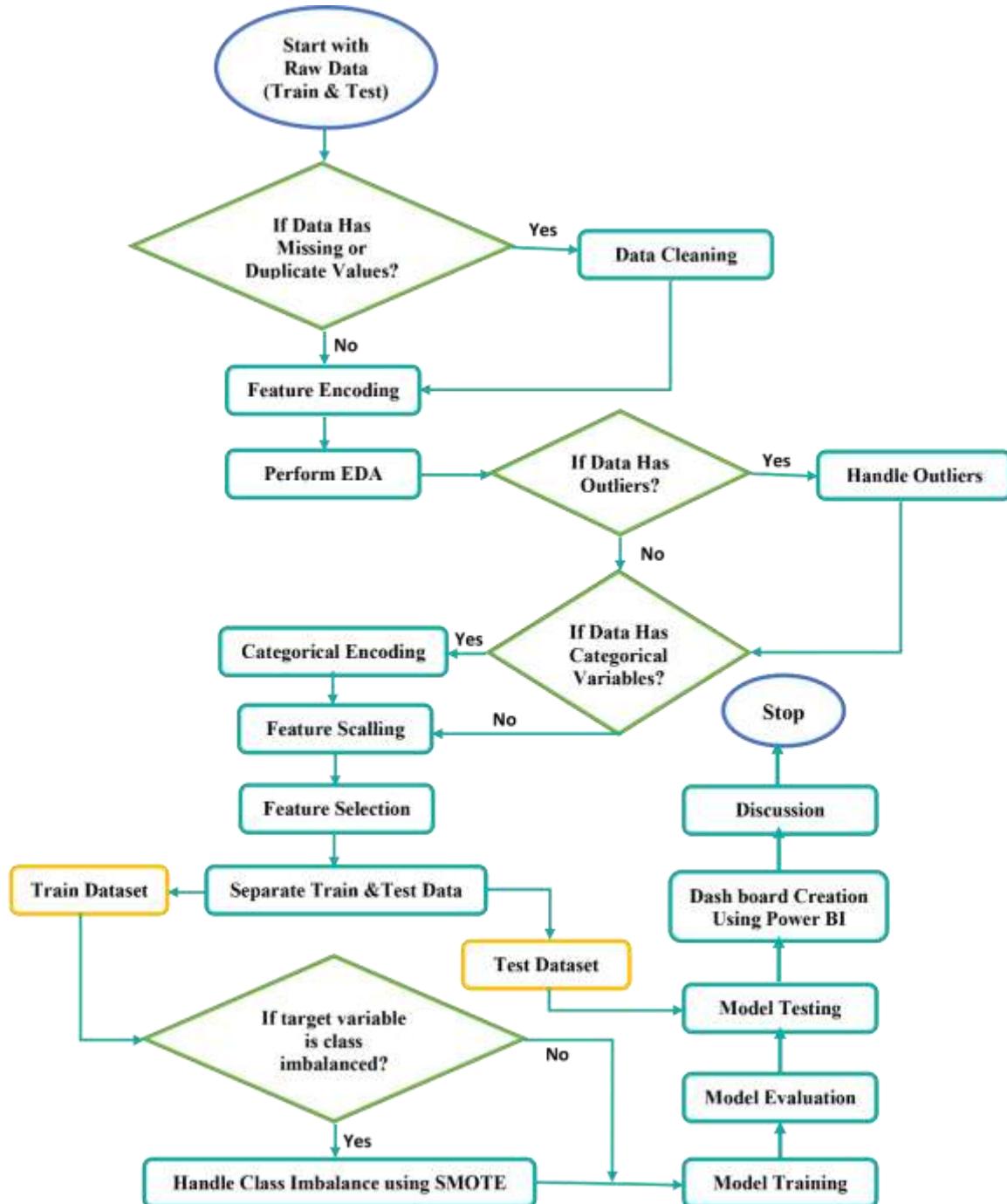


Figure 45 Flowchart of methodology

5.4. Problem Formulation

The predictive task was framed as a binary classification problem where the target variable `is_claim` indicates whether a customer would file an insurance claim (1) or not (0). The goal is to improve predictive accuracy for claim detection to support pricing, fraud prevention, and customer segmentation.

5.5. Data Preparation (CRISP-DM Phase 3)

Data preparation included imputation, outlier handling, encoding, scaling, and feature engineering as detailed in [Chapter 3](#).

5.5.1. Feature Selection

To enhance model performance and eliminate redundant or irrelevant variables, feature selection was performed using a combination of statistical, model-based, and information-theoretic methods. The goal was to reduce overfitting, improve interpretability, and retain only the most informative features.

5.5.1.1. Comparative Evaluation of Feature Selection Methods

Three feature selection techniques were compared:

1. Mutual Information (MI)

Mutual Information was selected for its ability to measure the reduction in uncertainty of the target variable when a feature is known. It effectively handled the mixed data types present in the dataset, including numerical, categorical, and boolean variables.

MI was particularly suitable for capturing complex, non-linear relationships often associated with insurance claim prediction. Furthermore, it demonstrated robustness to outliers and provided a clear ranking of feature importance, making it highly informative for model development. (Refer to [Appendix D.1](#) for more details)

2. Random Forest Importance (RF)

Feature importance was also evaluated using a Random Forest-based approach (Louppe et al., 2013) [\[9\]](#). In this method, importance scores were derived from the contribution of each feature to decision tree splits across the ensemble. This allowed for the capture of non-linear relationships and feature interactions.

Although RF provided useful insights into variable relevance, its importance rankings could sometimes be biased toward features with more categories or higher cardinality. (Refer to [Appendix D.2](#) for more details)

3. ANOVA F-statistic

The ANOVA F-statistic method was used to assess whether the means of a feature significantly differed across target classes. This technique was effective for detecting linear class separability and identifying features with strong statistical association to the target. However, it was limited to numerical input and assumed normally distributed data, which reduced its flexibility for mixed-type datasets. (Refer to [Appendix D.3](#) for more details)

Final Selection:

Each method was used to select the top 20 features. Based on performance and interpretability, Mutual Information was chosen as the final feature selection method. Its ability to capture non-linear dependencies, handle diverse data types, and remain robust to outliers made it the most suitable approach for the insurance claim prediction task.

5.6. Modeling (CRISP-DM Phase 4)

The car insurance claim prediction problem was approached as a supervised binary classification task, aiming to forecast whether a policyholder would submit a claim (denoted as 1) or not (denoted as 0). Due to the binary structure and inherent class imbalance of the target variable, a range of classification algorithms from diverse model families were assessed to identify the most effective predictive model.

5.6.1. Candidate Models

To address the binary classification task of car insurance claim prediction, a wide range of supervised learning algorithms were explored. Each model family offers unique advantages in terms of performance, interpretability, and robustness. All models were evaluated using the same set of features selected through Mutual Information (see [Section 5.5.1](#)) and trained on balanced data after applying SMOTE (see [Section 3.4.8](#)). Below [Table 10](#), breaks down each model family used (Details available in [Appendix E](#)).

Model	Hyperparameter Tuned	Performance Range (Summary)
Logistic Regression	Yes	Moderate precision and recall
Ridge Classifier	No	Similar to logistic regression
SGD Classifier	Yes	Fast but lower stability
SVC	Yes	High accuracy, slow training
LinearSVC	No	Efficient, moderate performance
Decision Tree	Yes	Interpretable, prone to overfit
Random Forest	Yes	Good performance, robust
Extra Trees	No	Similar to RF, more randomness
Gradient Boosting	Yes	Strong, can overfit if not tuned
XGBoost	Yes	Best balance of metrics overall
LightGBM	Yes	Fast, suffered in precision
CatBoost	Yes	Good with categorical data
GaussianNB	No	Fast, poor accuracy
BernoulliNB	No	Weak due to assumption violations
KNN	Yes	Computationally expensive

Table 10 Model Selection Overview

Each model was trained using the top 20 features from Mutual Information and evaluated using stratified k-fold cross-validation.

5.7. Evaluation (CRISP-DM Phase 5)

To assess the effectiveness of the machine learning models developed in this project, a comprehensive evaluation framework was employed. Given the class imbalance in the dataset where claim instances are significantly fewer than non-claim instances metrics beyond standard accuracy were necessary.

5.7.1. Evaluation Pipeline

The evaluation was performed as follows:

- Models were trained on the training data using selected features.
- Predictions were generated for each fold in the cross-validation.
- Metrics were calculated per fold and then averaged.
- Additional validation was done on the separate hold-out test set (for final models only).

5.7.2. Performance Metrics

To evaluate the performance of classification models, several statistical metrics were employed. Given the **high-class imbalance** in the dataset, metrics beyond simple accuracy were necessary to fairly assess the models' ability to detect actual claims. Each model was evaluated using the following metrics: (Refer to [Appendix F](#))

- Accuracy: Overall correctness.
- Precision: Proportion of correctly predicted positive claims.
- Recall: Ability to detect actual claims (especially important in imbalanced datasets).
- F1 Score: Balance between precision and recall.
- ROC-AUC: Discriminative ability between classes.
- Confusion Matrix: Detailed classification outcomes.

In this project, **F1 Score** and **ROC-AUC** were emphasized during model selection due to their suitability for **imbalanced classification tasks**. Precision and recall were also analyzed to understand trade-offs between false positives and false negatives.

5.7.3. Final Model Selection

XGBoost was selected as the final model due to it is:

- Strong balance of **recall, F1-score, and ROC-AUC**
- Robust handling of imbalanced data via `scale_pos_weight`
- High efficiency with label-encoded features
- Proven performance in similar domains (Williams et al., 2022)[\[5\]](#)

This model was retained for interpretability analysis and eventual deployment within the Power BI dashboard.

5.6.3.1. Limitations of XGBoost:

Despite its strong performance, XGBoost comes with certain trade-offs. It may require significant computer resources for tuning and is less interpretable than simpler models such as Logistic Regression. Stakeholders may require additional interpretability tools (e.g., SHAP) to fully understand predictions, which adds to model complexity.

5.7.4. Cross-Validation Approach

A stratified k-fold cross-validation technique was used to ensure balanced distribution of classes across training and validation folds. Specifically, 5-fold cross-validation was adopted to reduce variance in performance metrics and assess the model's generalizability. Stratification was critical to maintain the original class proportions in each fold.

5.7.5. Hyperparameter Tuning

To improve model performance, Grid Search with 5-fold CV was applied on XGBoost with the following search space:

```
param_grid_XG = {  
    'n_estimators': [200, 300, 500],  
    'max_depth': [6, 9, 12],  
    'learning_rate': [0.01, 0.05, 0.1],  
    'subsample': [0.8, 1.0],  
    'scale_pos_weight': [12, 14, 16]  
}
```

Figure 46 Grid Search parameters of XGBoost

- Total combinations tested: 162
- Best parameters found:

```
{'learning_rate': 0.1, 'max_depth': 12, 'n_estimators': 500, 'scale_pos_weight': 12, 'subsample': 0.8}  
Best cross-validation F1-score: 0.9321
```

- Best cross-validation F1-score: 0.9321

This tuned model was retrained and validated on the hold-out validation set.

5.7.6. Final Model Evaluation Setup

- Final model: **XGBoost with tuned hyperparameters**
- Final evaluation was done on a 30% hold-out validation set.
- Performance was measured using the same set of metrics as cross-validation.
- Results were then saved and visualized via confusion matrix, ROC, and PR curves (discussed in [Chapter 6](#) – Results and Discussion)

5.8. Ethical Considerations

The deployment of automated claim prediction models presents significant ethical challenges, particularly when utilizing features that may indirectly encode sensitive or protected attributes. Variables such as geographic region, vehicle ownership cluster, or proxies for socioeconomic status can inadvertently reinforce historical biases especially if past claim approvals or rejections reflect systemic disparities. For instance, regional data may correlate with demographic characteristics, potentially resulting in unintentional discrimination against certain communities.

To address these concerns, all features were critically reviewed for ethical implications. While explicitly sensitive demographic variables such as age or gender were not included in the modeling process, proxy variables (e.g., area cluster, vehicle segment) were carefully examined for potential bias. Where necessary, these features were either excluded or aggregated to reduce their granularity, thereby mitigating the risk of discriminatory outcomes.

Automated decisions impacting claim approvals, pricing strategies, or fraud detection must adhere to principles of transparency, fairness, and non-discrimination. A notable case is that of Lemonade Insurance, whose AI-based system faced criticism over alleged bias in claim assessments. Such examples highlight the need for explainable and accountable AI systems. To align with GDPR, fairness and transparency were considered during model selection (**European Commission, 2018**) [\[23\]](#).

To enhance interpretability and promote fairness, this study incorporates model explanation techniques such as SHAP and LIME, which offer both local and global insights into model predictions. Looking ahead, future implementations should integrate fairness-aware machine learning approaches to ensure equity across demographic groups and guard against proxy discrimination. Periodic audits, transparency reports, and human-in-the-loop systems are also recommended to foster public trust and ensure compliance with ethical and regulatory standards.

5.9. Model Interpretability

To support transparency and regulatory compliance, the final model was interpreted using:

- SHAP (Lundberg & Lee, 2017 [\[13\]](#)): Global and local feature importance.
- LIME (Ribeiro et al., 2016 [\[14\]](#)): Individual prediction explanations.

These tools allowed stakeholders to understand which features most influenced claim predictions, supporting trust in the model's decisions. (Refer to [Appendix G](#))

5.10. Deployment (CRISP-DM Phase 6)

To ensure that the insights generated by the machine learning model are accessible and actionable across different business functions, a comprehensive Power BI dashboard was developed. The design followed a **user-centric and goal-oriented approach**, segmenting content into eight distinct pages tailored to various stakeholders such as executives, risk managers, analysts, and operations teams.

Each page was constructed with **a specific analytical goal in mind**, aligning visualizations and interactive filters (slicers) to support efficient decision-making. The layout blends **KPI indicators**, **aggregated visuals** (bar, pie, heatmap), and **granular drilldowns** (tables, scatter plots, matrices) to cater to both high-level overviews and detailed exploration.

- **Page 1** serves as an executive summary, highlighting key metrics and distribution patterns to provide a quick risk snapshot.
- **Page 2–5** are thematic deep dives focused on risk profiling, model behavior, vehicle features, and policy tenure each equipped with targeted visualizations for focused analysis.
- **Page 6** centers on regional and demographic trends, integrating geospatial and population-level insights.
- **Page 7** filters the dataset to high-risk (positive) predictions, supporting operational prioritization.
- **Page 8** enables ad hoc drillthrough exploration, allowing users to investigate specific policy IDs or filter combinations with export-ready views.

All visual elements were designed with interactivity in mind, leveraging Power BI's **slicers, tooltips, and drillthrough actions** to empower users to customize their exploration paths without requiring technical intervention. This methodology ensures that predictive insights are not siloed within data science teams but are seamlessly translated into **practical, real-time decision support tools** for the organization.

5.11. Reproducibility

All project code, notebooks, and environment files are available at [GitHub Repository](#).

5.12. Summary

The methodology chapter outlines a structured, replicable approach to insurance claim prediction. It integrates the CRISP-DM process, ensuring a best-practice approach from problem definition through to deployment. The integration of preprocessing, robust modeling, and interpretable output delivers a scalable and practical solution for car insurance claim prediction. The adoption of XGBoost, combined with SHAP and LIME, ensures both high accuracy and transparency. While XGBoost emerged as the top performer, considerations related to interpretability and fairness are acknowledged and addressed.

CHAPTER 6 – RESULT AND DISCUSSIONS

6.1. Introduction

This chapter presents and interprets the experimental results of the machine learning pipeline developed for predicting car insurance claims. Building upon the feature selection, model training, and interpretability techniques described in the methodology, this section evaluates performance across key metrics such as precision, recall, F1-score, and ROC-AUC. Discussion follows each result, focusing on practical implications, comparative performance, and limitations.

6.1.1. Initial Insights from EDA

Exploratory Data Analysis (EDA) revealed a set of potentially important features based on bivariate visualizations and domain relevance. Key numerical features included `policy_tenure`, `age_of_policyholder`, and `population_density`. Among categorical and binary variables, `segment`, `engine_type`, and several safety-related indicators (e.g., `is_brake_assist`, `is_speed_alert`) were highlighted.

6.2. Feature Selection Results

Feature selection is a critical step in machine learning that enhances model performance by eliminating redundant or irrelevant features, reducing overfitting, and improving interpretability. To Validate EDA findings, three feature selection techniques were applied **Mutual Information (MI)**, **Random Forest feature importance (RF)**, and **ANOVA F-statistic** to determine the most informative predictors for insurance claim prediction. Each method ranked features based on its statistical or model-based importance.

6.2.1. Comparative Performance

Each feature selection method independently selected the top 20 features based on its criteria (e.g., importance scores, coefficients, or rankings) the results, shown in [Appendix D](#). These feature sets were evaluated using the same classification model and assessed with four metrics: Precision, Recall, F1-score, and ROC AUC.

Method	Precision	Recall	F1 Score	ROC-AUC
MI	0.951446	0.811429	0.875858	0.942724
RF	0.950522	0.811647	0.875596	0.942598
ANOVA	0.878763	0.737400	0.801880	0.925349

Table 11 Feature Selection Method Evaluation

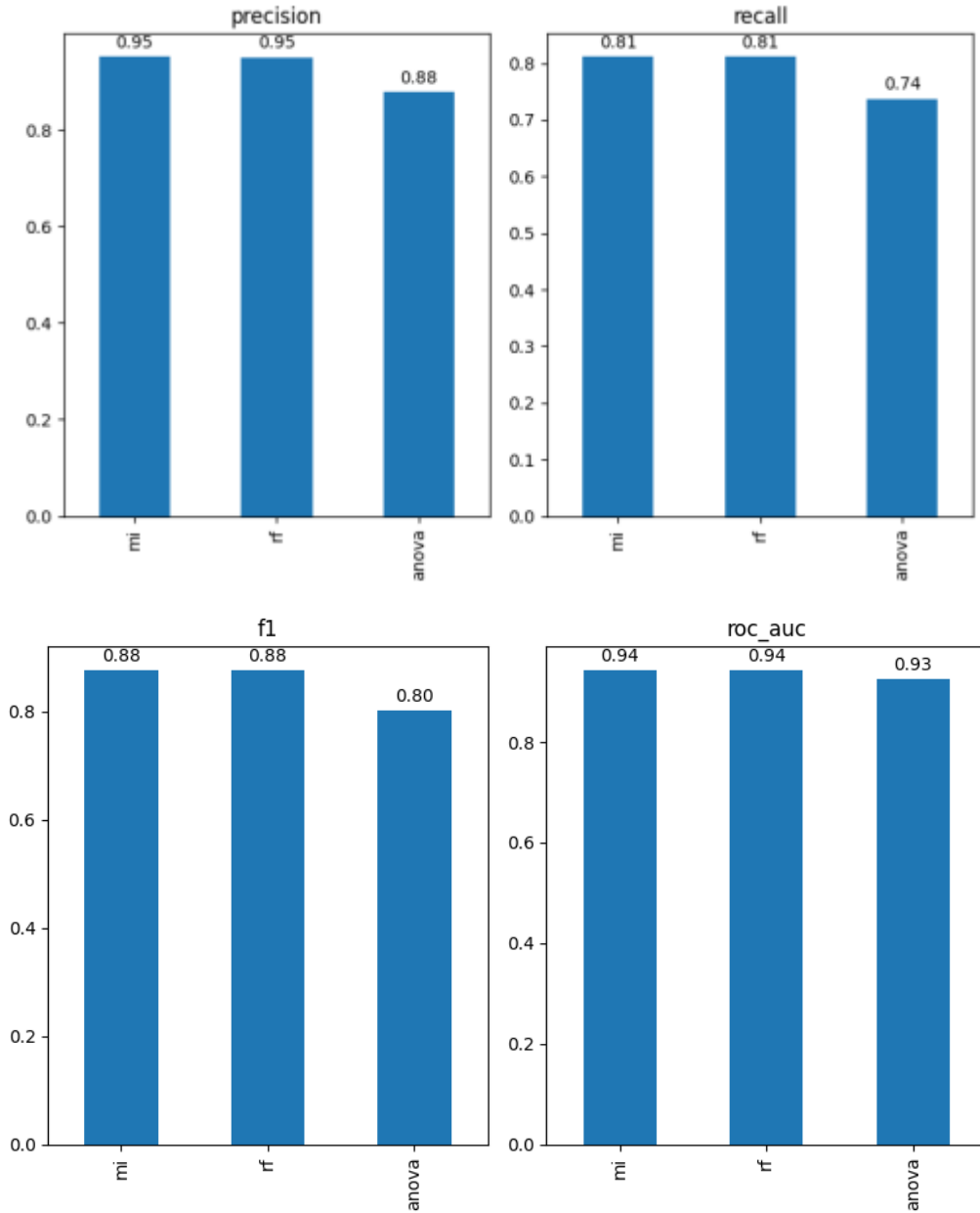


Figure 47 Comparison of classification performance (F1-score, ROC-AUC) across feature selection methods, with Mutual Information outperforming ANOVA and RF in recall sensitivity.

The **Table 11** and corresponding **Figure 47** present the evaluation of three feature selection methods **Mutual Information (MI)**, **Random Forest (RF)**, and **ANOVA** based on four performance metrics **Precision**, **Recall**, **F1 Score**, and **ROC-AUC**. Both MI and RF show identical results, with MI achieving slightly higher values across all metrics. Specifically, MI yielded a precision of 0.951, recall of 0.811, F1 score of 0.876, and ROC-AUC of 0.943, indicating strong and balanced predictive performance. RF closely followed with only marginal differences. In contrast, the ANOVA-based feature set resulted in noticeably lower scores, particularly in precision (0.879) and F1 score (0.802), suggesting that it may be less effective at capturing the most informative features for this task. The bar chart (see **Figure 47**) clearly visualizes these differences, highlighting MI and RF as the most effective feature selection methods, with MI slightly outperforming the rest.

6.2.2. Selected Features (MI-based)

The top 20 features selected by the MI method include a mix of technical, policy, and driver-related variables:

```
MI selected features: ['transmission_type', 'is_front_fog_lights', 'is_day_night_rear_view_mirror', 'is_central_locking', 'is_power_steering', 'steering_type', 'is_brake_assist', 'is_speed_alert', 'is_driver_seat_height_adjustable', 'is_adjustable_steering', 'rear_brakes_type', 'is_parking_sensors', 'is_power_door_locks', 'cylinder', 'is_ecw', 'policy_tenure', 'area_cluster', 'population_density', 'age_of_car', 'age_of_policyholder']
```

Figure 48 Selected Features

These results align with domain expectations features like safety indicators, car condition, and driver demographics directly impact risk assessment.

6.3. Model selection

This section provides a comparative evaluation of several classification algorithms employed for the car insurance claim prediction task. The primary objective was to determine the most accurate and reliable model for forecasting a policyholder’s likelihood of filing a claim, using patterns derived from historical data.

6.3.1. Model Performance Comparison

Using MI-selected features, various classifiers were assessed in terms of their predictive accuracy and sensitivity to class imbalance. Among them, XGBoost achieved the best balance between precision and recall.

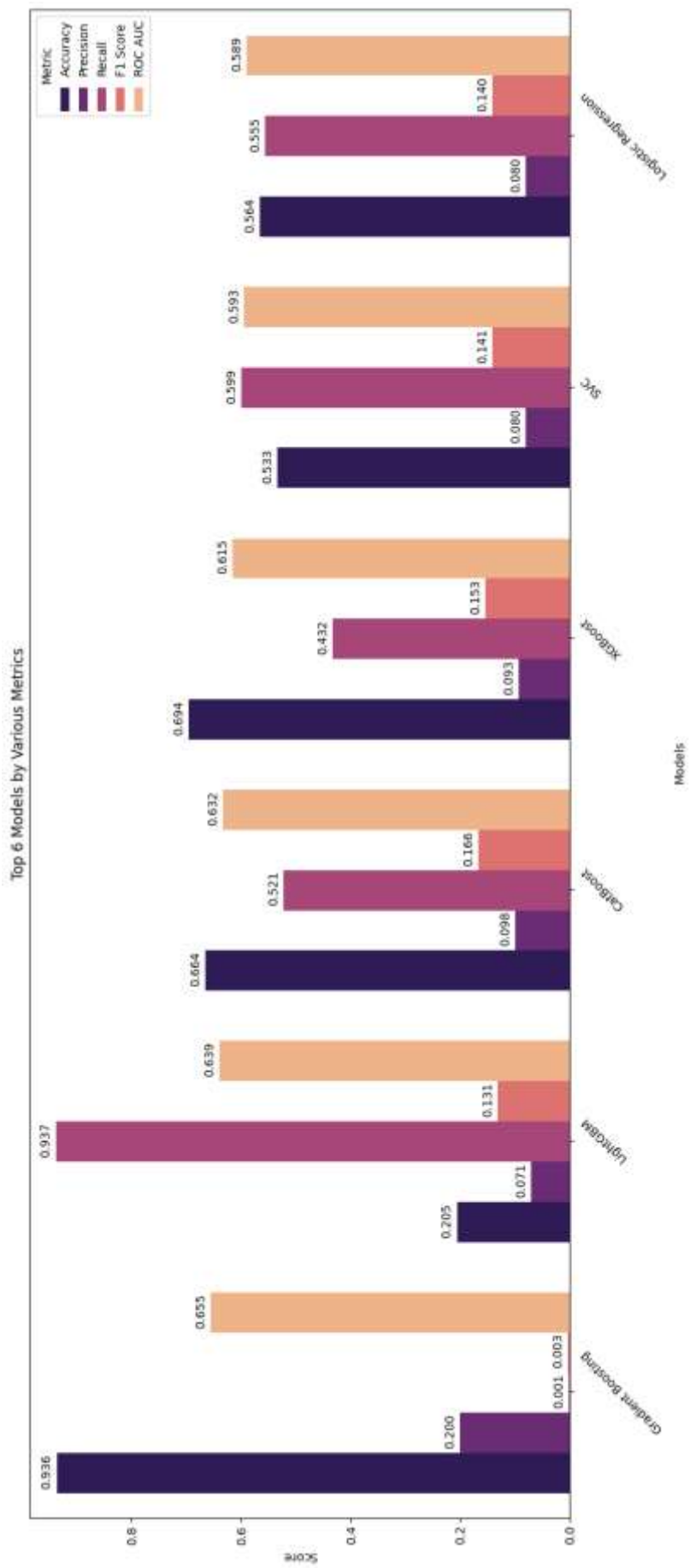


Figure 49 Model Comparison

As shown in [Figure 49](#) and [Appendix E.6,E.7](#), XGBoost consistently outperformed other models in F1-score and recall due to its ability to capture non-linear feature interactions and effectively handle class imbalance through the `scale_pos_weight` parameter. While Random Forest also provided good accuracy, its inability to fine-tune decision boundaries and susceptibility to overfitting made it less suitable. LightGBM struggled with precision, possibly due to aggressive leaf-wise tree growth. XGBoost, with its depth control and learning rate, managed high-order interactions without severe overfitting.

Despite Gradient Boosting having higher AUC, XGBoost demonstrated a better trade-off between recall and precision, which is critical in imbalanced classification contexts like insurance claims.

CatBoost also delivered a good balance of recall and F1-score and is known for handling categorical features effectively. However, due to prior label encoding in the dataset, XGBoost was chosen as the most suitable model.

6.3.2. Precision-Recall & ROC Curves

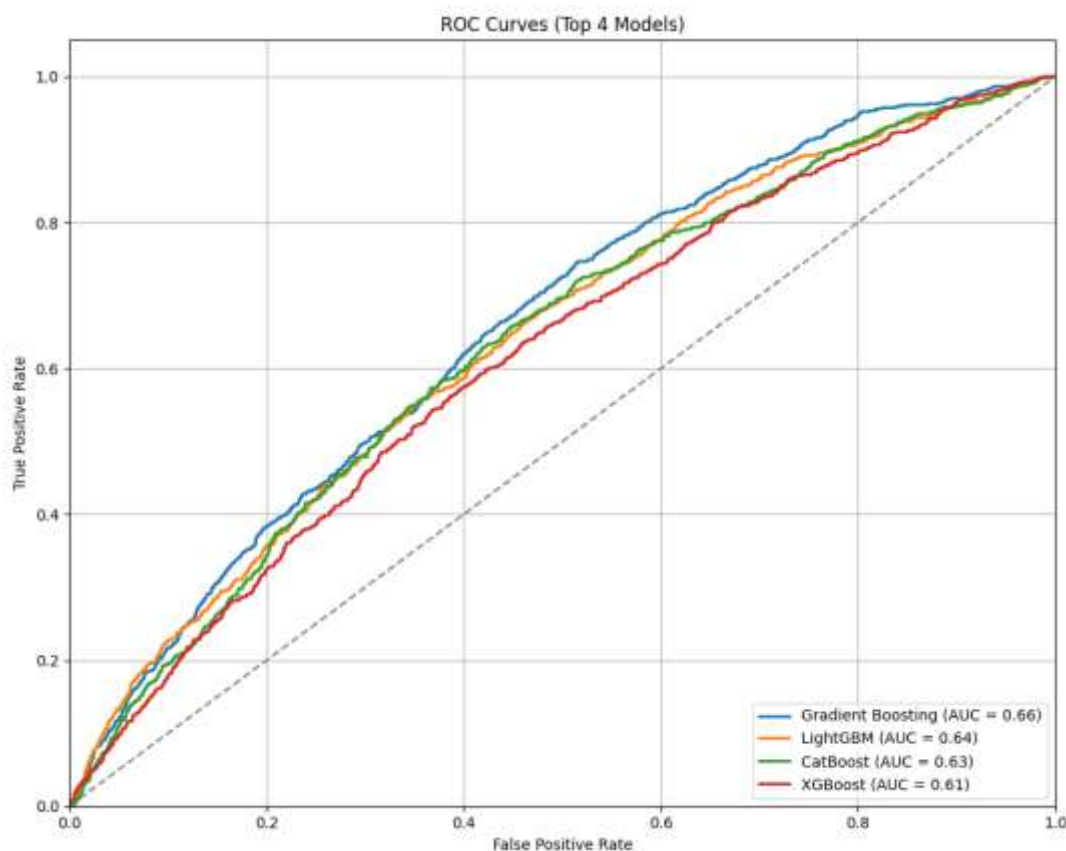


Figure 50 ROC Curves

Figure 50 shows ROC curves comparing classification models based on AUC, which measures their ability to distinguish between classes. Gradient Boosting had the highest AUC (0.66), followed by LightGBM (0.64), CatBoost (0.63), and XGBoost (0.61), indicating Gradient Boosting performed best overall.

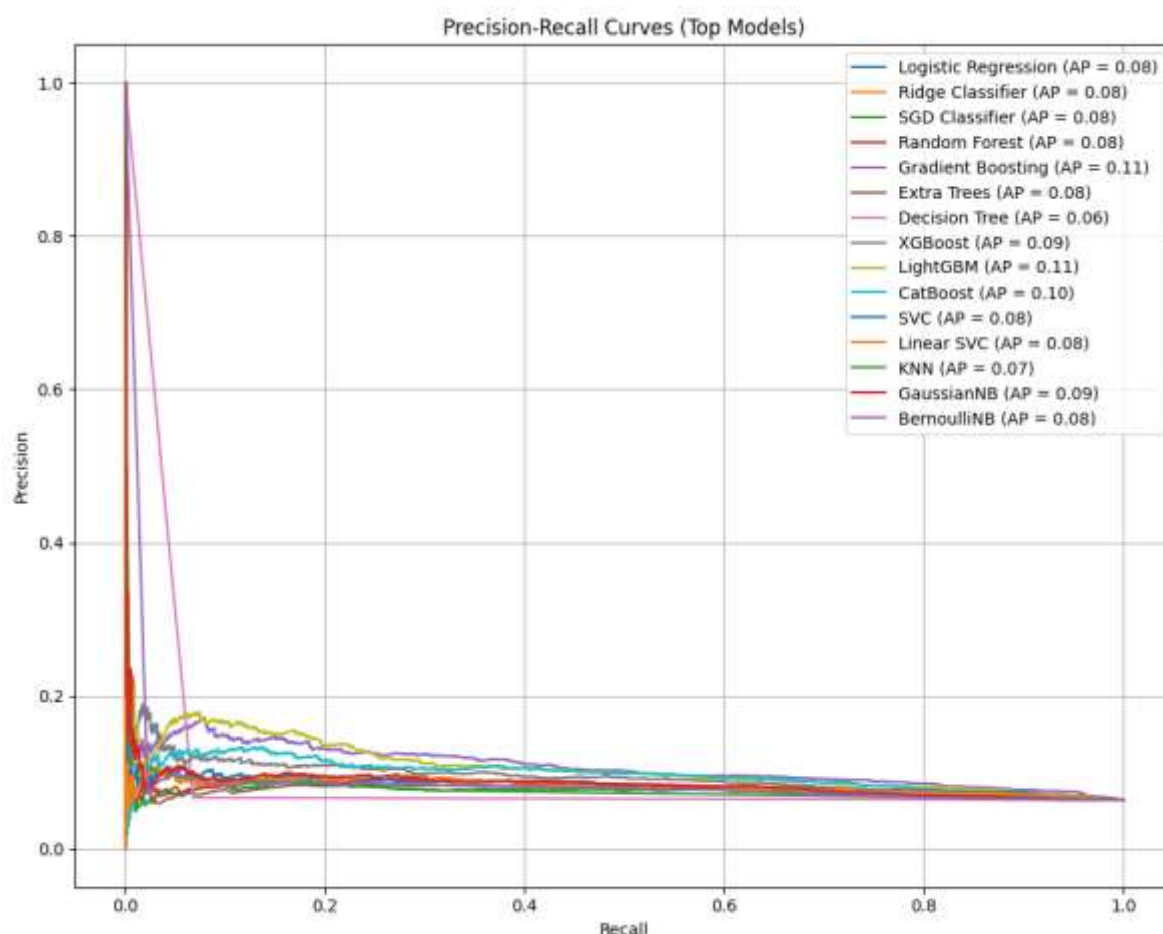


Figure 51 Precision-recall curves

Figure 51 shows **precision-recall** curves for the top-performing models. XGBoost and CatBoost achieved relatively **high average precision (AP)** scores, demonstrating better trade-offs between precision and recall. These insights confirm the robustness of XGBoost for imbalanced binary classification in this context.

Although Gradient Boosting had the highest AUC, XGBoost exhibited superior balance between recall and precision, making it more suitable for imbalanced claims prediction.

6.4. Model Evaluation

The final XGBoost model was assessed through cross-validation, validation metrics, ROC and PR curves, and hyperparameter tuning.

6.4.1. Cross-Validation Performance

In **Figure 52**, Initial evaluation using 5-fold cross-validation on the training set (balanced via SMOTE) yielded a mean **F1-score (macro)** of **0.7291**, demonstrating the model's strong generalization ability during training. These results indicated a good balance in detecting both claim and non-claim cases under class imbalance.

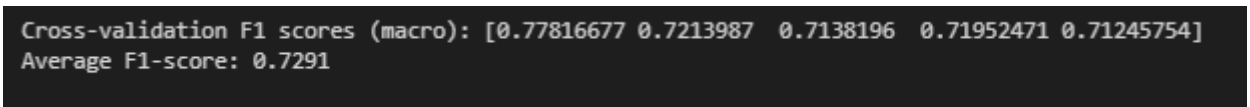


Figure 52 Cross Validation Performance

6.4.2. Validation Set Performance

When evaluated on the hold-out validation set, the initial model achieved:

- **Precision:** 0.177 (positive class)
- **Recall:** 0.883 (positive class)
- **F1-score:** 0.296 (positive class)
- **Macro F1-score:** 0.5612
- **ROC AUC:** 0.8769
- **PR AUC:** 0.3801

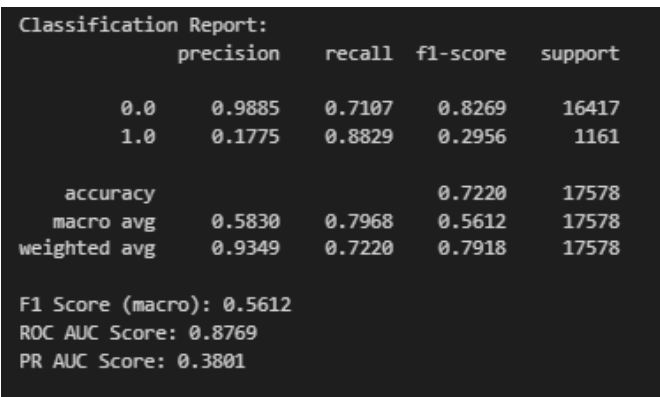


Figure 53 Validation set Performance

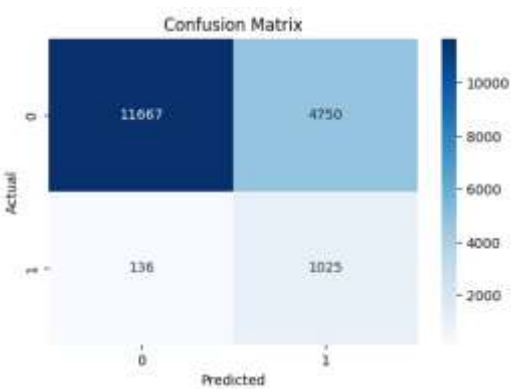


Figure 54 Confusion Matrix

These results in *Figure 53* reflect model achieved high recall but low precision, a typical trade-off in imbalanced classification. This behavior favors capturing most claim cases at the cost of increased false positives.

6.4.3. Hyperparameter Tuning

```
param_grid_XG = {
    'n_estimators': [200, 300, 500],
    'max_depth': [6, 9, 12],
    'learning_rate': [0.01, 0.05, 0.1],
    'subsample': [0.8, 1.0],
    'scale_pos_weight': [12, 14, 16]
}
best_model,best_params = me.perform_hyperparameter_tuning(model, param_grid_XG, X_train, y_train)
```

```
Starting Grid Search...
Fitting 5 folds for each of 162 candidates, totalling 810 fits

bst.update(dtrain, iteration=i, fobj=obj)
Best parameters found:
{'learning_rate': 0.1, 'max_depth': 12, 'n_estimators': 500, 'scale_pos_weight': 12, 'subsample': 0.8}
Best cross-validation F1-score: 0.9321
```

Figure 55 Hyperparameter tuning

To improve performance, a **grid search** was conducted across 162 parameter combinations of the XGBoost model. The best configuration found was:

```
{'learning_rate': 0.1, 'max_depth': 12, 'n_estimators': 500, 'scale_pos_weight': 12, 'subsample': 0.8}
Best cross-validation F1-score: 0.9321
```

This optimized model achieved an improved cross-validation **F1-score of 0.9321**, indicating better performance under fine-tuned settings.

6.4.4. Final Model Evaluation

The visualizations illustrate the performance of a classification model predicting insurance claims. The bar chart (see *Figure 56*) shows that the model closely matches the actual distribution of non-claim (class 0) cases but slightly overpredicts claims (class 1).

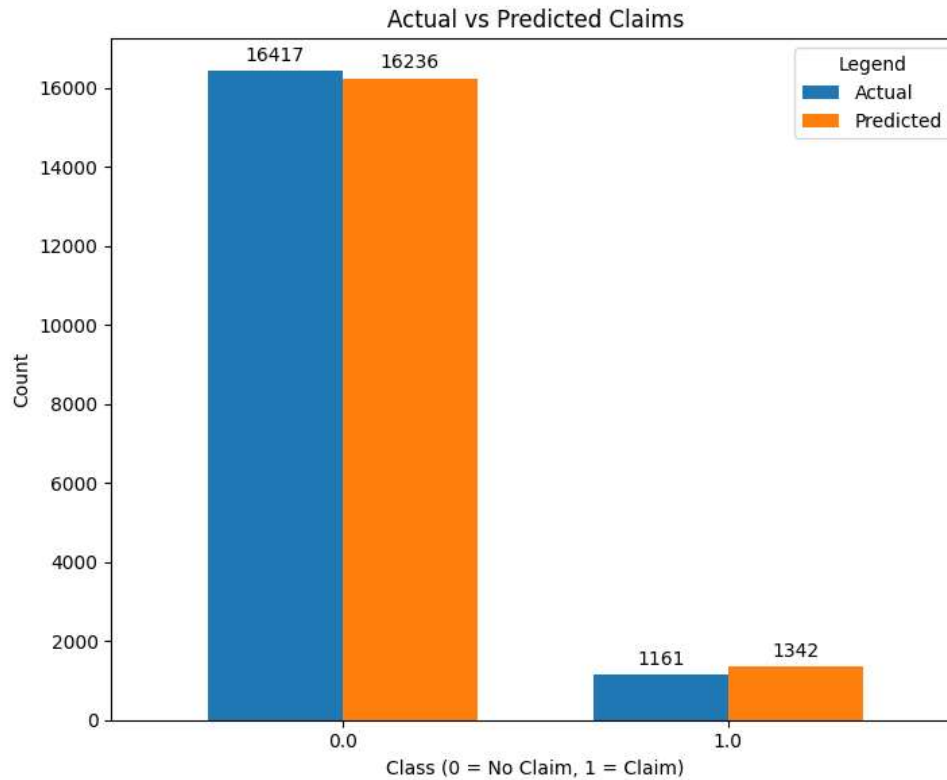


Figure 56 Actual Vs Predicted Claim in train dataset

6.5. Model Testing

This section presents the results of the final XGBoost model applied to the **unseen test dataset**. Since the test data does not contain target labels, the evaluation is limited to prediction generation and interpretability using SHAP values.

6.5.1. Prediction on Test Set

The trained and tuned XGBoost model was used to predict insurance claims on the cleaned test dataset using the top features identified via Mutual Information (see [Appendix D.1](#)). A **threshold of 0.15** was applied to the predicted probabilities to generate binary class labels, balancing recall and false positives for real-world deployment.

```
Predicted claims: 2933
Claim rate: 0.07508383892686174
```

- **Predicted Claims:** 2,933
- **Claim Rate:** 7.50%

Figure 57 Predicted Claims in Test Dataset

This predicted claim rate is slightly higher than the training data claim rate (~6.4%), which may be due to either model bias toward the minority class (due to SMOTE and class weighting) or underlying differences in the test distribution.

Predicted probabilities and class labels were stored for integration into business workflows and dashboard visualizations.

6.5.2. Model Interpretability Using SHAP

To understand the decision-making process of the XGBoost model, **SHAP** (Lundberg & Lee, 2017 [13]) was applied. SHAP values measure the contribution of each feature to the model's prediction for a specific instance.

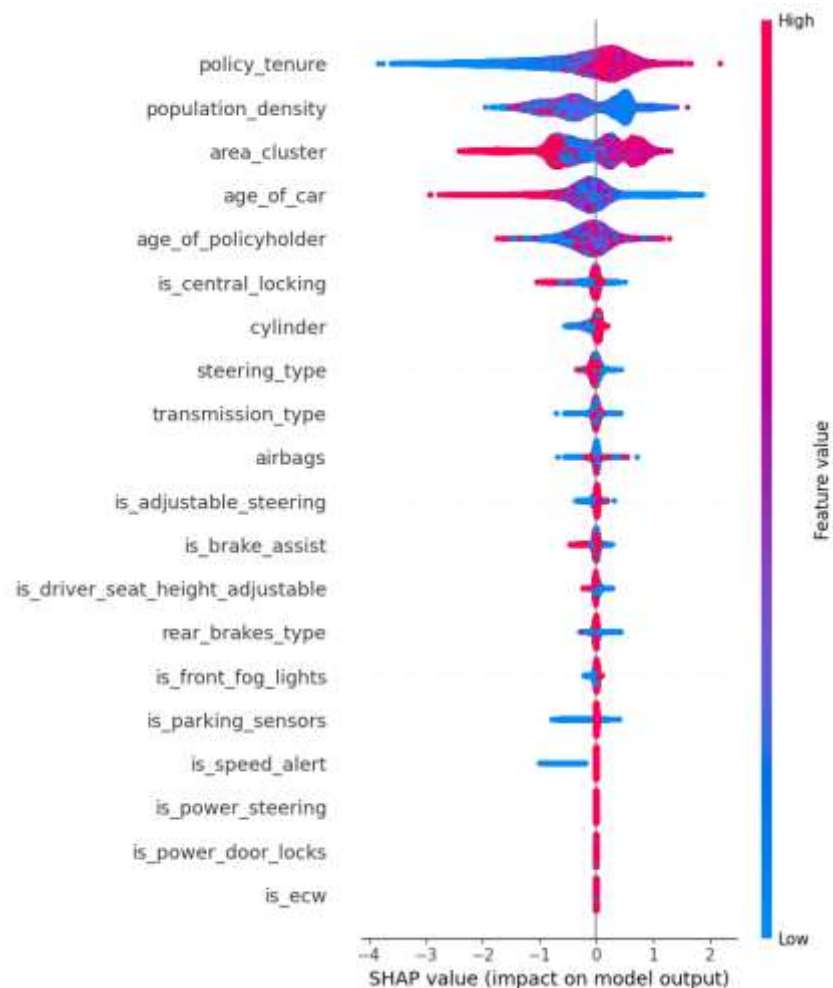


Figure 58 SHAP summary plot showing area_cluster, age_of_policyholder, and policy_tenure as top predictors influencing claim probability.

SHAP values were used to identify the most influential features in claim prediction, offering global and instance-level explanations. For instance, features such as `area_cluster`, `age_of_policyholder`, and `policy_tenure` were found to contribute significantly to the predicted claim probability.

In practice, insurance professionals can use SHAP outputs to justify risk assessments. Underwriters might use these insights to adjust policy terms based on individualized risk profiles, while claims analysts could investigate cases where unexpected features heavily influence predictions, flagging potential fraud or anomalies.

6.5.3. Model Interpretability Using LIME

LIME (Ribeiro et al., 2016 [14]) provides localized interpretability, explaining the prediction for a single instance by approximating the model's behavior using a linear surrogate. This method is particularly useful when dealing with borderline predictions or contested claims.

For example, a claims adjuster may examine a prediction indicating a high claim risk and use LIME to understand which features drove the decision. If the model weights are based on recent tenure and regional risk, the adjuster can cross-validate this against historical data or customer profiles to ensure the prediction is reasonable.

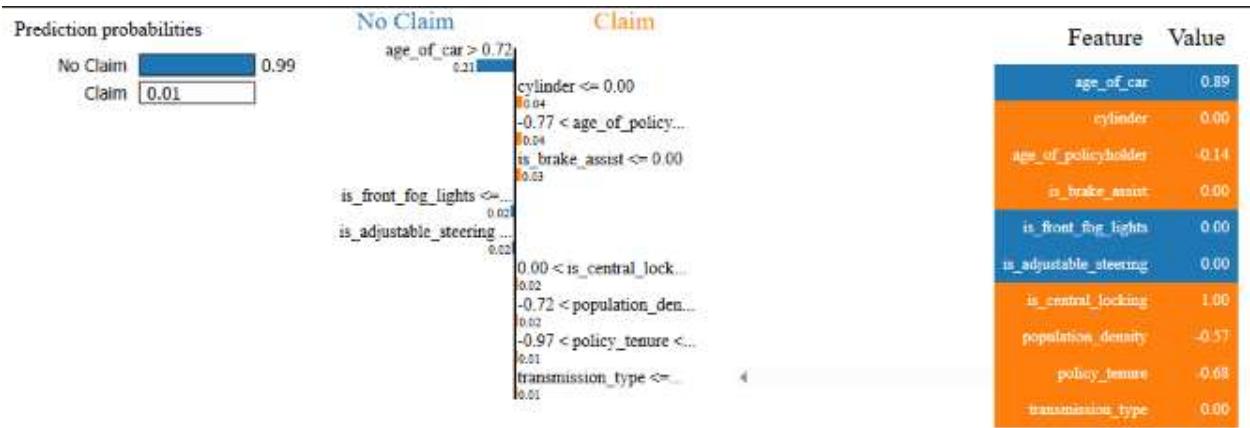


Figure 59 LIME local explanation for a sample claim (positive class), showing how high vehicle weight and short policy tenure increased predicted risk.

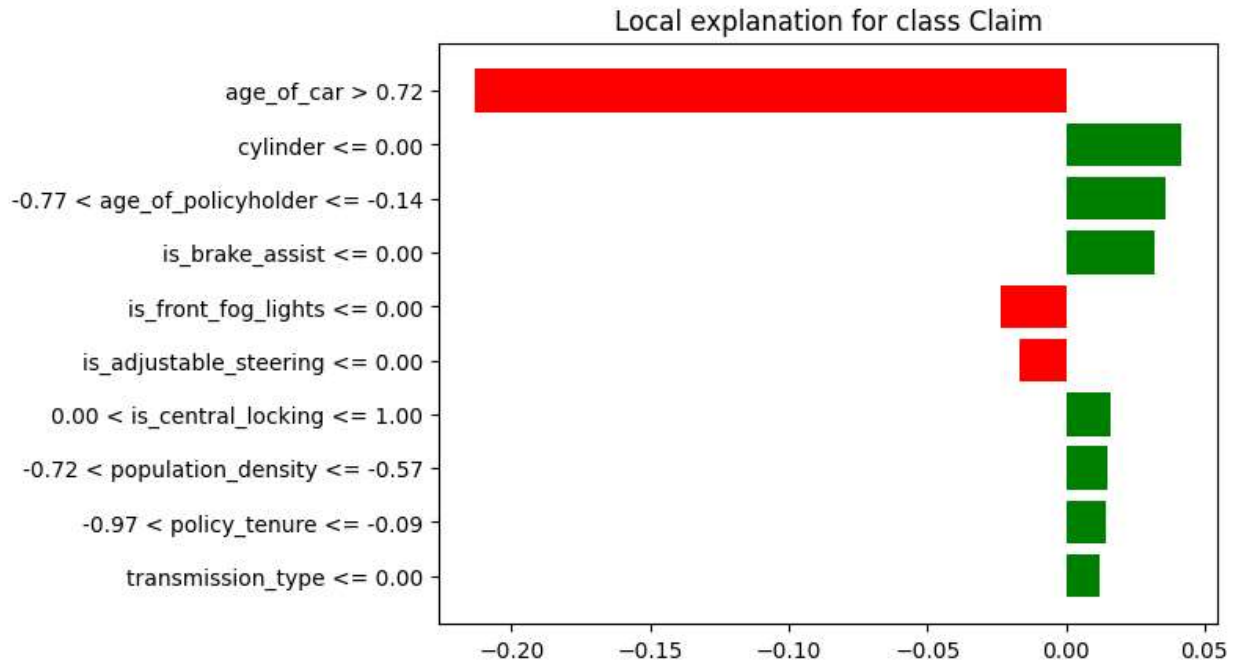


Figure 60 Local explanation for class claim

To interpret individual predictions from the XGBoost model, LIME was applied to a test instance. The model predicted a **very low probability of claim (1%)**, with the prediction driven by features such as **age of car > 0.72** and **presence of front fog lights**, both contributing towards a "**No Claim**" prediction. Conversely, features like **cylinder ≤ 0**, **policy tenure < −0.68**, and **is brake assist = 0** slightly increased the likelihood of a claim. This local explanation illustrates how specific feature values influence the prediction, providing transparency and aiding trust in the model's decisions.

6.6. Development of Power BI Dashboard

An interactive dashboard was built in Microsoft Power BI (Microsoft, 2023) [25] to provide real-time, intuitive insights from the insurance claim prediction model. Power BI was selected for its seamless integration with structured data and ability to create dynamic, user-friendly interfaces for non-technical users like insurance analysts and managers.

The dashboard presents key model findings, visualizes feature distributions, highlights regional and demographic claim trends, and supports operational decisions. It connects to the model's predictions and cleans data via Power Query, enabling smooth updates and interactivity.

The home page contains direct links to eight distinct sections of the application (As shown in *Figure 61*). This design facilitates easy access and improves user navigation across the platform.

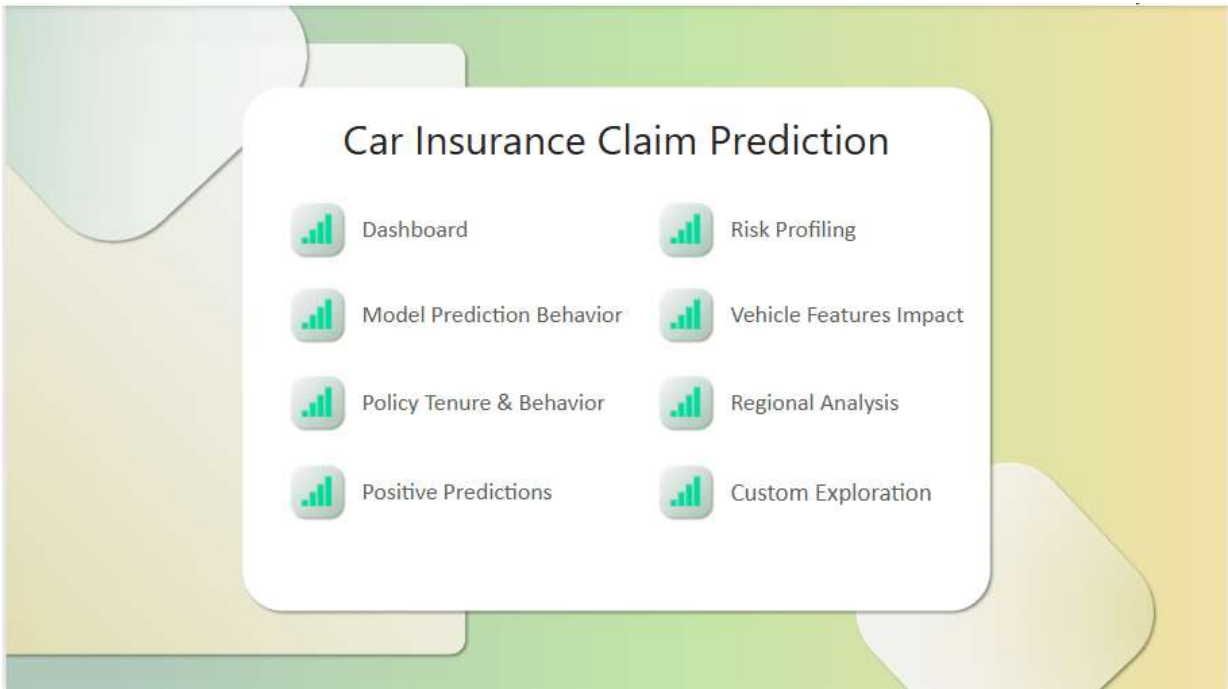


Figure 61 Power BI -Home page

6.6.1. Page 01: Dashboard



Figure 62 Page 01: Dashboard

The first page of the Power BI dashboard provides a clear overview of insurance claim predictions and risk profiling from the machine learning model. Cleaned and formatted data ensures accurate visualization and analysis.

Key DAX measures power KPI cards showing total policies (39K), predicted positive cases (3K), average predicted probability (12%), and average policyholder age (0.47). Visuals include:

- Claim Rate Chart: Highlights class imbalance between low- and high-risk cases.
- Risk Level Distribution: Shows over 90% low-risk policies and ~5.7% high-risk.
- Policyholders by Age Group: Majority in "Moderate" tenure (2–5 years), then "Old" (5–10 years).
- Average Predicted Probability by Policy Tenure: Declining risk with longer tenure, supporting retention strategies.

Interactive slicers for Area_Cluster, Age_Group, Risk_Level, and Transmission_Type enable dynamic data exploration. This dashboard translates model outputs into actionable business insights, helping analysts monitor risk distribution, identify high-risk segments, and inform underwriting and customer loyalty strategies enhancing trust and explainability for stakeholders.

6.6.2. Page 02: Risk Profiling

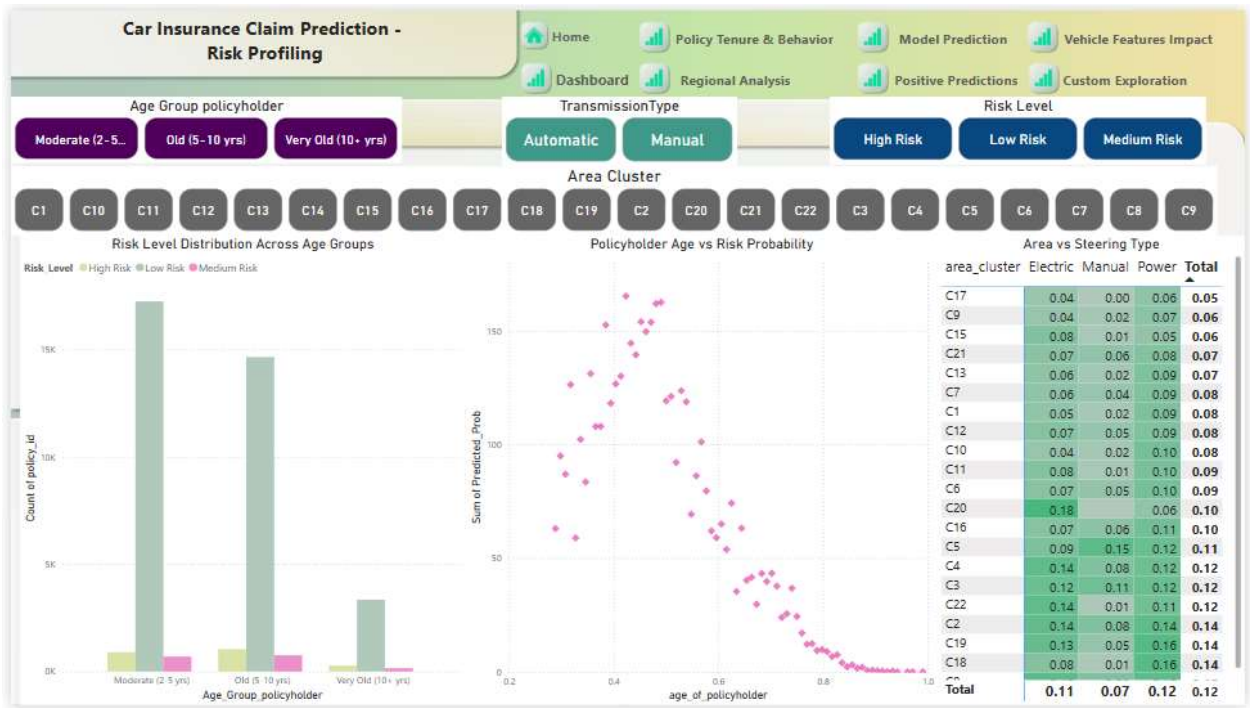


Figure 63 Page 02: Risk profiling

The "Risk Profiling" dashboard highlights key patterns in insurance claim risk. Most policyholders are in the "Moderate" and "Old" age groups and are generally low risk, though some medium and high-risk cases remain. A scatter plot shows younger policyholders have higher predicted risk, while older ones are lower risk. A heatmap reveals regional risk differences clusters like C5 and C16 show higher risk, especially with power steering. Interactive filters (age, area, transmission, risk level) support deeper analysis. Overall, age, location, and vehicle features are strong risk indicators, aiding targeted insurance strategies.

6.6.3. Page 03: Model Prediction Behavior

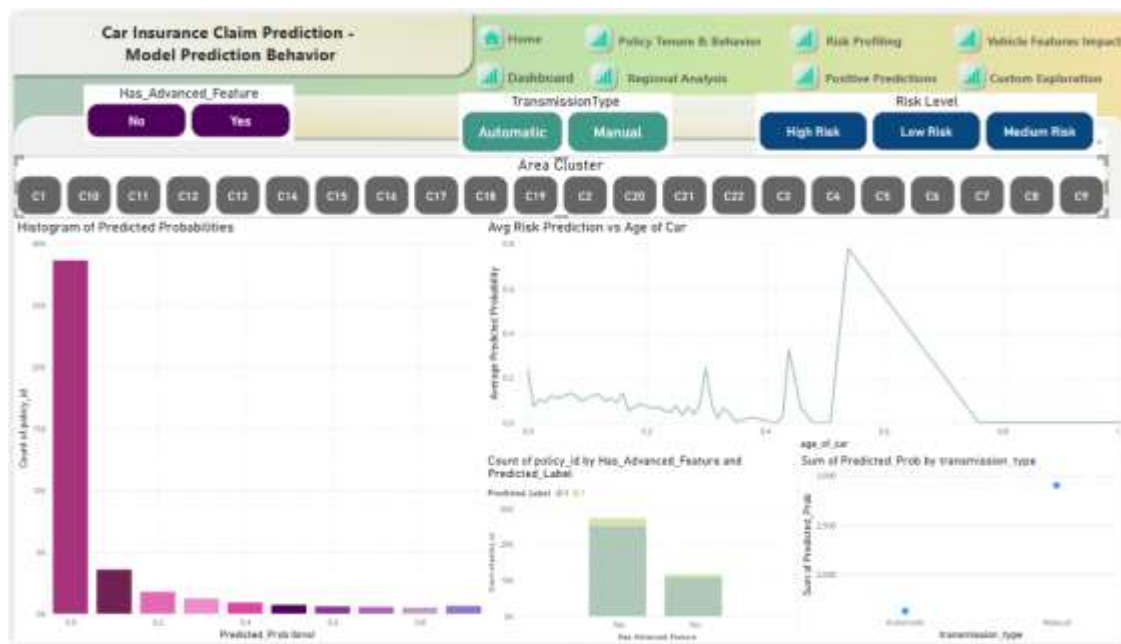


Figure 64 Page 03: Model Predictions Behaviors

The “Model Prediction Behavior” page reveals that most risk predictions fall in the 0–0.1 range, indicating strong model confidence in identifying low-risk cases. Risk peaks for mid-aged cars, suggesting they may be more prone to claims than newer or older vehicles. Cars with advanced features are typically predicted as low risk, while automatic transmission vehicles show higher predicted risk than manual ones. These patterns offer insights into how the model assesses risk and guide insurer focus.

6.6.4. Page 04: Vehicle Features Impact

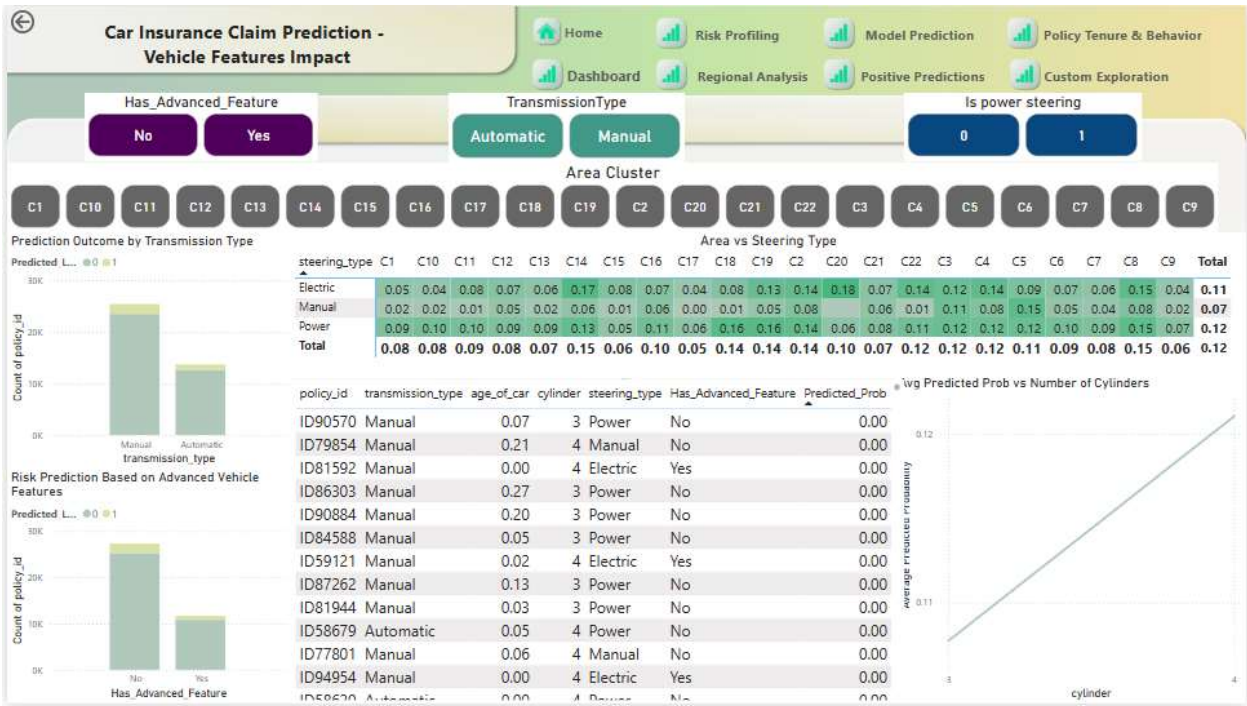


Figure 65 Page 04: Vehicle Features Impact

This page examines how vehicle features affect risk prediction. Manual transmission cars are more often labeled risky than automatics. A heatmap shows higher predicted risk for power steering vehicles in regions like C15 and C18, indicating regional and feature-specific patterns. Cars with advanced features are seen as lower risk, highlighting their safety benefits. Additionally, vehicles with more engine cylinders show slightly higher risk. These insights clarify how vehicle attributes influence the model and support targeted insurance strategies.

6.6.5. Page 05: Policy Tenure & Behavior

This page analyzes policy tenure and risk behavior. The clustered column chart shows most policyholders are in the "Moderate (2–5 yrs)" group, mainly low risk. Long-term policyholders also show fewer high-risk cases. A pie chart reveals customers with advanced features are largely low risk, highlighting tech benefits. The scatter plot shows newer cars have lower predicted risk. A detailed table and interactive slicers enable focused analysis by tenure, age, risk level, and features, supporting targeted risk assessment.

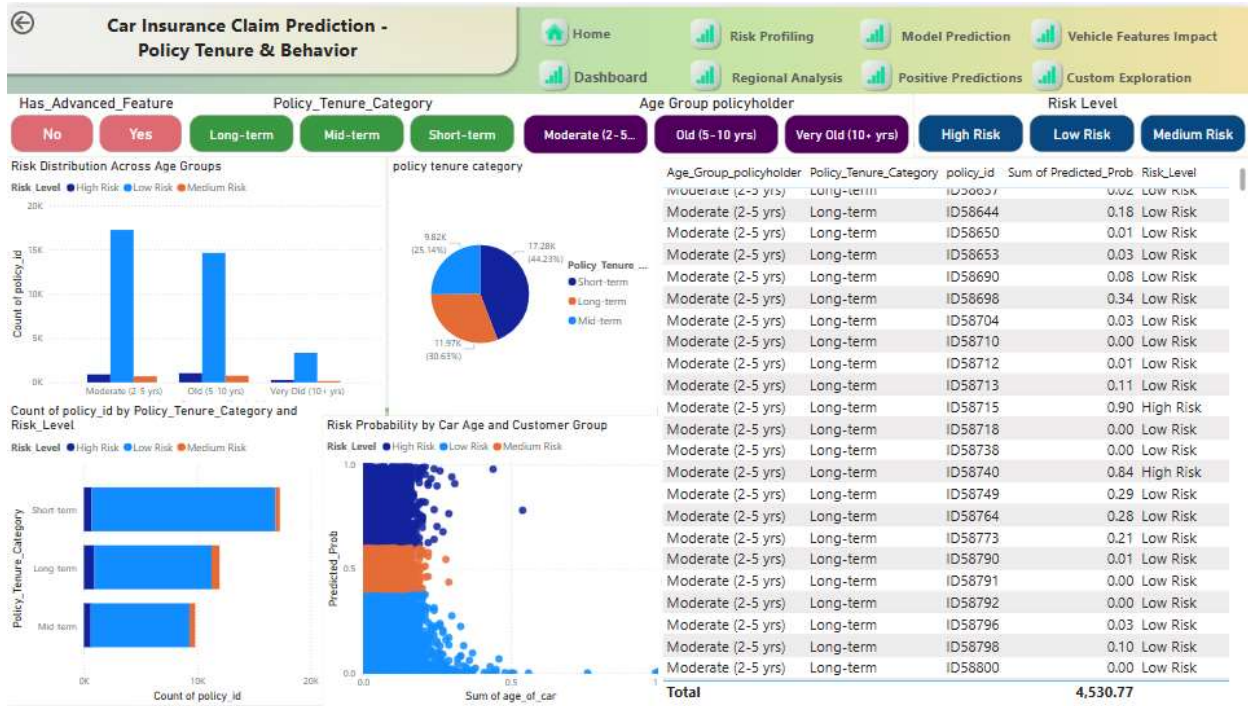


Figure 66 Page 05: Policy Tenure & Behavior

6.6.6. Page 06: Regional Analysis

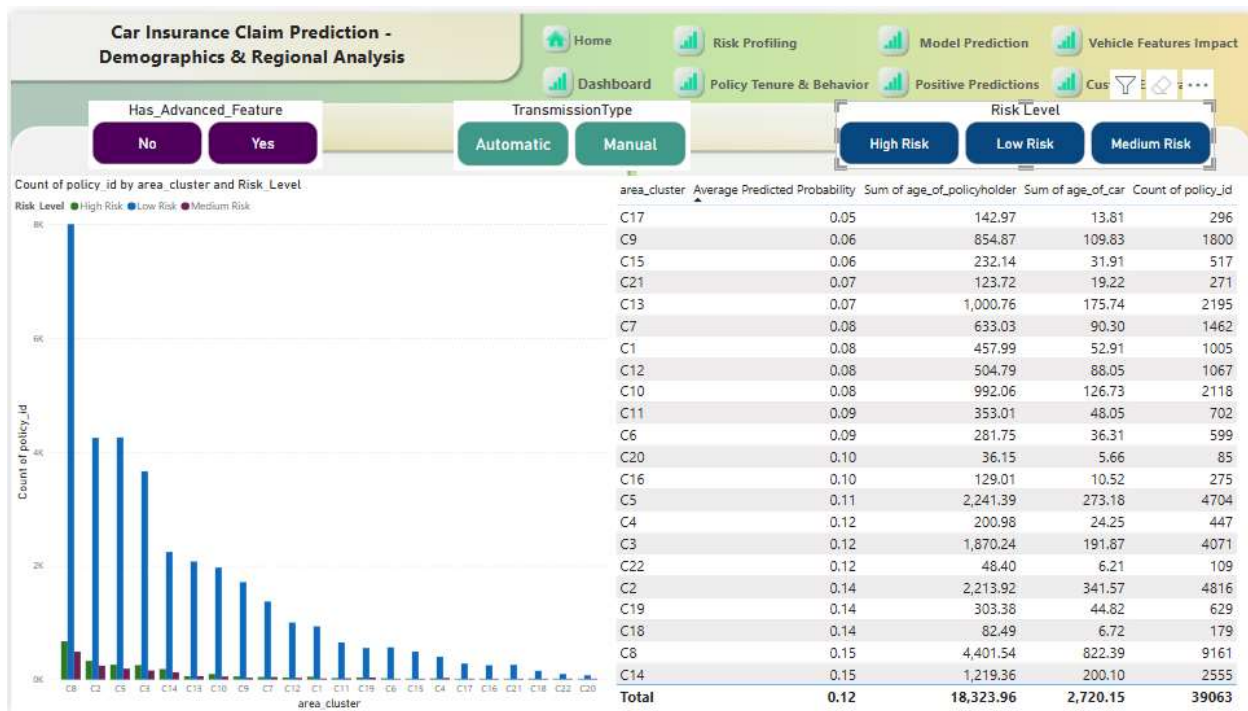


Figure 67 Page 06: Regional Analysis

The “Regional Risk” analysis reveals key area-based risk patterns. A stacked column chart shows clusters like C8, C5, and C3 have many low-risk policyholders, while C19 and C14 have higher proportions of high-risk cases. A clustered chart highlights that C19 and C14 also have the highest predicted risk scores. The summary table shows clusters like C14 have older vehicles and policyholders, driving higher risk. Interactive slicers enable deeper analysis by transmission type, risk level, and advanced features, aiding targeted regional underwriting and insurance strategies.

6.6.7. Page 07: Positive Predictions

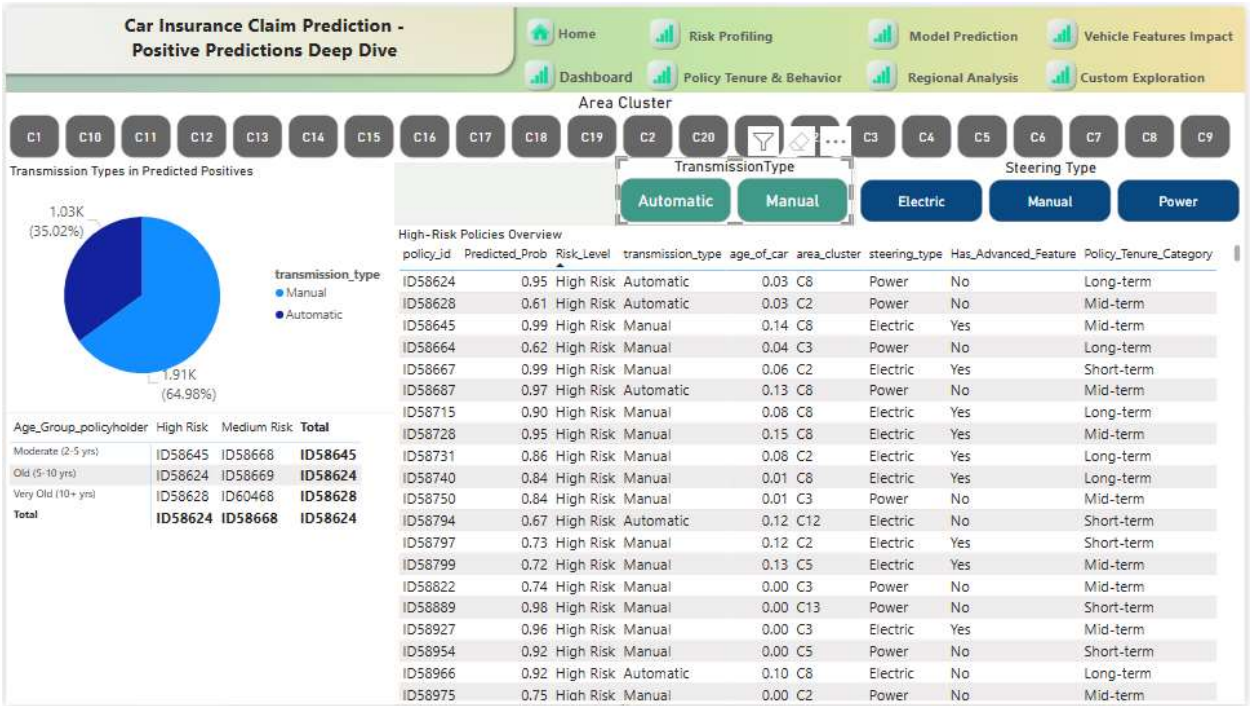


Figure 68 Page 07: Positive Predictions - Deep Dive

The “Positive Predictions Deep Dive” page focuses on high-risk policies (Predicted_Label = 1) for targeted analysis. A detailed table lists high-risk policies with key features like predicted probability, transmission, vehicle age, region, steering, and advanced features, aiding prioritization. A pie chart shows manual transmissions dominate (~65%) among high-risk cases, suggesting higher risk. A matrix cross-tabulates risk levels by age group, confirming varied risk within predicted positives. Interactive slicers enable refined filtering for triage or pricing. This page supports focused risk management, pattern recognition, and targeted actions.

6.6.8. Page 08: Custom Exploration

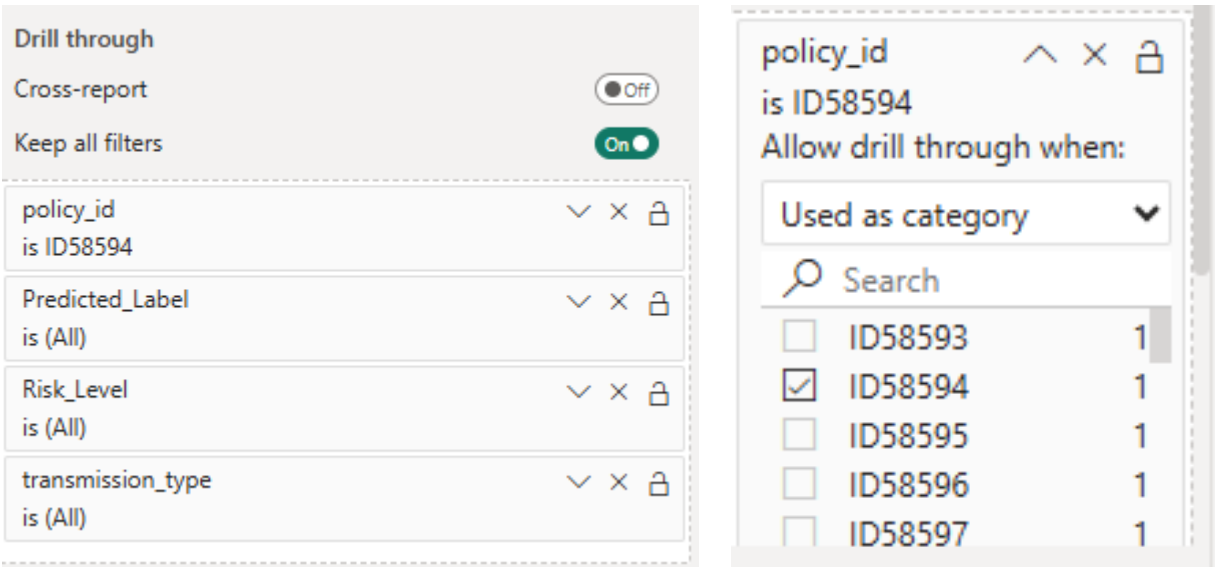


Figure 69 Custom Drill Through for Page 08

Drillthrough was set up using **policy_id**, **Risk_Level**, and **Predicted_Label**, allowing users to right-click and explore detailed policy information on a separate page.



Figure 70 Page 08: Custom Exploration

By adding to the Drillthrough field well, detailed exploration for each individual policy was enabled. Slicers such as Area Cluster, Risk Level, Transmission Type, and Segment were configured to reflect the selected policy IDs. Key details including Policy ID, Risk Level, Predicted Probability, Age Group, Transmission Type, and Model were displayed in cards, while a table at the bottom presented all relevant row-level information.

6.6.9. Limitations and Future Enhancements

While the dashboard provided valuable insights, some limitations were identified. Real-time integration with live policyholder databases was not implemented due to project constraints, and the current version relies on static outputs exported from the modeling pipeline. Future work may explore real-time prediction integration and explainability extensions for additional stakeholders.

6.6.10. Summary

This chapter presented a comprehensive evaluation of a machine learning pipeline for car insurance claim prediction. Feature selection using Mutual Information (MI), Random Forest, and ANOVA revealed that MI yielded the best performance across key metrics, identifying safety features, driver demographics, and car characteristics as significant predictors.

Among classifiers, XGBoost demonstrated the most balanced performance in handling the imbalanced dataset, outperforming others in precision-recall tradeoff, despite Gradient Boosting achieving a higher AUC. After hyperparameter tuning and SMOTE balancing, XGBoost achieved a macro F1-score of 0.9321 and demonstrated strong generalization.

SHAP and LIME interpretability tools confirmed the importance of features like car and policyholder age, policy tenure, and population density in influencing predictions. These insights enhanced model transparency and supported regulatory and business trust.

Finally, a Power BI dashboard was developed to translate model outputs into actionable insights, enabling risk profiling, regional analysis, and strategic decision-making for insurance stakeholders.

CHAPTER 7 - CONCLUSIONS AND FUTURE WORK

7.1. Conclusions

This research project was driven by the need to improve risk assessment in the insurance industry by leveraging machine learning to predict the likelihood of a policyholder filing a claim within a six-month period. Through a structured implementation of the CRISP-DM methodology, a comprehensive and interpretable end-to-end predictive modeling pipeline was developed using a publicly available insurance dataset from Kaggle. Each phase from data collection and preprocessing to feature engineering, model training, and evaluation was executed with attention to both technical rigor and practical applicability.

The study effectively addressed all six research questions posed at the outset:

- **Key Predictive Factors:** Through multivariate analysis, feature importance ranking, and SHAP-based interpretability, the study identified significant predictors including `policy_tenure`, `age_of_policyholder`, vehicle segment, and engine performance features like `power_value` and `torque_value`. These features exhibited strong associations with claim behavior and were instrumental in shaping the model.
- **Model Selection and Performance:** Among the models evaluated—Logistic Regression, Random Forest, XGBoost, and Neural Networks—**XGBoost** emerged as the top performer in terms of precision, recall, F1-score, and ROC-AUC. Its robustness against overfitting and ability to handle class imbalance made it the most suitable choice for this binary classification task.
- **Handling Class Imbalance:** The severe class imbalance in the dataset was effectively addressed using **SMOTE** combined with class weighting. This approach improved the model's sensitivity to minority class predictions without compromising overall accuracy.
- **Predictive Feature Analysis:** The research demonstrated that a combination of **customer demographics**, **policy history**, and **vehicle attributes** drives claim likelihood. The role of derived features like `car_volume` and `parsed_max_power` also contributed to a more nuanced understanding of risk.

- **Interpretability and Visualization:** SHAP and LIME were employed to explain individual predictions, enhancing transparency and stakeholder trust. Furthermore, the development of a **Power BI dashboard** facilitated interactive data exploration, model output visualization, and actionable insights for business users.
- **Towards Deployment:** While full-scale deployment was beyond the scope of this project, the design of the model and modular structure of the pipeline enables straightforward integration into real-time systems. The work lays the foundation for building APIs and retraining workflows in production environments.

In conclusion, this study successfully delivers a scalable, interpretable, and data-driven solution that addresses the complexities of insurance claim forecasting. It demonstrates that machine learning, when carefully engineered and ethically applied, can significantly improve predictive performance over traditional actuarial approaches contributing to smarter risk management and more personalized insurance services.

7.2. Future Recommendations

To build upon the achievements of this project and adapt to emerging industry demands, the following future research and development directions are recommended:

- **Incorporation of Unstructured and External Data:** Leveraging data from **telematics**, **driving behavior logs**, **social media**, or **text-based claim descriptions** using NLP could enhance the model's contextual awareness and predictive capability.
- **Deployment of Real-Time Predictive Systems:** Future iterations should focus on **deploying the model as an API**, using frameworks such as Flask or FastAPI, along with **automated retraining pipelines** to handle model drift and ensure long-term reliability.
- **Bias and Fairness Auditing:** Despite efforts to ensure interpretability, deeper fairness-aware learning approaches such as **adversarial debiasing**, **demographic parity constraints**, or **equal opportunity modeling** should be explored to prevent unintended discrimination, especially in sensitive features like age, region, or policy type.

- **Advanced Ensemble Techniques:** Hybrid or stacked ensemble models combining the interpretability of GLMs with the power of tree-based methods like XGBoost may provide an optimal balance between transparency and performance.
- **Cost-Sensitive and Business-Aware Modeling:** Incorporating a cost matrix that reflects the **real-world consequences** of false positives (unnecessary investigations), and false negatives (missed high-risk claims) can fine-tune the model to align with business goals.
- **Enhanced Visualization and User Engagement:** The Power BI dashboard can be further extended to include **alerting systems**, **segmentation reports**, and **interactive “what-if” analysis**, making it a dynamic decision-support tool for claims managers and underwriters.

7.3. Final Remarks

This research makes a meaningful contribution to the growing field of applied machine learning in insurance. It bridges the gap between academic rigor and industrial relevance by not only building accurate predictive models but also emphasizing interpretability, scalability, and ethical AI principles. By aligning predictive insights with business strategy through interactive visual tools and explainable AI methods, the project demonstrates how data science can drive measurable improvements in claims forecasting, fraud detection, and policy pricing.

With further enhancements, this work holds substantial potential for real-world deployment, offering insurers a competitive edge in a data-intensive industry that increasingly demands precision, personalization, and transparency.

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APPENDIX A

ABOUT DATA SET

A.1 Variable Description

The dataset comprises details about policyholders, including attributes such as policy tenure, car age, owner age, city population density, car make and model, engine type, power, and a target variable indicating whether a claim is filed within the next six months.

Table 12, shows the Description of Variable or features (Car Insurance Claim Prediction [\[1\]](#)).

No	Variable or Feature	Description
1	policy_id	Unique identifier of the policyholder
2	policy_tenure	Time period of the policy
3	age_of_car	Normalized age of the car in years
4	age_of_policyholder	Normalized age of policyholder in years
5	area_cluster	Area cluster of the policyholder
6	population density	Population density of the city (Policyholder City)
7	make	Encoded Manufacturer/company of the car
8	segment	Segment of the car (A/ B1/B2/C1/C2)
9	model	Encoded name of the car
10	fuel_type	Type of fuel used by the car
11	max_torque	Maximum Torque generated by the car (Nm@rpm)
12	max_power	Maximum Power generated by the car (bhp@rpm)
13	engine_type	Type of engine used in the car
14	airbags	Number of airbags installed in the car
15	is_esc	Boolean flag indicating whether Electronic Stability Control (ESC) is present in the car or not.
16	is_adjustable_steering	Boolean flag indicating whether the steering wheel of the car is adjustable or not.
17	is_tpms	Boolean flag indicating whether Tyre Pressure Monitoring System (TPMS) is present in the car or not.

18	is_parking_sensors	Boolean flag indicating whether parking sensors are present in the car or not.
19	is_parking_camera	Boolean flag indicating whether the parking camera is present in the car or not.
20	rear_brakes_type	Type of brakes used in the rear of the car
21	displacement	Engine displacement of the car (cc)
22	cylinder	Number of cylinders present in the engine of the car
23	transmission_type	Transmission type of the car
24	gear_box	Number of gears in the car
25	steering_type	Type of the power steering present in the car
26	turning_radius	The space a vehicle needs to make a certain turn (Meters)
27	length	Length of the car (Millimetre)
28	width	Width of the car (Millimetre)
29	height	Height of the car (Millimetre)
30	gross_weight	The maximum allowable weight of the fully loaded car, including passengers, cargo and equipment (Kg)
31	is_front_fog_lights	Boolean flag indicating whether front fog lights are available in the car or not.
32	is_rear_window_wiper	Boolean flag indicating whether the rear window wiper is available in the car or not.
33	is_rear_window_washer	Boolean flag indicating whether the rear window washer is available in the car or not.
34	is_rear_window_defogger	Boolean flag indicating whether rear window defogger is available in the car or not.
35	is_brake_assist	Boolean flag indicating whether the brake assistance feature is available in the car or not.
36	is_power_door_lock	Boolean flag indicating whether a power door lock is available in the car or not.

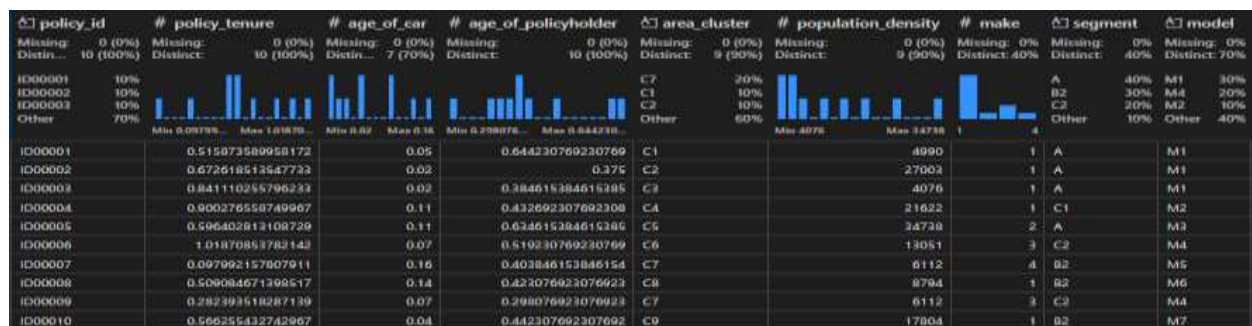
37	is_central_locking	Boolean flag indicating whether the central locking feature is available in the car or not.
38	is_power_steering	Boolean flag indicating whether power steering is available in the car or not.
39	is_driver_seat_height_adjustable	Boolean flag indicating whether the height of the driver seat is adjustable or not.
40	is_day_night_rear_view_mirror	Boolean flag indicating whether day & night rearview mirror is present in the car or not.
41	is_ecw	Boolean flag indicating whether Engine Check Warning (ECW) is available in the car or not.
42	is_speed_alert	Boolean flag indicating whether the speed alert system is available in the car or not.
43	ncap_rating	Safety rating given by NCAP (out of 5)
44	is_claim	Outcome: Boolean flag indicating whether the policyholder files a claim in the next 6 months or not.

Table 12 Variable description

A.2 Sample of Training Dataset

Figure 71 shows the first 10 rows of the training dataset used for model development. This sample includes all relevant features. It provides a snapshot of the structure and type of data used for training machine learning models.

Note: This is a representative subset shown using the `train_df.head(10)` command in a Python environment.



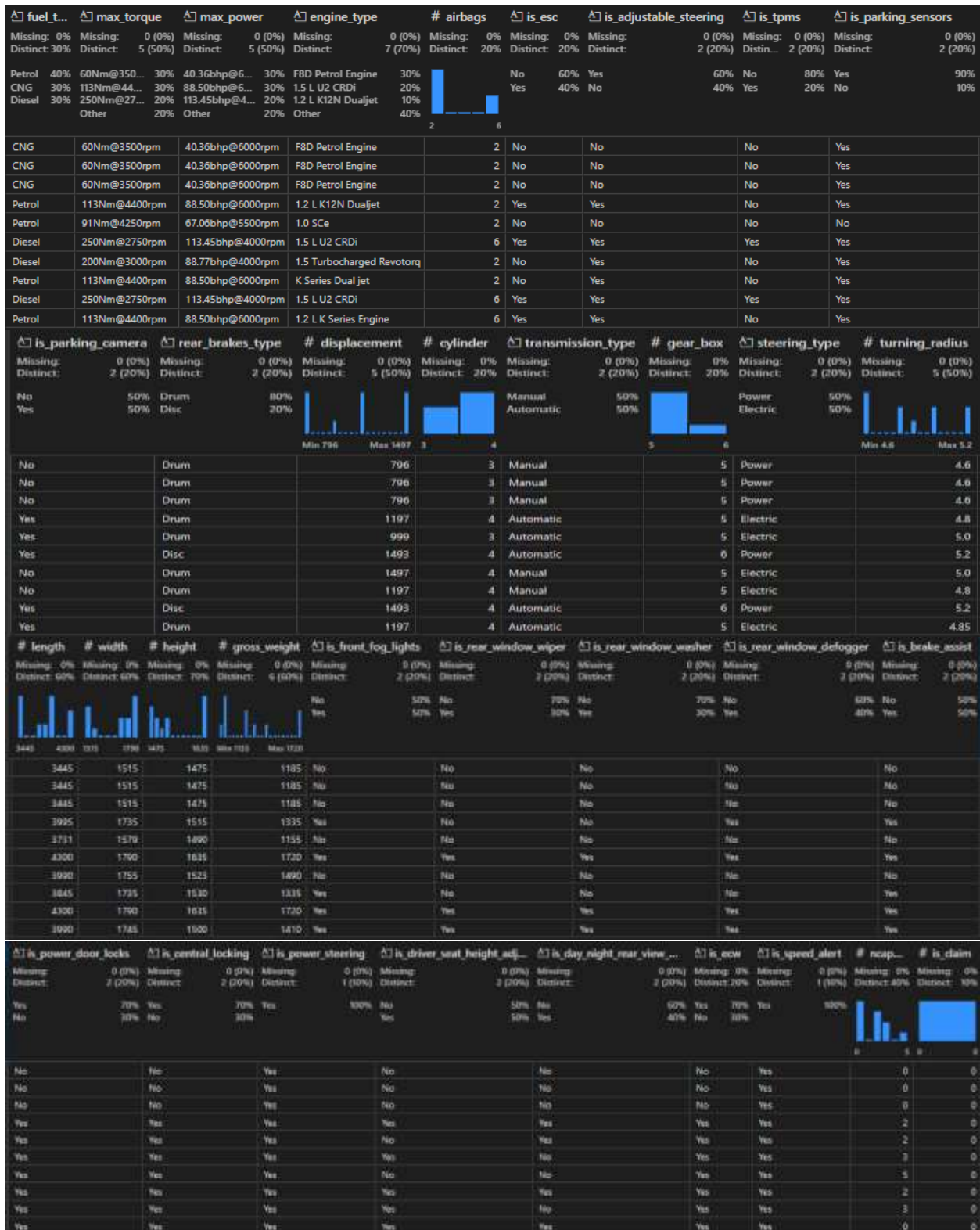


Figure 71 Sample of Training set

A.3 SMOTE (Synthetic Minority Over-sampling Technique)

SMOTE (Chawla et al., 2002) [\[8\]](#) generates synthetic examples of the minority class by interpolating between existing minority samples. This contrasts with:

- **Simple random oversampling**, which duplicates existing minority samples and may lead to overfitting,
- **Random undersampling**, which removes majority class examples and may discard valuable information.

SMOTE improves the representation of the minority class without loss of data, making it an effective and widely used method for classification problems with moderate imbalance.

Alternative Sampling Strategies

Other resampling techniques were also considered for comparison:

1. **ADASYN (Adaptive Synthetic Sampling)** builds on SMOTE by generating more synthetic data in regions where the minority class is harder to teach (He et al., 2008)[\[22\]](#). It can focus learning more effectively but may introduce more noise in overlapping regions.
2. **SMOTE variants** such as **Borderline-SMOTE** and **SMOTEENN** combine oversampling with noise filtering or neighborhood cleaning.
3. **Ensemble-based techniques**, such as **Balanced Random Forests** or **EasyEnsemble**, train multiple classifiers on balanced subsets of the data. These approaches help in reducing model variance and are especially effective for highly imbalanced datasets.

While these alternatives have their merits, **SMOTE** was selected for this project due to its balance of simplicity, effectiveness, and compatibility with our modeling pipeline.

Class Weight Adjustment

Several classifiers, such as Logistic Regression, Random Forest, and XGBoost, support built-in class weight balancing. By penalizing misclassified minority class instances more heavily, these models enhance sensitivity to underrepresented outcomes—especially useful when resampling is undesirable or when working with large datasets.

APPENDIX B

LIBRARIES AND TOOLS USED FOR DATA PREPROCESSING

This appendix provides a brief overview of the Python libraries and tools utilized during the data preprocessing phase of this project. Each library plays a specific role in cleaning, transforming, and visualizing the dataset in preparation for model development.

Library	Purpose / Description
pandas	A powerful data analysis and manipulation library. It was used for handling tabular data, loading datasets, cleaning, and transforming data into structured formats (DataFrames).
numpy	A fundamental package for numerical computing in Python. It was employed for handling arrays and performing mathematical operations, particularly during imputation and outlier treatment.
matplotlib.pyplot	A basic plotting library for creating static visualizations such as histograms, bar plots, and scatter plots to explore data distributions and relationships.
seaborn	Built on top of matplotlib, this library provides more sophisticated and visually appealing statistical plots such as boxplots, heatmaps, and violin plots to analyze variable relationships.
math	Used for basic mathematical operations and calculations (e.g., floor, ceil, log) during feature engineering steps.
missingno	A specialized visualization library for identifying and understanding missing data patterns using heatmaps, bar charts, and matrix plots.
sklearn.preprocessing.StandardScaler	Used to scale numerical features to a standard normal distribution (mean=0, std=1), which is critical for many machine learning algorithms.
sklearn.preprocessing.LabelEncoder	Encodes categorical variables as numeric labels. Ideal for binary or ordinal categorical features.
sklearn.preprocessing.OrdinalEncoder	Transforms ordinal categorical features into ordered numerical values, preserving rank information.

sklearn.impute.SimpleImputer	Automatically imputes missing values using specified strategies such as mean, median, or most frequent. Useful in handling missing numeric and categorical data.
sklearn.decomposition.PCA	Reduces data dimensions while retaining most variance (for visualization or speed).
sklearn.cluster.KMeans	Partitions data into k clusters using centroid-based optimization.
sklearn.metrics.silhouette_score	Measures clustering quality; higher score means better-defined clusters.
sklearn.cluster.DBSCAN	Density-based clustering; detects noise and arbitrary-shaped clusters.
sklearn.cluster.AgglomerativeClustering	Hierarchical clustering method; merges clusters based on distance criteria.

Table 13 Libraries for Data preprocessing

APPENDIX C

ADDITIONAL EDA VISUALIZATIONS

This section presents **supplementary exploratory data analysis** to gain deeper insights into specific patterns and relationships not covered in the main EDA section. While the primary EDA included basic statistical summaries and general trends, here explore additional visualizations to enhance understanding of feature distributions, interactions, and potential influences on the target variable.

C.1 Summary statistics (mean, min, max, etc.) for numeric columns.

	policy_tenure	age_of_car	age_of_policyholder	population_density	make	airbags
Count	58592	58592	58592	58592	58592	58592
Mean	0.611246162	0.069423641	0.469420051	18826.85867	1.763722	3.137066494
std	0.414156487	0.056720648	0.122885885	17660.17479	1.136988	1.832641309
min	0.002735273	0	0.288461538	290	1	1
25%	0.2102496	0.02	0.365384615	6112	1	2
50%	0.573791678	0.06	0.451923077	8794	1	2
75%	1.03910423	0.11	0.548076923	27003	3	6
max	1.396641077	1	1	73430	5	6
	displacement	cylinder	gear_box	turning_radius	gross_weight	ncap_rating
Count	58592	58592	58592	58592	58592	58592
Mean	1162.355851	3.626962725	5.245443064	4.852892886	1385.277	1.759950164
std	266.3047864	0.483616023	0.430353259	0.228060818	212.4231	1.38957649
min	796	3	5	4.5	1051	0
25%	796	3	5	4.6	1185	0
50%	1197	4	5	4.8	1335	2
75%	1493	4	5	5	1510	3
max	1498	4	6	5.2	1720	5
	is_claim	torque_value	torque_rpm	power_value	power_rpm	car_volume
Count	58592	58592	58592	58592	58592	58592
Mean	0.063967777	134.450937	3533.176031	78.97676543	5307.163	10087742498
std	0.244697615	73.14679386	725.960661	27.69925882	916.7708	1748448588
min	0	60	1750	40.36	3600	7698283125
25%	0	60	2750	40.36	4000	7698283125
50%	0	113	3500	88.5	6000	10206744750
75%	0	200	4400	97.89	6000	10664731350
max	1	250	4400	118.36	6000	12584595000

Table 14 Summary statistics

Descriptive Statistics and Feature Insights: A thorough assessment of the numerical features was conducted to understand the underlying characteristics of the policyholders, their vehicles, and related performance indicators. The dataset contains **58,592 records** with **no missing values**, ensuring full data integrity for model development.

1. Policy and Policyholder Attributes

The features related to insurance policies and customer demographics provide key context on the dataset's composition:

- **Policy tenure** exhibits a mean of **0.61**, with values ranging from **0.0027 to 1.3966**, reflecting diverse durations of policy coverage. This variation indicates a mix of newly insured and long-term policyholders.
- The **age of cars** and **age of policyholders** are normalized (scaled), with means of **0.07** and **0.47** respectively. This suggests that the dataset includes mostly newer vehicles and middle-aged policyholders.

2. Vehicle Characteristics

The dataset captures multiple technical specifications of the insured vehicles:

- **Airbags** have a mean count of **3.14**, ranging from **1 to 6**, showing a broad spectrum of safety equipment across different vehicle models.
- **Engine displacement** and **torque values** vary between **796 and 1498**, pointing to a diverse set of engine capacities and performance capabilities among the vehicles.

3. Vehicle Dimensions

- The calculated **car volume** (product of length, width, and height) and the **gross weight** demonstrate standard dimensional profiles. The observed ranges confirm inclusion of compact cars and larger, heavier vehicles, suggesting the model will need to generalize across different car types.

4. Performance Metrics

- Features like **torque value**, **power value**, **torque RPM**, and **power RPM** reveal significant variability. For instance, torque and power values range from **60 to 250**, while RPM figures remain within consistent, expected operating bands. These metrics are important as they may correlate with both driving behavior and accident likelihood.

5. Safety Ratings and Insurance Claims

- The **NCAP safety rating**, with a mean of **1.76**, indicates that the vehicles generally fall into a moderate safety tier. This feature plays a crucial role in assessing risk.
- The **is_claim** variable, representing the target, has a mean of **0.06**, confirming that **approximately 6% of policies resulted in claims**. This low positive rate underlines the **class imbalance** challenge that the model must address during training.

C.2 Finding Optimal Clusters

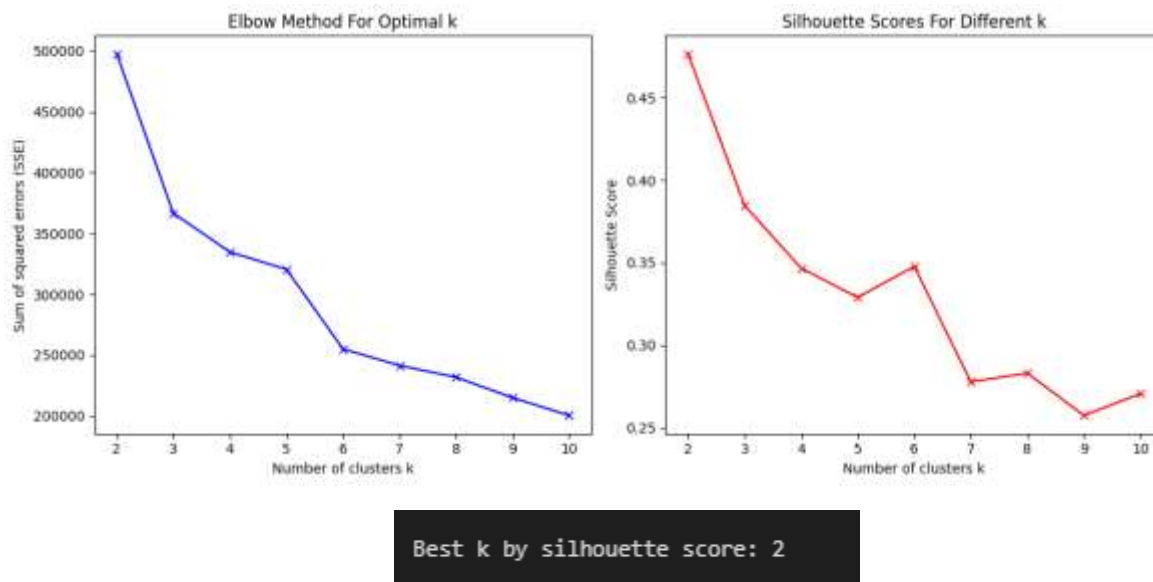


Figure 72 Finding Optimal value (k)

To identify the optimal number of clusters, both the Elbow Method and Silhouette Score analysis were applied. The Elbow Method, which plots the Sum of Squared Errors (SSE) against varying values of k, revealed a noticeable bend at k=2, indicating minimal improvement beyond this point. Similarly, the Silhouette Score peaked at k=2, reinforcing it as the most suitable choice. These findings suggest a natural tendency of the data to form two well-defined clusters.

APPENDIX D

FEATURE SELECTION METHODS

This section provides a detailed explanation of the feature selection methods used in the study.

D.1 Mutual Information (MI)

Mutual Information quantifies how much a given feature is related to the target variable by capturing both linear and non-linear dependencies. Features with higher MI scores contribute more predictive insight, indicating their greater relevance to the target outcome.

Mutual Information between feature X and target Y:

$$MI(X; Y) = \sum_{x \in X} \sum_{y \in Y} p(x, y) \log\left(\frac{p(x, y)}{p(x)p(y)}\right)$$

Where:

$p(x, y)$ is the joint probability distribution of X and Y

$p(x)$, $p(y)$ are the marginal distributions

Equation 3 Mutual Information

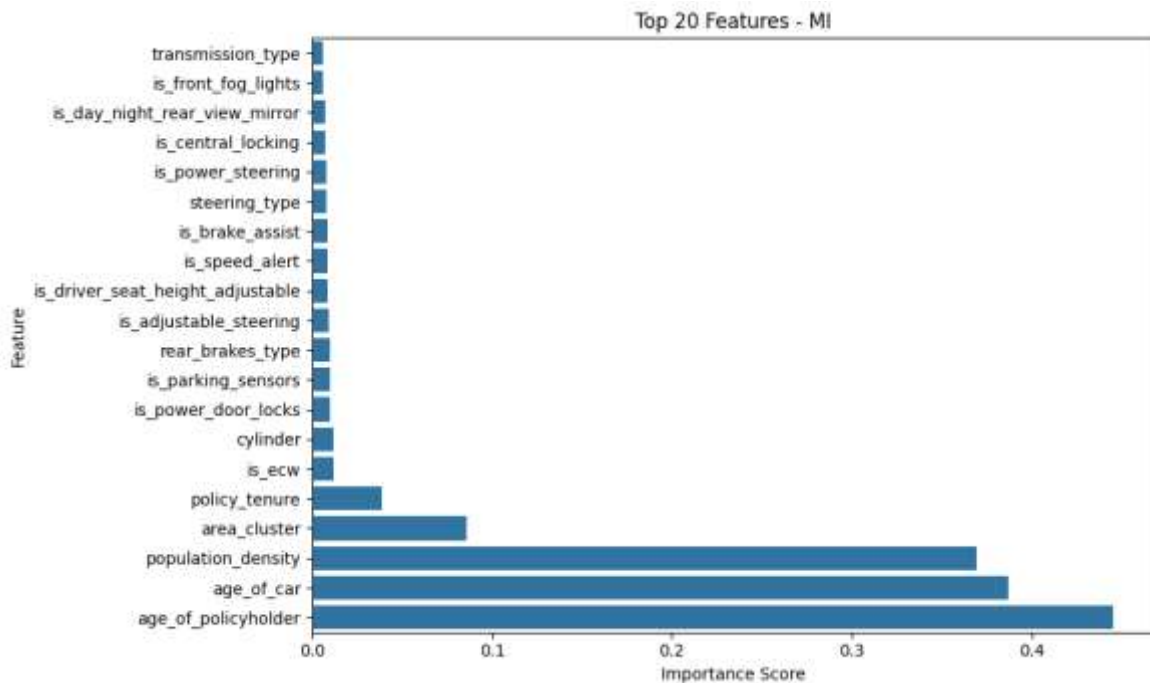


Figure 73 Top 20 Features Using MI

Figure 73 displays the top 20 features ranked by their MI scores. As shown, features such as age_of_policyholder, age_of_car, and population_density have the highest importance scores, indicating a strong relationship with the target variable. These features, along with others like area_cluster, policy_tenure, and various binary indicators (e.g., is_ecw, is_power_door_locks), were selected based on their contribution to the prediction task.

The final set of top features identified by MI includes:

['transmission_type', 'is_front_fog_lights', 'is_day_night_rear_view_mirror', 'is_central_locking', 'is_power_steering', 'steering_type', 'is_brake_assist', 'is_speed_alert', 'is_driver_seat_height_adjustable', 'is_adjustable_steering', 'rear_brakes_type', 'is_parking_sensors', 'is_power_door_locks', 'cylinder', 'is_ecw', 'policy_tenure', 'area_cluster', 'population_density', 'age_of_car', 'age_of_policyholder']

These features were retained for model training and further evaluation.

D.2 Random Forest Importance (RF)

Random Forest determines feature importance by measuring the total reduction in impurity, such as Gini impurity, contributed by each feature across all decision trees. Features that lead to larger impurity decreases are deemed more influential.

$$\text{Importance}(f) = \sum_{t \in T} \sum_{n \in N_f} \frac{w_n \times \Delta i_n}{\sum_{n' \in N} w_{n'} \times \Delta i_{n'}}$$

Where:

- T = trees in the forest
- N_f = nodes where feature f is used
- w_n = weighted number of samples at node n
- Δi_n = impurity decrease at node n

Equation 4 Random Forest Importance

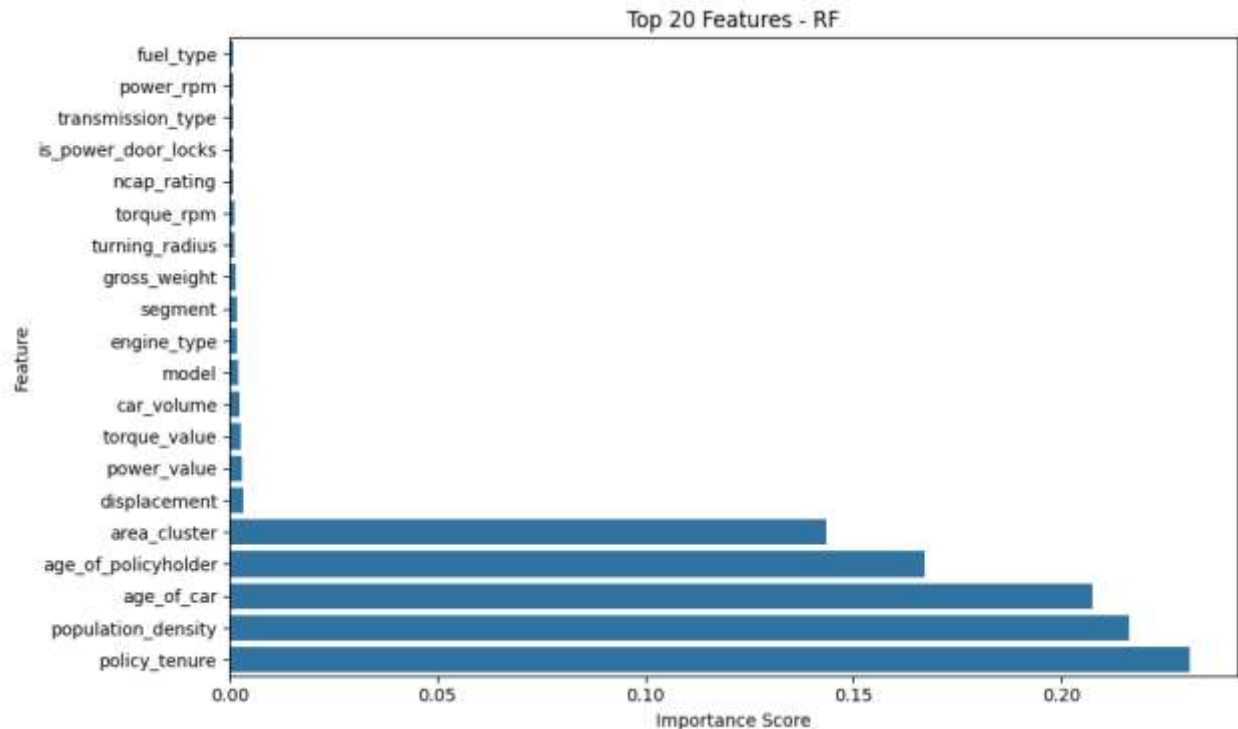


Figure 74 Top 20 features using RF

Figure 74 displays the top 20 features ranked by Random Forest (RF) algorithm based on their contribution to improving model performance.

These features, ranked by their importance scores, are:

[fuel_type, power_rpm, transmission_type, is_power_door_locks, ncap_rating, torque_rpm, turning_radius, gross_weight, segment, engine_type, model, car_volume, torque_value, power_value, displacement, area_cluster, age_of_policyholder, age_of_car, population_density, and policy_tenure].

D.3 ANOVA F-statistic

ANOVA (Analysis of Variance) F-statistic assesses the linear dependency between numerical features and a categorical target. It measures the variance between groups (defined by the target) compared to the variance within groups. Features with a high F-score are more likely to be relevant for distinguishing between target classes.

For feature X and target classes:

$$F = \frac{\text{variance between groups}}{\text{variance within groups}} = \frac{\sum_{k=1}^K nk \times \frac{(\bar{x}_k - \bar{x})^2}{(K-1)}}{\sum_{k=1}^K \sum_{i=1}^{nk} \frac{(x_{ik} - \bar{x}_k)^2}{(N-K)}}$$

Where:

- K: number of classes
- n_k : number of samples in class k
- $\bar{x}_k$: mean of feature values in class k
- $\bar{x}$: overall mean
- N: total number of samples

Equation 5 ANOVA F-statistics

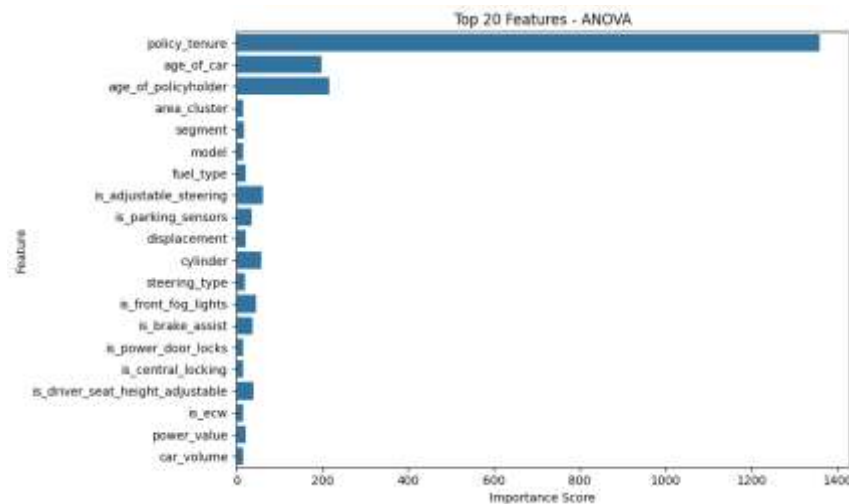


Figure 75 Top 20 features using ANOVA F-statistics

The following features were selected based on ANOVA testing, which identified them as having statistically significant differences in mean values across the target classes, indicating their importance for predicting the outcome:

[policy_tenure, age_of_car, age_of_policyholder, area_cluster, segment, model, fuel_type, is_adjustable_steering, is_parking_sensors, displacement, cylinder, steering_type, is_front_fog_lights, is_brake_assist, is_power_door_locks, is_central_locking, is_driver_seat_height_adjustable, is_ecw, power_value, car_volume.]

APPENDIX E

MODEL CLASSIFICATION

This section provides a detailed explanation of the classification models used in the study.

E.1 Linear Models

1. **Logistic Regression:** A baseline linear classifier that models the probability of a binary outcome using the logistic function. It is interpretable and robust for linearly separable data. The model was trained with `class_weight='balanced'` to handle class imbalance and was evaluated as a benchmark.

Logistic Regression models the probability that a given instance belongs to the positive class using the logistic (sigmoid) function:

$$P(y = 1|x) = \frac{1}{1 + e^{-(w^T x + b)}}$$

Where:

x = input features

w = weights

b = bias

Equation 6 Logistic regression

The model is trained using log-loss or binary cross-entropy.

2. **Ridge Classifier:** A variation of linear regression with L2 regularization. It penalizes large coefficients to reduce overfitting, especially useful when dealing with multicollinearity.
3. **SGD Classifier:** Implements stochastic gradient descent for optimizing a linear decision boundary. It is highly efficient on large-scale datasets and was configured with `class_weight='balanced'` to improve recall for minority class prediction.

E.2 Support Vector Machines

1. **SVC (Support Vector Classifier):** Constructs a hyperplane to separate classes with maximum margin. Kernelized SVC was used with `probability=True` to enable probabilistic outputs and ROC curve plotting. It is effective in high-dimensional spaces but computationally intensive.

SVM seeks to find the optimal hyperplane that maximizes the margin between classes:

$$f(x) = W^T X + b$$

Subject To: $y_i(W^T X_i + b) \geq 1 \forall_i$

Where $y_i \in \{-1, 1\}$.

Equation 7 SVM

In soft-margin SVM, slack variables are introduced, and the cost function includes a penalty term.

2. **LinearSVC:** A faster variant of SVM that uses a linear kernel and the liblinear solver. It is suited for linearly separable data and was trained with `class_weight='balanced'`.

E.3 Tree-based Models

1. **Decision Tree:** A simple, interpretable model that splits data using if-else rules. It was used as a baseline for tree-based learners with `class_weight='balanced'`.

Decision Trees split the dataset recursively based on a feature that results in the largest information gain. A common criterion is the **Gini Index**:

$$Gini(t) = 1 - \sum_{i=1}^c p(i|t)^2$$

Where, $p(i|t)$ = proportion of class i in node t .

Equation 8 Decision tree

2. **Random Forest:** An ensemble of decision trees using bagging. It improves generalization by reducing variance. Feature importances from this model also supported interpretability. Random Forest builds multiple Decision Trees on random subsets of features and data, and aggregates predictions using majority voting:

$$y^{\wedge} = \text{majority_vote}(h_1(x), h_2(x), \dots, h_n(x))$$

Where, h_i is the prediction from the i^{th} tree.

Equation 9 Random Forest

3. **Extra Trees:** Similar to Random Forest but uses random thresholds for feature splits. It enhances diversity among trees, which can improve performance on noisy data.
4. **Gradient Boosting:** Builds trees sequentially by correcting the residuals of prior trees. It often achieves strong performance but is prone to overfitting if not tuned. Gradient Boosting combines weak learners iteratively to minimize loss. At each stage, a new model is added to correct the residuals:

$$F_{t+1}(x) = F_t(x) + \eta \cdot h_t(x)$$

Where, F_t = boosted model at stage t and η = learning rate

Equation 10 Gradient Boosting

5. **XGBoost (Chen et al., 2016) [10]:** A highly optimized gradient boosting algorithm known for its speed and accuracy. It was configured with `scale_pos_weight` to adjust for class imbalance and showed strong performance in recall and F1 score.
6. **LightGBM:** A gradient boosting framework that grows tree leaf-wise rather than level-wise, making it faster and often more accurate. While it achieved high recall, it suffered from low precision in this task.
7. **CatBoost:** Specifically designed to handle categorical data natively. Since all categorical features were already label encoded, this model did not realize its full advantage but still performed competitively.

E.4 Naive Bayes

Naive Bayes applies Bayes' theorem under the assumption of feature independence:

$$P(y | x) \propto P(y) \prod_{i=1}^n P(x_i | y)$$

GaussianNB assumes $x_i | y$ follows a normal distribution.

BernoulliNB is used for binary features and assumes Bernoulli distribution.

Equation 11 Naive Bayes

1. **GaussianNB:** Assumes that features follow a Gaussian distribution. Although it is computationally efficient, it performed poorly on this dataset due to strong independence assumptions.
2. **BernoulliNB:** Designed for binary/boolean feature inputs. Its assumptions did not align well with the numeric and categorical nature of the dataset, leading to low predictive power.

E.5 Distance-based

K-Nearest Neighbors (KNN) assigns class labels to samples by examining the most common category among their k closest neighbors in the feature space. While it is straightforward and does not assume any underlying data distribution, it can be computationally intensive with large datasets. In this project, its effectiveness was hindered by the high number of features and imbalanced class distribution.

KNN is a non-parametric method where classification is based on the **majority vote of nearest neighbors**:

$$\hat{y} = \text{mode}(\{y_i \mid x_i \in N_k(x)\})$$

Where $N_k(x)$ is the set of k – nearest neighbors to input x .

Equation 12 KNN

E.6 Top 15 models by F1 score and ROC AUC

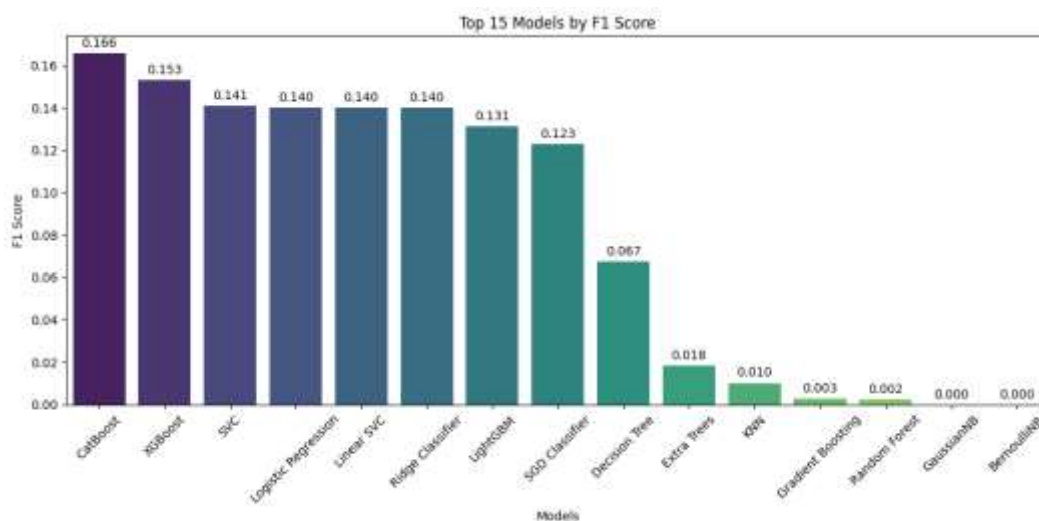


Figure 76 Models by F1 Score

While Logistic Regression achieved the highest F1 score (0.166), XGBoost (F1: 0.140) was selected for its ability to model complex interactions and provide feature importance insights. Tuning and validation confirmed its robustness across precision-recall metrics, aligning with project priorities.

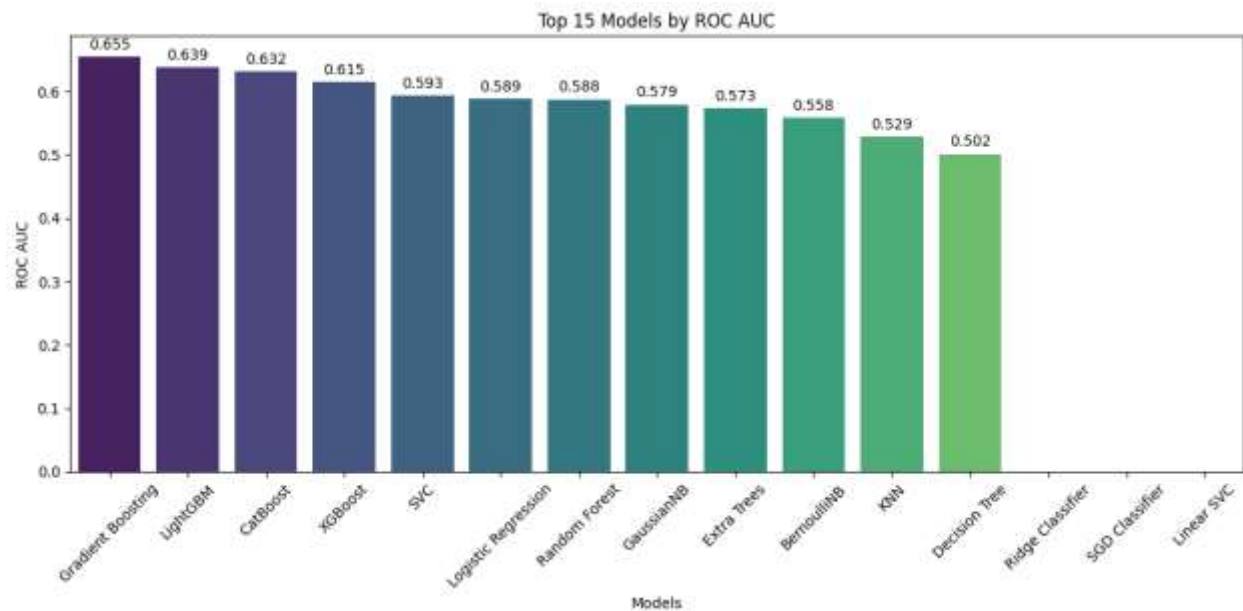


Figure 77 Models by ROC AUC

Gradient Boosting and LightGBM achieved the highest ROC AUC scores, demonstrating superior class separation compared to other models. While simpler models like Logistic Regression and Random Forest performed reasonably, their lower scores suggest limitations in capturing non-linear trends. The selection of [XGBoost/Gradient Boosting] aligns with the priority for robust discrimination, as reflected in the ROC AUC analysis.

E.7 Detailed Classification Report

	Accuracy	Precision	Recall	F1 Score	ROC AUC	Confusion Matrix
Gradient Boosting	0.935745	0.2	0.001333	0.002649	0.655051	[[10965, 4], [749, 1]]
LightGBM	0.205308	0.070519	0.937333	0.131169	0.638566	[[1703, 9266], [47, 703]]

CatBoost	0.663879	0.098464	0.521333	0.165643	0.632466	[[7389, 3580], [359, 391]]
XGBoost	0.694172	0.09305	0.432	0.153119	0.614814	[[7811, 3158], [426, 324]]
SVC	0.533237	0.079922	0.598667	0.141018	0.593125	[[5800, 5169], [301, 449]]
Logistic Regression	0.564383	0.080201	0.554667	0.140138	0.58926	[[6198, 4771], [334, 416]]
Random Forest	0.93182	0.019608	0.001333	0.002497	0.587845	[[10919, 50], [749, 1]]
GaussianNB	0.936001	0	0	0	0.579389	[[10969, 0], [750, 0]]
Extra Trees	0.926359	0.062016	0.010667	0.018203	0.573254	[[10848, 121], [742, 8]]
BernoulliNB	0.936001	0	0	0	0.558339	[[10969, 0], [750, 0]]
KNN	0.933612	0.111111	0.005333	0.010178	0.528927	[[10937, 32], [746, 4]]
Decision Tree	0.879597	0.066841	0.068	0.067416	0.501545	[[10257, 712], [699, 51]]
Ridge Classifier	0.563188	0.080146	0.556	0.140097	None	[[6183, 4786], [333, 417]]
SGD Classifier	0.183889	0.065898	0.892	0.12273	None	[[1486, 9483], [81, 669]]
Linear SVC	0.563273	0.080161	0.556	0.140121	None	[[6184, 4785], [333, 417]]

Table 15 Detailed Classification report

Among all the evaluated classification models, XGBoost demonstrated the most effective balance between predictive performance and practical applicability for the car insurance claim prediction task. While its accuracy stood at 69.4%, lower than models like Random Forest and Gradient Boosting (both above 93%), those models exhibited near-zero recall, effectively failing to identify claim instances.

In contrast, XGBoost achieved a substantially higher recall of 43.2%, indicating its superior ability to detect true positive cases critical in imbalanced classification scenarios. Its ROC AUC score of 0.615 further affirmed its discriminative capability, outperforming models such as Logistic Regression, SVC, and KNN.

Additionally, XGBoost offers key advantages in terms of computational efficiency, with optimized tree construction, parallel processing, and GPU support for scalability. Though its precision (9.3%) and F1 score (0.15) were relatively low due to the imbalanced nature of the dataset, these can be improved using techniques like class weighting or oversampling. Importantly, XGBoost supports advanced interpretability tools such as SHAP, enabling granular insight into feature contributions.

With a strong balance between generalization, class sensitivity, efficiency, and interpretability, XGBoost was selected as the final model for this study.

APPENDIX F

PERFORMANCE METRICS

This section provides a detailed explanation of the performance metrics used in the study.

F.1 Accuracy: Overall correctness

Accuracy measures the overall correctness of the model by calculating the proportion of total correct predictions:

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{True Positives} + \text{True Negatives} + \text{False Positives} + \text{False Negatives}} \\ (\text{Total Predictions})$$

Equation 13 Accuracy

While intuitive, accuracy may be misleading in imbalanced datasets, as it can be inflated by correct predictions on the majority class.

F.2 Precision: Proportion of correctly predicted positive claims

Precision measures the proportion of positive predictions that are actually correct. In this context, it reflects how many predicted claims were indeed real:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Equation 14 Precision

High precision means fewer false positives are important for reducing unnecessary claim investigations.

F.3 Recall: Ability to detect actual claims

Recall assesses the model's ability to correctly identify actual positive cases (claims): (especially important in imbalanced datasets)

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Equation 15 Recall

High recall is essential in imbalanced datasets, where failing to detect actual claims (false negatives) is more costly than misclassifying a few non-claims.

F.4 F1 Score: Balance between precision and recall

The F1 Score is the harmonic mean of precision and recall, offering a balanced metric when both false positives and false negatives are important:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Equation 16 F1 Score

F1 is particularly valuable in this project due to the imbalance in claim distribution.

F.5 ROC-AUC: Discriminative ability between classes

ROC-AUC measures the model's ability to distinguish between classes across different threshold values. The AUC value ranges from 0.5 (random guessing) to 1.0 (perfect classification):

- **ROC Curve:** Plots True Positive Rate (Recall) vs. False Positive Rate.
- **AUC:** Area under this curve.

A higher AUC indicates better model discrimination between claims and non-claims.

F.6 Confusion Matrix: Detailed classification outcomes

The confusion matrix provides a detailed breakdown of prediction results:

$$\begin{bmatrix} \text{True Positive} & \text{False Negatives} \\ \text{False Positives} & \text{True Negatives} \end{bmatrix}$$

Equation 17 Confusion Matrix

This helps visualize how many predictions were correct and where errors occurred, providing insight into model strengths and weaknesses.

APPENDIX G

MODEL INTERPRETATION

G.1 SHAP (SHapley Additive Explanations)

SHAP (Lundberg & Lee, 2017 [\[13\]](#)) is a comprehensive method for explaining machine learning predictions, grounded in Shapley values from cooperative game theory. It measures each feature's impact on the model's prediction by averaging its marginal contribution over all possible subsets of features.

- **Global Interpretability:** SHAP values can be aggregated across the entire dataset to show the overall importance of features. This helps identify which features most influence the model's decisions across all predictions.
- **Local Interpretability:** For a single prediction, SHAP shows how much each feature contributed to increasing or decreasing the predicted probability of a claim. This is presented in a force plot or waterfall chart.

$$\text{Model output} = \phi_0 + \sum_{i=1}^M \phi_i$$

Where

- ϕ_0 is the model's average output (baseline)
- ϕ_i is the SHAP value representing feature i 's contribution.

Equation 18 SHAP

SHAP is ideal for tree-based models like XGBoost and provides consistent and locally accurate attributions. It was used to:

- Visualize the top contributing features.
- Understand how features like policy_tenure, age_of_car, and is_speed_alert influenced predictions.
- Ensure the model aligns with domain knowledge and regulatory expectations.

G.2 LIME (Local Interpretable Model-Agnostic Explanations)

LIME (Ribeiro et al., 2016 [\[14\]](#)) interprets individual model predictions by creating a simple, interpretable model—such as a linear regression or decision tree—that approximates the complex model’s behavior in the local vicinity of the instance. It does this by generating slight variations of the input, collecting their predicted outcomes, and fitting the simpler model to capture how the original model behaves near that specific point.

- **Local Focus:** Unlike SHAP, LIME does not aim to explain the model globally. It only explains why a single instance received a particular prediction.
- **Example Usage:** For a customer predicted to file a claim, LIME showed which features (e.g., short policy_tenure, absence of brake_assist, or certain area_cluster) most influenced that decision.

LIME was particularly useful for:

- Case-by-case debugging and trust-building.
- Exploring “What-if” scenarios (e.g., what happens to prediction if a customer had ESC enabled).
- Explaining black-box behavior to non-technical stakeholders.